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**RAPID DIFFERENTIATION OF THE PLASMA
SAMPLES FROM ISCHEMIC STROKE
PATIENTS AND HEALTH CONTROL GROUP
AND INVESTIGATION OF THE EFFECTS OF
HYPOXIA ON VESICULAR LIPID
TRAFFICKING**

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PhD

The Hong Kong Polytechnic University

2022

The Hong Kong Polytechnic University

**Department of Applied Biology and Chemical
Technology**

**Rapid Differentiation of the Plasma Samples from
Ischemic Stroke Patients and Health Control
Group and Investigation of the Effects of Hypoxia
on Vesicular Lipid Trafficking**

Wang Jianying

**A thesis submitted in partial fulfilment of the
requirements for the degree of Doctor of Philosophy**

August 2021

CERTIFICATE OF ORIGINALITY

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Abstract

Stroke is one of the leading causes of death and disability in the world, and ischemic stroke (IS) accounts for around 85% of the total stroke cases. IS should be treated with tissue plasminogen activator within 3-4.5 hours of the stroke onset, and this time window is called the “golden window” of stroke treatment. However, conventional methods for diagnosis of IS, such as computed tomography and magnetic resonance image, are time-consuming and sometimes they cannot provide reliable results, hereby the discovery of biomarkers for rapid diagnosis of IS is a strategy to solve the problem. However, most of the biomarkers reported so far are proteins, which cannot readily cross the blood-brain barrier to cause significant changes in blood within 3-4.5-hour “golden window”. Lipids account for about 10% wet weight of the brain and they alter immediately once ischemic injury happens, thus they are potential biomarkers of IS. Therefore, we aimed to discover lipid biomarkers for rapid and reliable diagnosis of IS utilizing lipidomics approach. In this project, plasma samples from IS patient group and health control (HC) group were collected and analyzed after optimization of the sample preparation procedures and liquid chromatography mass spectrometry (LC-MS) protocols. Principal component analysis (PCA) based on the obtained LC-MS data showed good distinguishment of the IS and HC groups in the

discovery dataset, indicating that IS patients could be differentiated from HC based on their lipid profiles in the discovery phase. Subsequently, based on the marker discovery in discovery dataset and the evaluation of validation dataset, a total of 20 lipids were selected as the potential biomarkers. Among these 20 potential markers, the levels of FA(20:0), LPC(O-20:0), LPS(O-15:0), nordeoxycholic acid, 23:0 sterol, and 14:0 cholesterol ester were significantly increased in the plasma of IS patients compared with their levels in the health controls. In contrast, the remaining 14 lipids showed reductions in IS patients, including n-butyl arachidonoyl amine, FA(20:3), TG(14:0/18:2/18:3), TG(15:1/19:1/20:5), PS(P-20:0/20:5), PI(17:0/22:1), PC(O-38:9), PI(O-20:0/15:1), PE(P-16:0/20:3), PC(18:2/22:6), PS(O-18:0/19:1), PS(20:0/22:4), glycinoprenol-9, and Cer(m18:1/24:1). Furthermore, after binary logistic regression analysis, a combination of three lipids, i.e., FA(20:3), PS(O-18:0/19:1) and LPS(O-15:0), showed distinctive differentiation of IS and HC with area under the curve (AUC) value, sensitivity, and the specificity of 1.000, 100% and 100%, and 0.963, 100% and 87.5% for the discovery dataset and the validation dataset, respectively. Therefore, the combination of FA(20:3), PS(O-18:0/19:1) and LPS(O-15:0) was defined as a “biomarker panel” and could be applied in the diagnosis of IS. The development of LC-MS-based methods allowed rapid analysis of

plasma samples and diagnosis of IS with high accuracy, thereby shortening the diagnosis time before tpA treatment and reducing the mortality and morbidity of IS patients. In addition, we could find that the correlation network among lipid biomarkers and phospholipid and sphingolipid pathways were altered upon IS onset, indicating that these potential IS-related lipid markers may be the targets of IS prevention and treatment. Furthermore, the levels of six lipid biomarkers including GlcCer(t18:1/18:0), GlcCer(d16:2/22:0), LPE(O-14:0), PG(O-20:0/22:0), PI(O-18:0/19:1) and PC(20:5/20:4) were significantly changed in IS patients when compared to HC, which suggested that these six lipids might help with the assessment of the severity of IS patients and evaluate the prognosis of IS treatment.

To further shorten the analysis time required for IS diagnosis, direct analysis in real time-mass spectrometry (DART-MS), a rapid and simple technique that allows direct analysis of samples with minimal sample pretreatment, was attempted for direct analysis of the plasma samples from the IS patients and HC group. The spectral data from 36 IS samples and 19 health controls were used as the discovery dataset to build an orthogonal projection to latent structures discriminant analysis (OPLS-DA) model, and the result of response permutation testing indicated that the established OPLS-DA model was valid

without overfitting. The score plots of OPLS-DA model showed good differentiation of the IS patient group from the HC group with prediction power for both the HC group and IS group of 100%. Subsequently, an additional spectral data from 17 IS samples and 9 HC samples were employed to further validate the established OPLS-DA model with the prediction abilities of 100% and 94.1% for the healthy group and the disease group, respectively. Therefore, the established OPLS-DA could be employed for the rapid differentiation between IS and HC groups using DART-MS.

Vesicle trafficking is closely related to the progress of diseases. However, the studies of vesicle trafficking have been mainly focused on the transport of proteins, with very few on vesicular trafficking of small molecules such as lipids. In this project, the lipidomics studies of semi-intact (SI) cells and vesicle samples were carried out to respectively reflect the alterations of lipid compositions of intracellular membrane system and vesicle after hypoxia, an important characteristic for development of tumor cell. Interestingly, the level of glycerophosphocholine (PC) was increased significantly in SI cell treated with insufficient oxygen. However, the levels of glycerophosphoglycerol (PG), glycerophosphoserine (PS) and lysophosphocholine (LPC) decreased markedly after hypoxia, and the levels of diacylglycerol (DG) and

glyceroethanolamine (PE) were not changed in SI after hypoxic cultivation, indicating that the hypoxia inhibited the catabolism pathway of PC, instead of PC anabolism pathway. Since sphingolipid and lysophospholipid, bioactive lipids that were related to the vesicular transportation, were significantly altered in the SI cell after oxygen deprivation, we further investigated the relationship between hypoxia and vesicular lipid composition. Interestingly, different from other glycerophospholipid (GP), the level of PE was decreased in vesicles from the hypoxic cells, so we presumed that the rigidity of vesicle membrane might be increased by hypoxia. Strikingly, our data showed that significantly more long fatty acyl chains ($C > 40$) were observed in vesicles under hypoxia compared to normal condition, which further confirmed that stimulation of hypoxia increased the stiffness of vesicle membrane. Moreover, the transmission electron microscope (TEM) result showed that membrane of hypoxic vesicle became thicker than that of normal vesicle, it might be due to the longer fatty acyl chain accumulated in hypoxic vesicle. In addition, the Nanosight and TEM results showed that vesicle size was significantly increased under hypoxia. This study revealed that lipid composition of intracellular membrane system and vesicle were significantly influenced by oxygen deprivation, and hypoxia increased the rigidity of vesicle membrane

and vesicle size, which might shed a new light on the study on mechanism and treatment of cancer.

Research publications

Journal papers

1. Wu, Q.; Wang, J. Y.; Han, D. Q.; Yao, Z. P. Recent Advances in Differentiation of Isomers by Ion Mobility Mass Spectrometry. *Trends Anal. Chem.* **2020**, *124*, 115801.
2. Wang, J. Y.; Wu, Q.; Yao, Z. P. Recent Advances in Applications of Mass Spectrometry in Discovering Biomarkers of Ischemic Stroke (Chinese). *Fenxi Ceshi Xuebao (J. Instrum. Anal.)* **2020**, *39*, 44-49. (Invited contribution)
3. Chao, Y.; Wang, X.; Wang, J. Y.; Wang, X. J.; Chen, W. T.; Lu, N.; Siniossoglou, S.; Yao, Z. P.; Liu K. Rewiring Neuronal Glycerolipid Metabolism Determines the Extent of Axon Regeneration. *Neuron* **2020**, *105*, 276-292.

Conference papers

1. Wang, J. Y.; Yin, Y. H.; Xin, G. Z.; Yao, Z. P. Least Absolute Shrinkage and Selection Operator (LASSO)-based Prediction of Collision Cross Section (CCS) Values for Lipid Isomers. The 8th Asia-Oceania Mass Spectrometry Conference (AOMSC). Macau, 3 Jan **2020**. (Poster)
2. Wang, J. Y.; Yin, Y. H.; Xin, G. Z.; Yao, Z. P. Lasso-based Prediction of Collision Cross Section Values for Lipids: Differentiation of lipid

- isomers. HKSMS Symposium 2019 & the 21st Annual General Meeting of The Hong Kong Society of Mass Spectrometry. Hong Kong, 22 June **2019**. (Poster)
3. Wang, J. Y.; Yin, Y. H.; Xin, G. Z.; Yao, Z. P. Lasso-based Prediction of Collision Cross Section Values for Analysis of Lipids by Ion Mobility Mass Spectrometry. The 26th Symposium on Chemistry Postgraduate Research in Hong Kong. Hong Kong, 4 May **2019**. (Poster)
4. Wang, J. Y.; Yin, Y. H.; Xin, G. Z.; Yao, Z. P. Lasso-based Prediction of Collision Cross Section Values for Lipids. 2018 China Mass Spectrometry Society (CMSS) Annual Conference, China, 23 Nov **2018**. (Poster)

Acknowledgements

I would like to express my sincere gratitude to my honorable supervisor, Prof. Zhongping Yao, for his patient guidance, advice, and supervision. I have gained abundant valuable experience and comprehensive knowledge of mass spectrometry and lipidomics under Prof. Yao's support. He also gave me the freedom to explore new knowledge and provide me with the resources, suggestions, and inspiration when I established a new platform for vesicle research.

My sincere thanks are given to my team members for their kind help. I would like to specially thank Dr. Pui Kin So and Dr. Yee Man Wong for their valuable and thoughtful suggestions on my research. Dr. So and Dr. Wong have a lot of experience in the usage of Agilent Q-Tof and Orbitrap that I used frequently for my projects. I learnt a lot about not only the use but the setup and troubleshooting of mass spectrometry, which always makes me excited and feel like I get the treasure. I am truly fortunate to have my groupmates Dr. Qian Wu, Dr. Haidi Yin, Miss Dong Zhang, Mr. Tsz Fung Wong, Mr. Zhen Yan Li, Miss Xuwei Lin, Mr. Ho Yin Ma, and Mr. Jun Dai. Dr. Wu and Dr. Yin gave me many suggestions about the ischemic stroke study and lipidomic analysis of vesicles, respectively. Miss Zhang helped with the cell culture, Mr.

Wong, Mr. Li, Miss Lin, Mr. Ma, and Mr. Dai helped with the rat liver cytosol preparation. Furthermore, Miss Lin and Mr. Ma also helped with the data processing of raw data produced from DART. Finally, I would like to express a sincere thanks to Miss. Hongman Leung and Dr. Suying Li for proofreading this thesis.

The support and inspiration from my family and friends are also acknowledged. The love from them always supports me to persist and to be positive when I was suffering from any difficulties.

My sincere and heartfelt thanks go to all the collaborators, Dr. Zhao Han, Dr. Haixing Wang, and Mr. Zicheng Cheng from the First Affiliated Hospital of Wenzhou Medical University for the provision and collection of the clinical plasma samples, and Prof. Yusong Guo and Dr. Yan Huang from The Hong Kong University of Science and Technology for their help with the establishment of the platform for vesicle research. I also acknowledge the supports by Alice and Heiman from Centralized Animal Facility for the dissection of the rats.

Last but not the least, I also appreciate all those people who give me support or help. I will remember all your help even though some of them are not mentioned.

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List of abbreviations

Full form	Abbreviation
Ambient ionization mass spectrometry	AIMS
Area under the curve	AUC
Atmospheric pressure chemical ionization	APCI
Blood-brain barrier	BBB
Carnitine palmitoyl transferase	CPT
Ceramide	Cer
Ceramide phosphoethanolamine	Cer-PE
Ceramide phosphoinositol	Cer-PI
Clear cell renal carcinoma	ccRCC
Coat protein complex I	COPI
Coat protein complex II	COPII
Computed tomography	CT
Desorption electrospray ionization	DESI
Deuterated-ceramide	d-ceramide
Diacylglycerol	DG
Direct analysis in real time	DART
Disialoganglioside	GD

Distance between dart source and sample supporter	DS
Dulbecco's modified eagle medium	DMEM
Electrospray ionization	ESI
Endoplasmic reticulum	ER
ER exit site	ERES
ER-Golgi intermediate compartment	ERGIC
False discovery rate	FDR
Fatty acyl	FA
Fetal bovine serum	FBS
Fold change	FC
Free fatty acid	FFA
Gas chromatography	GC
Glycerolipid	GL
Glycerophosphocholine	PC
Glycerophosphoethanolamine	PE
Glycerophosphoglycerol	PG
Glycerophosphoinositol	PI
Glycerophospholipid	GP
Glycerophosphoserine	PS
Glycosphingolipids	GSL

Glycerol-3-phosphate acyltransferase	GPAT
Hemorrhagic stroke	HS
Hydrophilic-interaction liquid chromatography	HILIC
Internal standard	InterS
Ischemic stroke	IS
Liquid chromatography-mass spectrometry	LC-MS
Low-density lipoprotein	LDL
Lysophosphatidic acid	LPA
Lysophosphocholine	LPC
Lysophosphoethanolamine	LPE
Lysophosphoglycerol	LPG
Lysophosphoinositol	LPI
Lysophospholipid	LPL
Lysophosphoserine	LPS
Magnetic resonance image	MRI
Mass spectrometry	MS
Monoacylglycerol	MG
Monosialoganglioside	GM
Multiple reaction monitoring	MRM
Nanoparticle tracking analysis	NTA

Odds ratio	OR
Orthogonal projection to Latent structures- discriminant analysis	OPLS-DA
Paper spray ionization	PSI
Phosphatidic acid	PA
Plasma membrane	PM
Platelet activating factor	PAF
Polyketides	PK
Prenol lipid	PR
Quality control	QC
Rat liver cytosol	RLC
Receiver operating characteristic	ROC
Relative log abundance	RLA
Saccharolipid	SL
Selective reaction monitoring	SRM
Semi-intact	SI
Serine palmitoyltransferase	SPT
Sphingolipid	SP
Sphingomyelin	SM
Sphingosine 1-phosphate	S1P

Stearoyl-CoA desaturase	SCD
Stearoyl-Coenzyme A	CoA
Sterol lipid	ST
Systolic blood pressure	SBP
Tandem mass spectrometry	MS/MS
Tissue plasminogen activator	tPA
Total ion current	TIC
Triacylglycerol	TG
Triple quadrupole mass spectrometry	QqQ
Unsaturation index	UI
Variable importance in projection	VIP
Very-low-density lipoprotein	VLDL

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Chapter 1: Introduction

1.1 General introduction of lipids

1.1.1 Definition and classification of lipids

There are various definitions of lipids. Some define lipids as not only fatty acids and their derivatives but also the components that are involved in the functions and biosynthesis of fatty acids.¹ Fathy et al. describe lipids as small hydrophobic or amphiphilic molecules that may originate entirely or in part from carbanion-based condensations of thioesters (such as fatty acids, polyketides) and/or from carbocation-based condensations of isoprene units (such as prenols, sterols), and this definition is related to the solubility, structure and nomenclature of lipids.² Since the definitions mentioned above are vague and lack structural information about lipids, the LipidWeb put forward a new definition: a lipid is a relatively small (m/z 50~1200) and hydrophobic organic molecule; either fatty acyl chains or isoprene units are the building block of its structure; hydroxyl group and carboxyl group are the major substitutionally oxygenated groups in the lipid structure; polar groups such as glycerol, glycosyl and phosphate are commonly covalently connected to the acyl chains or isoprene units.³

According to the classification of LIPID MAPS consortium, there are eight categories of lipids, namely fatty acyl (FA), glycerolipid (GL), glycerophospholipid (GP), sphingolipid (SP), sterol lipid (ST), saccharolipid (SL), prenol lipid (PR), and polyketides (PK) (**Figure 1-1**). Every category can be subdivided into several classes and each class can be further subdivided into various subclasses, and many lipid species, whose specific lipid class together with the number of carbon atoms and C=C double bond are clearly known, are included in each subclass. In addition to fatty acid, FA consists of other classes such as fatty alcohol, fatty aldehyde, and fatty ether. GL is a type of lipid with glycerol, to which the ester linkage of the carboxylic acid from the fatty acyl chains are linked with the hydroxyl groups. GL is generally divided into three classes, monoacylglycerol (MG), diacylglycerol (DG) and, triacylglycerol (TG), according to how many fatty acyls are connected to the glycerol. As one of the most common lipids, GP is primarily composed of glycerophosphocholine (PC), glycerophosphoethanolamine (PE), glycerophosphoserine (PS), glycerophosphoglycerol (PG), glycerophosphoinositol (PI), and glycerophosphate (PA) (**Figure 1-2**). SP mainly includes sphingoid base, ceramide (Cer) and phosphosphingolipid (**Figure 1-3**), and there are three subclasses of the phosphosphingolipid class, which are ceramide phosphocholine/sphingomyelin (SM), ceramide

phosphoethanolamine (Cer-PE), and ceramide phosphoinositol (Cer-PI). Different from the aforementioned lipid categories, other lipid categories, such as ST, SL, PR and PK, constitute only several lipid classes, their further classifications are thus not shown here.

The nomenclature of the lipid specie used in this thesis is based on the principle used in LIPID MAPS, which indicates the number of carbon atom and C=C double bond for every individual fatty acyl chain in the structure. For instance, FA (18:1) means that the FA specie has only one fatty acyl chain with 18 carbon atoms (C18) and one C=C bond; TG (20:1/20:1/22:0) indicate that all three hydroxyl groups of glycerol are linked with fatty acids, two of them are C20 fatty acyl chain with one C=C double bond and the third fatty acid is saturated with 22 carbon atoms.

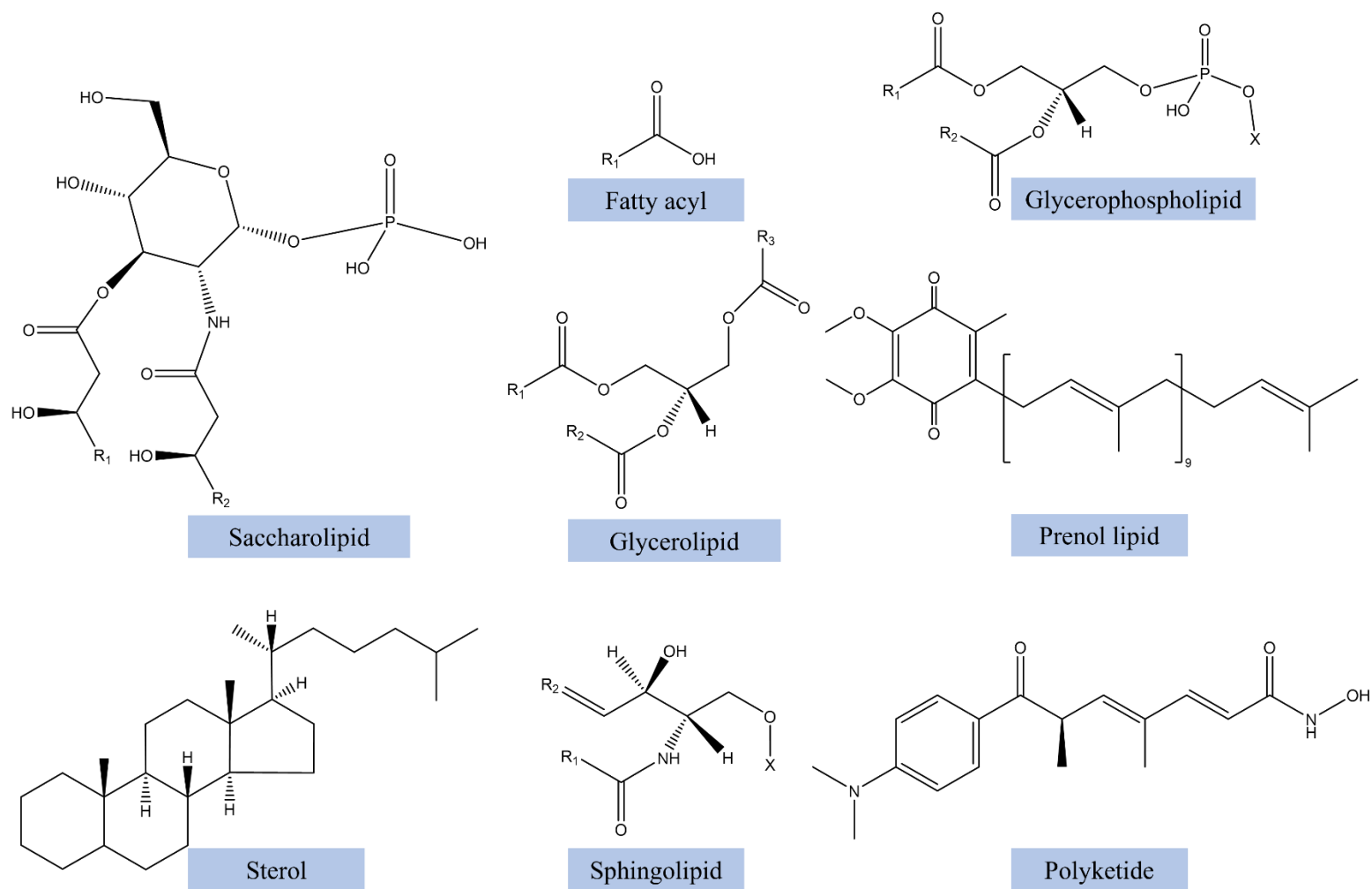


Figure 1-1. Eight lipid categories according to the LIPID MAPS consortium. One representative structure is shown for each category

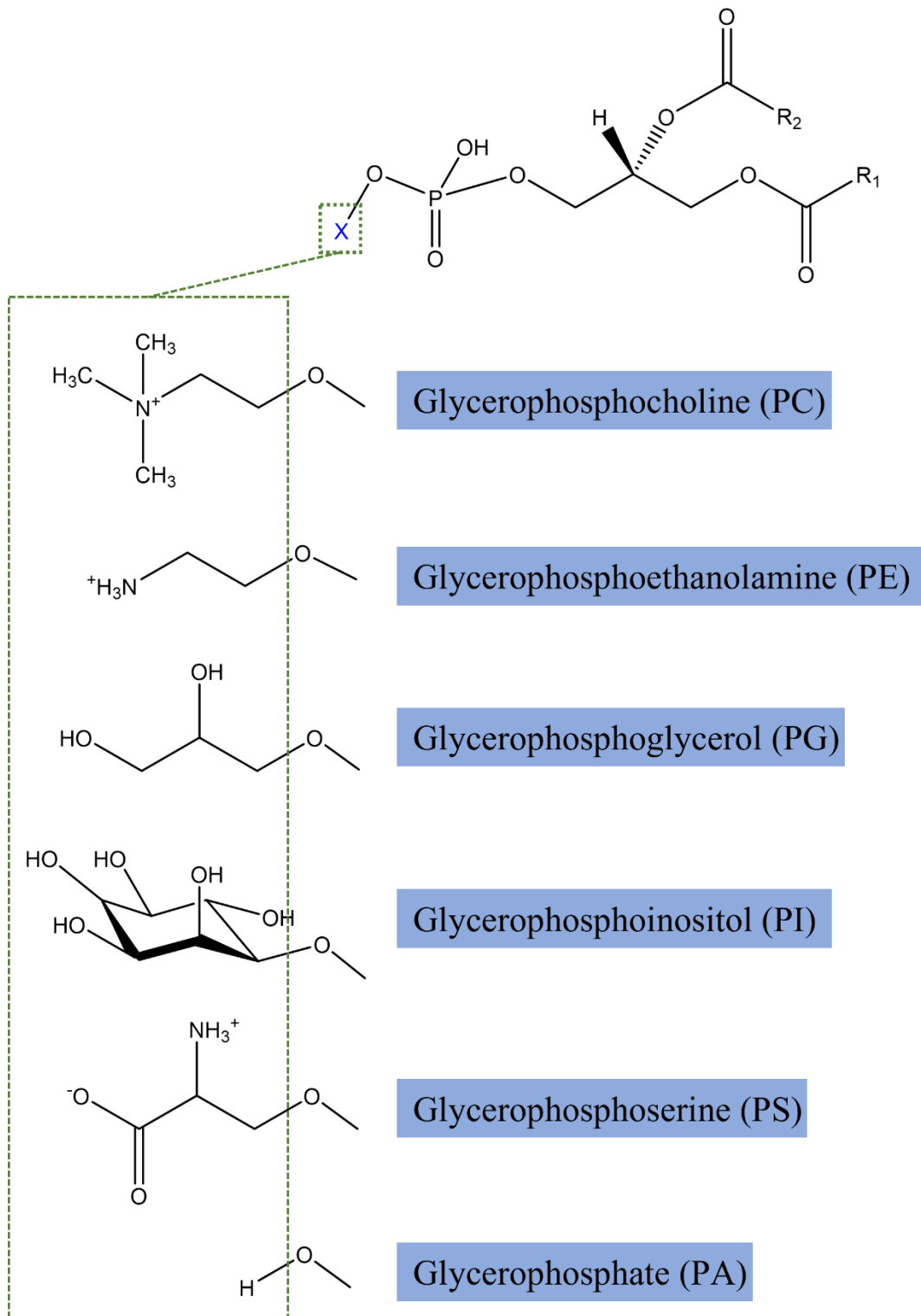


Figure 1-2. Structure of glycerophospholipids and their subclasses

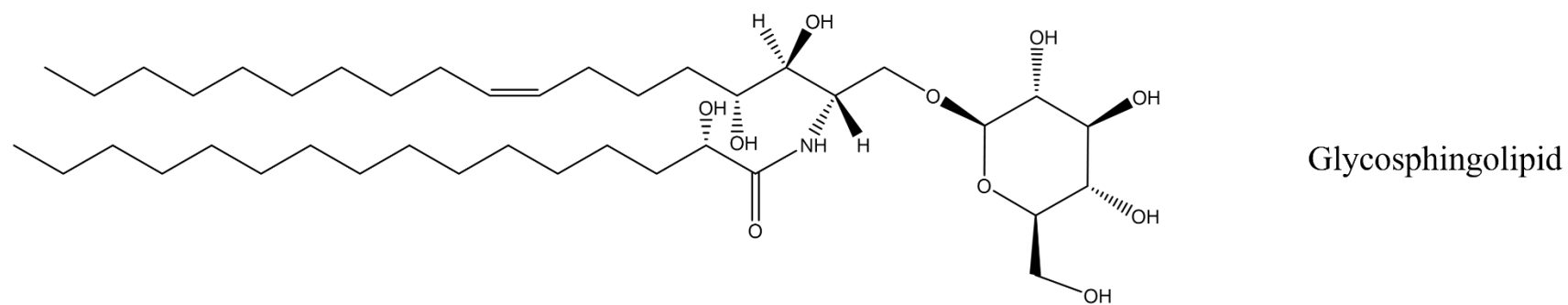
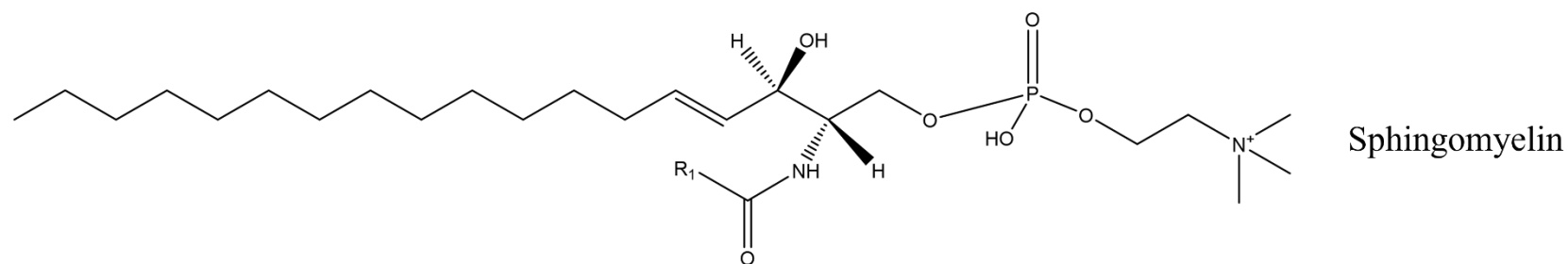
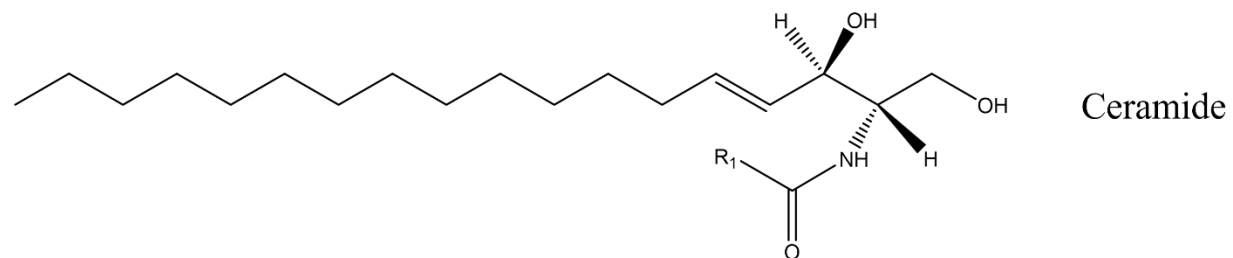
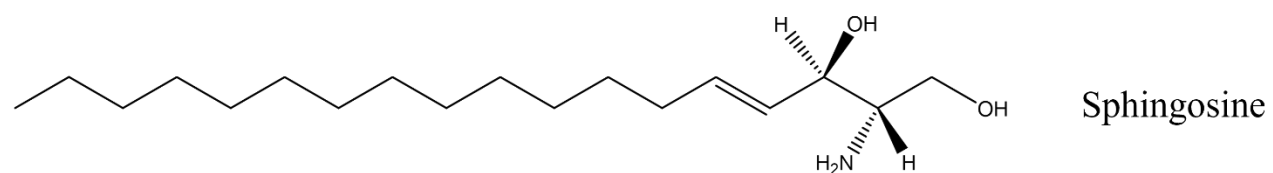


Figure 1-3. Representative structures of four major sphingolipid subclasses

1.1.2 The biological functions of lipids

Lipids are one of the most essential components of biological systems, and they play many crucial functions, including energy depots, membrane matrices and signaling.^{4, 5} First of all, triacylglycerol, a subclass of GP, is commonly stored in lipid droplets as a calory reservoir to provide energy. Second, the biological membranes predominantly comprise of amphiphilic phospholipids. The lipid bilayer structure in the plasma membrane separates the intracellular components from the extracellular environment, and the membrane system within the cell functions as the organelle barrier to recruit or restrict specific reaction products. Furthermore, lipid membrane is also strongly related to the vesicle budding, fission and fusion, which are central for the trafficking of important bioactive molecules between different compartments. The third function of lipids is that they play an essential role in signal transduction as the second messengers.

1.1.3 Lipid metabolism

The main metabolic networks in most organisms include fatty acid metabolism, glycerophospholipid metabolism, glycerolipid metabolism, and sphingolipid metabolism. The understanding of the metabolic pathways of

lipids facilitates the interpretation of lipid alteration during disease development and progress.^{6, 7}

1.1.3.1 Fatty acid metabolism

Catabolic pathway and anabolic pathway are two important processes in fatty acid metabolism (**Figure 1-4**) that are essential to the energy homeostasis and conversion to other lipid intermediates for the maintenance of membrane structure and function, energy storage and signal transduction. The key roles of fatty acid metabolism in the pathogenesis of several diseases, such as the central nervous system diseases (e.g., stroke, Alzheimer's disease, Parkinson's disease) and cancers, have been elucidated in recent years.

The β -oxidation of fatty acid in the mitochondria is a major process to produce energy, fatty acid is thus required for the initiation of β -oxidation when upon energy is needed. With the support of acyl-CoA synthetases, fatty acid is transferred to fatty acyl-CoA, which is subsequently conveyed to the mitochondria via carnitine-acylcarnitine translocase and carnitine palmitoyl transferase (CPT) for β -oxidation and Krebs cycle.⁸ CPT1 is one of the CPTs, and it functions as the rate-limiting enzyme for shipment of fatty acid to the mitochondria. The inhibitor of CPT1, etomoxir, and the direct inhibitor of

fatty acid oxidation reduce inflammation and demyelination in the central nervous system and may help to kill cancer cells.^{9, 10}

The anabolic process mainly refers to the synthesis of fatty acid, which occurs in the cytosol. Citrate produced from the Krebs cycle is transferred to acetyl-CoA, and acetyl-CoA is then converted into malonyl-CoA by acetyl-CoA carboxylase. As the original component of the fatty acid synthesis process, palmitate-CoA (also named C16:0-CoA) results from the reaction of malonyl-CoA and fatty acid synthase.^{11, 12} Likewise, malonyl-CoA is the initial product for the elongation of fatty acid. The elongation and desaturation of fatty acid are further conducted in endoplasmic reticulum (ER) by various elongases and desaturases, respectively. Emerging evidence has unveiled that rapamycin, the mediator of fatty acid synthesis, is significantly altered upon the development of Parkinson's disease and Alzheimer, and not surprisingly, the suppression of the expression of rapamycin indicates the alleviation the above-mentioned diseases.¹³ Cancer is fundamentally a disease related to cell proliferation, and cell growth requires many nutrients, such as proteins, saccharides, and nucleic acids. Lipid is another essential nutriment that cancer cells need for the membrane structure and energy provision, and fatty acid is included in the structures of most categories of lipids, therefore fatty acid synthesis is

important for the proliferation of cancer cells, which could be a target of cancer treatment.¹⁴ CD36, a transmembrane protein, has various biological functions and its function as the regulator of fatty acid uptake in breast cancer has been disclosed by several studies.¹⁵

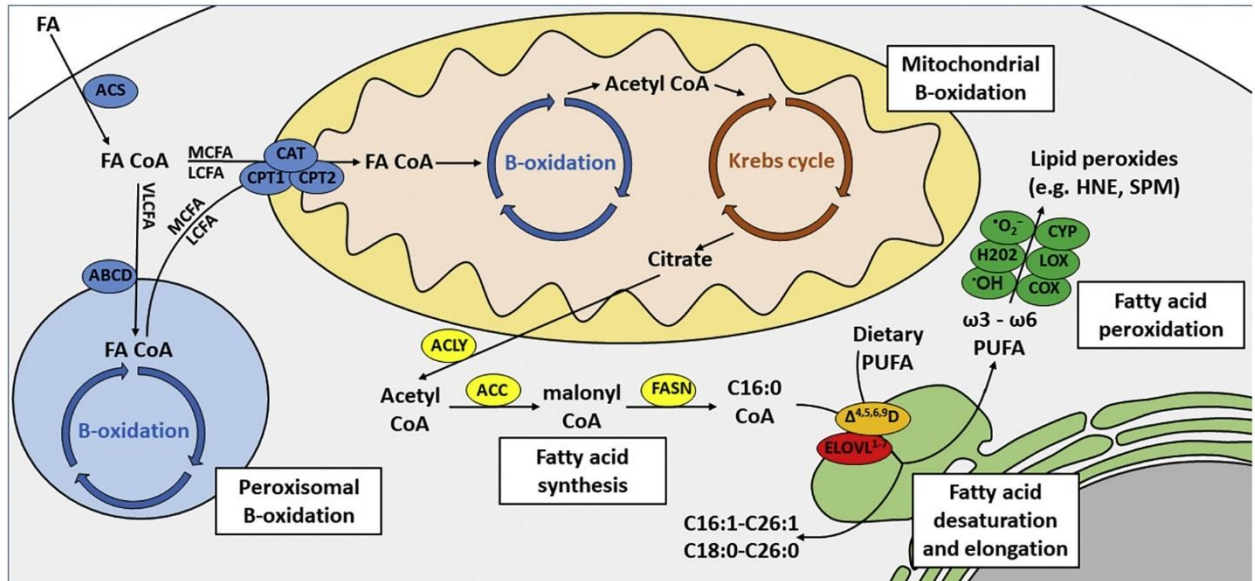


Figure 1-4. Fatty acid metabolism (Reprinted from Ref¹⁶)

1.1.3.2 Glycerophospholipid metabolism

The majority of synthesized fatty acids could be incorporated to the glycerol backbone to form diacylglycerol, which is the basis of GPs. GPs synthesized by the DG pathway are understood as the driving factors of many metabolic dysfunctions and diseases. The major metabolism pathway of phospholipids is illustrated in **Figure 1-5**.

Membrane glycerophospholipids, the most abundant lipid category, are classified into PC, PE, PI, PS, PG and PA based on the differences in their polar head groups and their corresponding lysophospholipids (LPLs), including lysophosphocholine (LPC), lysophosphoethanolamine (LPE), lysophosphoserine (LPS), lysophosphoglycerol (LPG), lysophosphoinositol (LPI), and lysophosphotidic acid (LPA). PC and PE are the most abundant phospholipids in the membrane of mammal cells. PC is commonly synthesized by condensation of CDP-choline with diacylglycerol scaffold for many cells. One of the synthesis pathways of PE is analogous to that of PC, the CDP-ethanolamine is added to diacylglycerol to form PE. And PE synthesis can be achieved by decarboxylating PS with the activation of phosphatidylserine decarboxylase.^{17, 18}

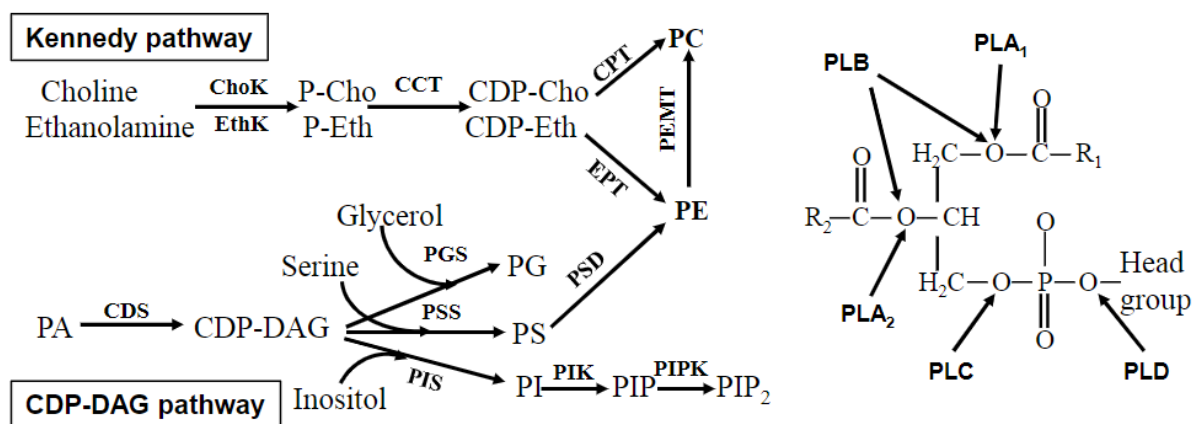


Figure 1-5. The glycerophospholipid metabolism (Reprinted from Ref¹⁷)

It is reported that phospholipids, instead of TG or DG, are the potential regulators of muscle insulin resistance. For example, the knockout of PE cytidylyltransferase, the enzyme linked with the PE *de novo* synthesis in mice muscle, contributed to the accumulation of TG and DG in tissues. Moreover, deficiency in the PE cytidylyltransferase facilitated the development of metabolic disorders.¹⁹ PC is indispensable for the synthesis and the secretion of very-low-density lipoprotein (VLDL), and perturbation of the secretory pathway was observed in people whose diet lacked choline and PC cytidylyltransferase A, which is an essential enzyme for PC synthesis in the CDP-choline pathway.^{20, 21}

Additionally, the PC/PE ratio was found to be involved in many physiological processes and this ratio could be imbalance in the obese overfed with fatty acids. An analysis of skeletal muscle in overweight men has underscored a strong correlation of PC/PE value to insulin sensitivity.^{22, 23} Moreover, the PC/PE ratios in hepatic cells have been reported to be inversely linked with the progression of nonalcoholic fatty liver disease.²⁴ In addition to the disease, the PC/PE ratio is also important for the membrane structure. For example, membrane integrity could be destroyed by low PC/PE ratio, which can ultimately change the membrane function, such as the permeability of

cytokines. Decreased PC:PE ratio induced by deficiency of phosphatidylethanolamine N-methyltransferase, the enzyme that catalyzes PC synthesis by PE methylation, led to ER stress and improved sensitivity to steatohepatitis in mice.²⁵

Plasmalogen, an ether phospholipid that predominantly occurs in nature and that is mainly classified as PC or PE, is a plasmenyl with the vinyl-ether function group linked with the *sn*-3 fatty acyl chain of lipid (**Figure 1-6**). The esterification of plasmalogens with polyunsaturated FA, such as arachidonic acid and docosahexaenoic acid, could be widely found in nature. Plasmalogens represent more than 50% of membrane phospholipids and their abundance is especially high in cardiac and neural cells. Endoplasmic reticulum and the peroxisome are the major organelles for the biosynthesis of plasmalogen, and the synthesis process requires many enzymes. The synthesis of the long-chain fatty acyl is mediated by the fatty acyl CoA reductase 1 and 2, which is the key step in the synthesis of plasmalogen. Serum plasmalogens are involved in the metabolomic dysfunction diseases, such as diabetes, obesity, cardiovascular disease, and ischemia, which are closely related to oxidative stress.²⁶ Therefore, it is speculated that plasmalogens, especially the vinyl-ether linkage of the plasmalogens, might play an important role in the

oxidative stress.²⁷ The inverse relation between plasmalogen and myocardial infarction was emphasized by the study conducted by Joseph V Moxon et.al.²⁸ However, investigation of the functions of plasmalogen in various metabolic diseases is still required.

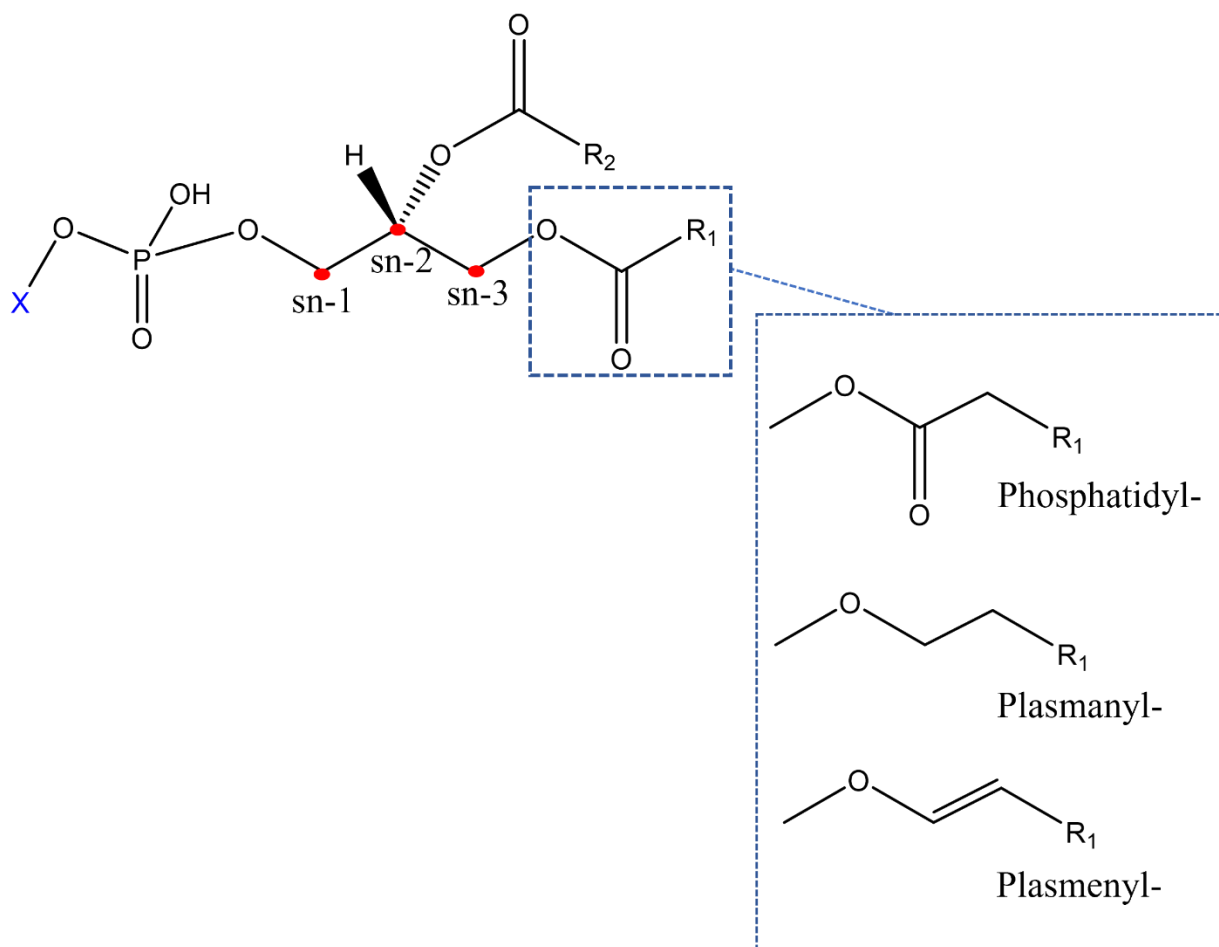


Figure 1-6. sn-1, sn-2, and sn-3 positions in the glycerol backbone and three subclasses of phospholipid classified based on the type of fatty acyl chain linked to the sn-3 position of the glycerol backbone

1.1.3.3 Glycerolipid (GL) metabolism

GL can be further subdivided into three subclasses, MG, DG, and TG. DG can be formed from the acylation of MG, the hydrolysis of TG via lipase, the dephosphorylation of PA by phosphatase, and the hydrolysis of the GP head group catalyzed by phospholipase C, which is the reason why DG is regarded as an important resource for the GP synthesis.²⁹⁻³² DGs could be produced in different organelles, such as ER and plasma membrane, and function as substrates for TG synthesis and signaling molecules, separately. TG, the dominant energy reservoir in organisms, can be synthesized by the addition of another fatty acyl chain to DG by diglyceride acyltransferases (**Figure 1-7**). The close relation between the modulation of TG synthesis and metabolic diseases has been reported. For example, glycerol-3-phosphate acyltransferase (GPAT), an enzyme that produces LPA from glycerol-3-phosphate, plays a role in the homeostasis of TG. Deficiency in GPAT not only induces the decline of TG synthesis and accumulation, but also decreases the possibility of liver cancer in mice treated with carcinogen.³³⁻³⁵

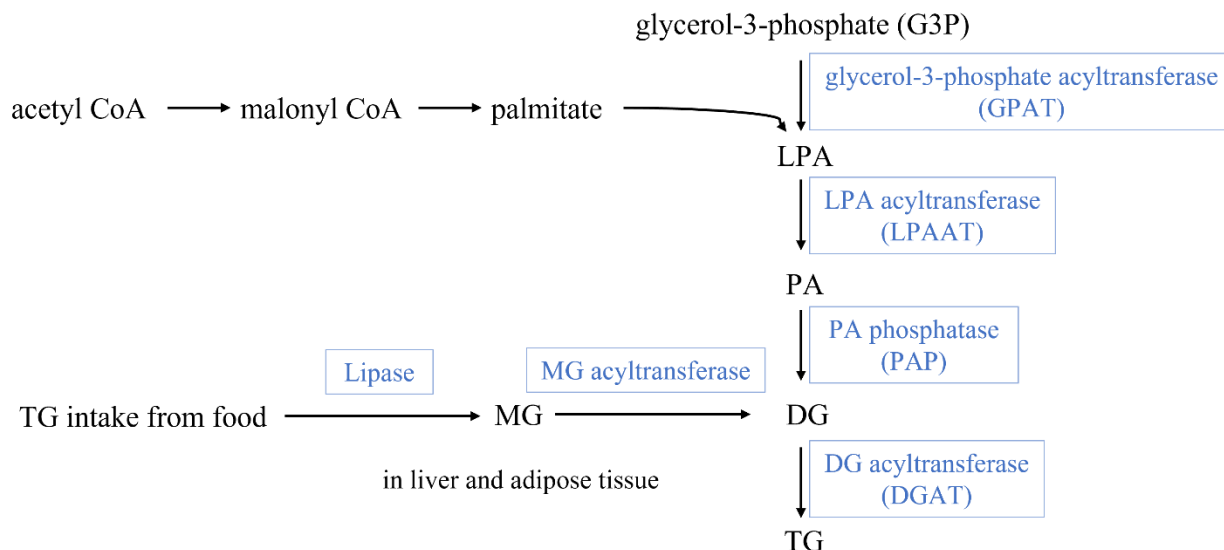


Figure 1-7. The biosynthesis and catabolism pathway of glycerolipids

Fatty acid is the major form by which energy is stored in complex organisms and animals and emerging evidence shows that perturbation of lipid metabolism, particularly those lipids related to GL/free fatty acid (FFA) cycling, is increasingly linked with metabolic syndrome diseases and the pathogenesis of cancers. Lipid droplets, an organelle with phospholipid monolayer surrounding the neutral lipids, e.g., glycerolipids and cholesterol esters, are the major intracellular source of GL formation for GL/FFA cycling, and they are essential to almost all lipid-related processes and function within almost all cells. There is increasing evidence indicating that many signaling lipids are manufactured by the GL/FFA cycling. The disturbance of these lipids with essential functions would affect many pathophysiological

processes, such as type 2 diabetes, obesity, non-alcoholic fatty liver disease and cancers.³⁶

1.1.3.4 Sphingolipid metabolism

Sphingolipids represent one of the major lipid categories in eukaryotic cells. The sphingoid bases, such as sphingosine, are the building block of all sphingolipids. The decarboxylated serine residue and the catalyzation of fatty acyl-CoA by serine palmitoyltransferase (SPT) take part in the reaction of the *de novo* synthesis pathway to manufacture ceramides.³⁷ Then enzymatic reactions, including the catalyzation by 3-ketoreductase, dihydroceramide synthases and desaturases, produce ceramide, which is the precursor for most complex sphingolipids. Subsequently, various sphingolipids, such as glycosphingolipids, ceramide-1-phosphate, and sphingomyelin, are predominantly synthesized by ceramides in the Golgi apparatus. Meanwhile, those sphingolipids can be hydrolyzed to ceramides in reverse within the lysosome. Besides, the diacylation of ceramide contributes to the formation of another important sphingolipid, the sphingosine, which could be further phosphorylated to sphingosine-1-phosphate. Additionally, sphingolipid basically consists of the sphingoid base and fatty acyl chain, and sphingolipid variations with slight difference in the sphingoid base or fatty acid could be

formed due to the catalyzation of various enzymes. Ceramide synthases introduce fatty acyl groups of distinct chain length, while SPT utilizes myristate or stearate instead of palmitate to introduce variations in the length of the sphingoid backbone. The pathway of sphingolipid is summarized in **Figure 1-8.**

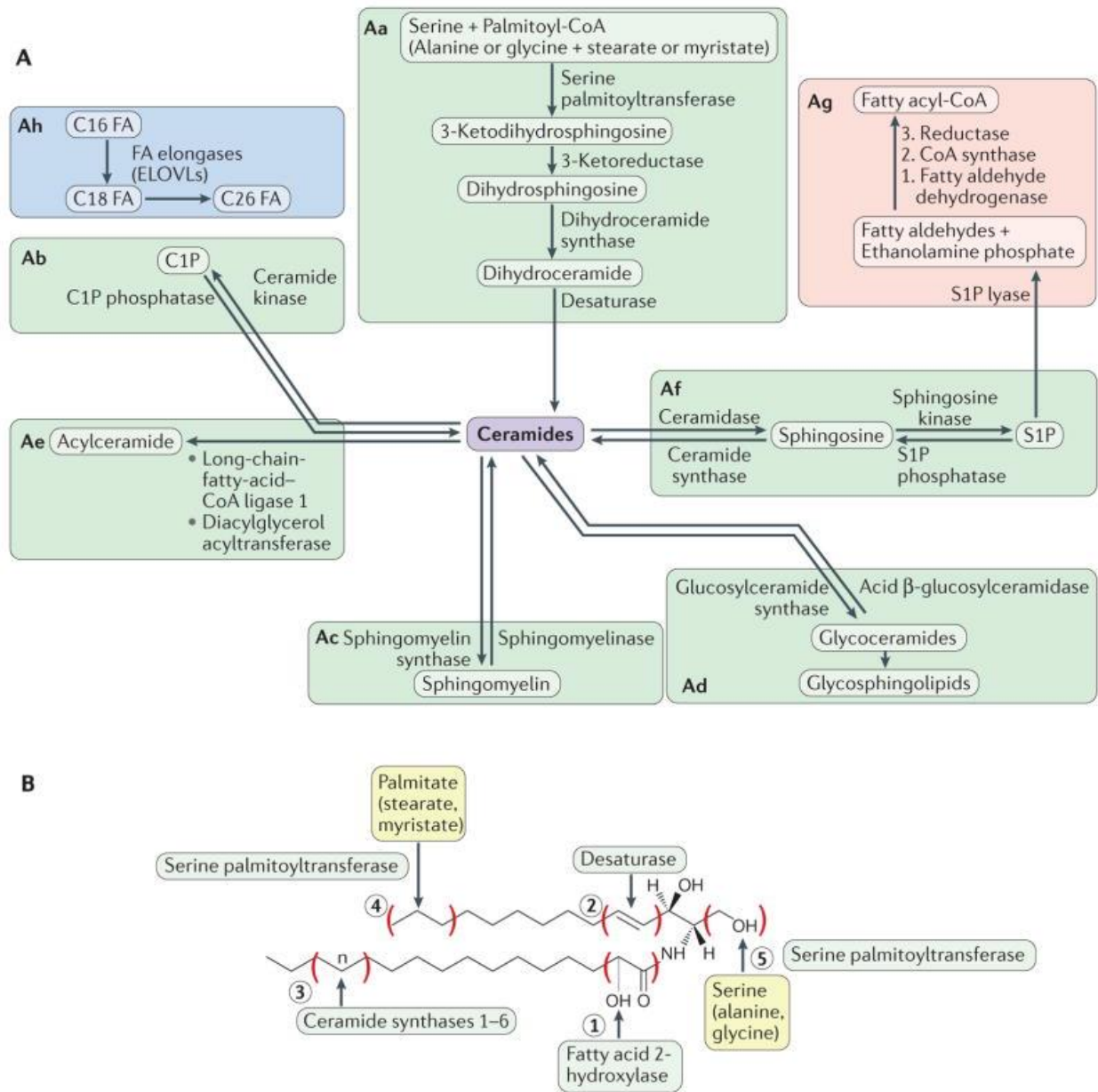


Figure 1-8. Overview of sphingolipid metabolism (Reprinted from Ref³⁷)

Glycosphingolipids (GSL) is a class of the SPs and many studies have shown that abnormal GSL level is associated with a variety of cancers, such as leukemia, colorectal cancer and lung cancer, and some specific GSLs have been chosen as the targeted antigens for tumors and the preclinical and clinical studies for their related antibodies have been carried out.³⁸⁻⁴⁰ The globotriaosylceramide are increased in human cancers, and glonotetraosylceramide facilitates cell proliferation and is overexpressed in cancer patients as well by mitogen-activated protein kinase pathway.⁴¹⁻⁴³ Comparative cohort studies by Li et al. between patients with benign breast tumor and healthy people suggest that monosialoganglioside (GM) 3 could be a potent marker for prognosis of breast cancer. Compared with GM3, GM2 has been used as the treatment target for many years due to its ability of tumor initiation^{44, 45}. And the overexpression of GM1 suppresses the tumor and alters the tumor phenotype^{46, 47}. In terms of disialoganglioside (GD) family, increased GD2 and GD3 are both connected with metastatic ability, proliferation, and angiogenesis of cancer cell.^{42, 48} However, GD1a is a good sphingolipid and abundant GD1a alleviated cancer metastasis by initiating caveolin-1 expression.^{49, 50} Accordingly, sphingolipid and glycosphingolipids could be the targeted compounds for therapy development. Besides, a metabolomics study based on a population of 491 on the linkage of plasma

sphingolipid with the development of prostate cancer demonstrates that the mechanisms of caveolin-1, which is strongly associated with prostate cancer, are the altered oncometabolomics of sphingolipids and extracellular vesicle biogenesis.⁵¹

The structure of SM, a type of sphingolipid, is similar to that of phosphatidylcholine. The shift of the phosphocholine moiety from phosphatidylcholine to the backbone of ceramide contributes to the production of SM, which is the most abundant in circulating low-density lipoprotein (LDL). Sphingomyelins in the lipid raft in the plasma membrane are in conjunction with insulin resistance, obesity, and dysfunction of liver and atherosclerosis.⁵² Higher level of SM 16:1 in plasma reduced the risk factor of diabetes, but the decline of plasma SM 24:1 occurred upon high-fat diet.^{53, 54} Another evident for the function of sphingomyelin is that the inhibition of SPT via myriocin to impede SM biosynthesis alleviated the symptom of atherosclerosis in mice.⁵⁵ Further study should be implemented to investigate the role of sphingomyelins in metabolic disorders.

1.2 MS-based Lipidomics analysis

Cellular lipids are highly complex and very dynamic. Tens to hundreds of thousands of possible lipid molecular species as predicted to be present in the cellular lipidome at the level of attomole to nanomole of lipids per mg of protein.⁵⁶ The physiological contributions of these diverse lipids and how their levels change in response to therapy remain largely unknown. In order to cover a wide range of lipid type, the chemically distinct lipid species in a cell, an organ, or a biological system were collected and the totality of lipids has been referred to as a lipidome.⁵⁷ By analogy to other “omics” disciplines, studying lipidomes in a large scale and at the levels of intact molecular species by utilizing analytical chemistry technologies and tools has been defined as lipidomics.⁵⁸ Direct infusion-based lipidomics, also known as shotgun lipidomics, and liquid chromatography-mass spectrometry (LC-MS)-based lipidomics are the most popular approaches in the lipidomic field.

1.2.1 Direct infusion mass spectrometry

As shotgun lipidomics allows samples to be introduced to MS directly, it can be used to quantitate bulks of lipids quickly.^{58, 59} It is also well-known to researchers in this field due to its high throughput when coupled with autosampler platform and the striking and compatible software for data

processing and data analysis. The major characteristic of this method is that all the lipids and other compounds, such as the matrix ions, are ionized concomitantly during the acquisition time by MS, which tends to contribute to ion suppression among the coelutes. However, since compounds are ionized under the same condition and time, internal standard can be spiked in samples for normalization, and better quantitative results can be obtained. However, it has several limitations, for instance, non-linear signal will be apparent if the concentration of sample is too high, and aqueous impurities, like salt and other polar components, should be eliminated to reduce ion suppression, and care should be taken especially for biological samples. Therefore, single-phase extraction before shotgun lipidomic analysis is usually employed.

1.2.2 Liquid chromatography-mass spectrometry (LC-MS)

The separation of lipid classes or species can be achieved when LC is coupled with MS. In addition to the separation, retention time could be obtained for the identification of lipid species. Although ion suppression effect is largely decreased by LC-MS, a longer analysis time will be required, usually at least 10 mins is needed for lipidomic analysis. Compared with shotgun lipidomics, LC-MS-based lipidomics is more tolerant to impurities. The separation result

depends on the properties of the analytical column and the mobile phase, hereby the optimization of the LC conditions should be optimized prior to analysis. Reverse-phase chromatography, C18 in particular, is commonly used because of the hydrophobic properties of most of the lipids. After samples were injected into LC, The C18 stationary phase adsorb lipids and then the lipids will be eluted with the mobile phase upon. To achieve better separation, the ratio of water to organic solvent in the mobile phase can be varied throughout the running time. A wide range of polarities for lipid species can be eluted if the gradient of the mobile phase is set from the high aqueous ratio to the high organic composition. And better separation for individual lipid species can be acquired if the mobile phase composition is constant. Another commonly used column is the hydrophilic-interaction liquid chromatography (HILIC), which shows powerful separation ability for some polar lipids due to its polar stationary phase. Naturally, this necessitates the reversal of mobile phase compositions, starting with the highest hydrophobicity and decreasing to elute analytes. By contrast, the retention of lipid species by C18 is based on the length and the saturation degree of fatty acyl chain, but lipid separation that relies on HILIC is commonly affected by the polarity, such as the polar head group of phospholipids, which can be employed for the differentiation of lipid classes. Therefore, the two-

dimensional liquid chromatography combining the columns illustrated above is popular to resolve lipid species, where those lipids co-eluted from the first dimension are injected to the second dimension. Furthermore, the second dimension provides increased confidence for lipid identification.

1.2.3 Tandem mass spectrometry (MS/MS)-based lipidomics

Tandem mass spectrometry (MS/MS)-based shotgun lipidomics is well-known to us due to its advantage in the good sensitivity, high efficiency, and simple operation. The commercially available technique, triple quadrupole mass spectrometry (QqQ), is commonly used to quantitate individual species of lipids in complicated biological samples by MS/MS acquisition. Multiple reaction monitoring (MRM, also known as selective reaction monitoring (SRM)) is a highly specific and sensitive mass spectrometry technique that can selectively quantify lipids within complex mixtures. The strategy can also be coupled with the shotgun or LC-MS lipidomics for high-throughput analysis or lipid separation. However, some concerns about this strategy should be kept in mind when we utilize MRM. Firstly, the characteristic fragments used in this method are commonly the polar head group fragments, whose fatty acyl composition could not be identified. Secondly, MS/MS scanning is not totally specific to the targeted lipids, for example, there are

other artifacts or lipid isomers that share the same m/z values with the analytes of interest. Finally, different fragmentation kinetics or thermodynamics may occur for the lipid species from the same lipid category, making the quantitative analysis might not be accurate as expected.

As illustrated above, LC-MS can be employed for the separation of lipids based on the stationary phase and the mobile phase, and it can be used for the separation of lipid isomers, which exert different biological functions and are commonly seen in biological samples. But LC is not the best choice for the differentiation of lipid isomers, particularly for geometrical isomers. Ion mobility mass spectrometry is famous for isomer separation, and it can be coupled with LC-MS to provide the second dimension for identification, further improving identification accuracy.

1.3 Ambient ionization mass spectrometry (AIMS)

Ambient ionization was defined as “the ionization of unprocessed or minimally modified samples in their native environment, and it typically refers to the ionization of condensed phase samples in air” by Prof. R. Graham Cook.⁶⁰ In the last two decades, innovations and progress in ambient ionization mass spectrometry have been made to enable the experimental

process labor-saving and time-saving, this technique has thus been applied widely in the scientific communities.

After the invention of two major ambient ionization techniques, solvent-based desorption electrospray ionization (DESI) and plasma-based direct analysis in real time (DART),⁶¹ a number of novel ambient ionization methods and their variations, such as thermal desorption, laser ablation, and vibrational excitation have been created for ionizing different samples under atmospheric condition. Based on the powerful potential of mass spectrometry in the field of analytical chemistry, AIMS has been extensively applied in the direct analysis of various samples in their native environments. AIMS techniques can be primarily categorized into three main classes according to the desorption method, and they are liquid extraction, laser ablation, and plasma desorption.

The liquid extraction approach typically refers to the desorbing of analytes by solvent and ionizing compounds by electrospray ionization (ESI), it is thus commonly employed for the analysis of those compounds which are able to be ionized by ESI and ESI-related ionization mechanism. Liquid extraction ambient ionization-based mass spectrometry basically includes spray-based

extraction, direct liquid extraction, and substrate spray. The molecules in samples can be desorbed by the solvent spray and then analyzed by MS in spray-based techniques, such as DESI. When direct liquid extraction techniques are applied, analytes are generally extracted directly by solvent, followed by detection by MS. However, ions can be directly created from samples or substrates for substrate spray techniques. The representative substrate spray technique is paper spray ionization (PSI), where liquid sample is allowed to be pipetted onto filter paper and then introduced to MS with the support of the spray solvent and voltages after the liquid samples are dried.

Although laser ablation methods, which primarily utilize UV or infrared laser sources for the enhancement of sample desorption, are regarded as AIMS techniques as well, low ionization efficiency is usually obtained by means of this method given that neutral ions represent the majority of desorbed ions. In order to promote its ionization efficiency, other sources, like ESI, DART, atmospheric pressure chemical ionization (APCI), are coupled with the laser ablation approach.

The principle of plasma-based desorption ambient ionization resembles that of APCI, where the electrical discharge electrodes produce the plasma, and

the analytes of interest will be ionized after the interaction of the plasma and the carrier gas. DART, the first plasma ambient ionization technique, is still the most widely used technique and it has been exploited in many fields. Gas molecules, such as helium, nitrogen, and argon, are induced from a DART source, followed by the excitation of the corona discharge needle into protonated water clusters. The excited gas molecules are subsequently exposed to the samples for desorption and ionization of the targeted analytes.^{62,}

⁶³ Although DART could be employed for the rapid analysis of many compounds in both positive and negative ion modes, it is particularly suitable for the detection of volatile compounds due to the vaporization of analytes prior to ionization. Moreover, another limitation of plasma ionization approach is that samples are desorbed by the thermal gas, resulting in the fragmentation or sample damage.

1.4 Vesicle trafficking of lipids

1.4.1 The definition of vesicle

Vesicle is a structure with contents surrounded by the lipid bilayer which is majorly composed of high abundance of phospholipids together with some sphingolipids and cholesterol (Figure 1-9). The hydrophobic parts of phospholipids consisting of fatty acyl chains are attracted and congregated

together and face inward. In contrast, two hydrophilic head groups of phospholipids are neatly exposed in the cytosolic and intra-vesicular environments. The majority of components, such as protein and lipid, are synthesized in ER or Golgi apparatus, and these components should be moved or secreted to other organelles to exert their functions. Besides, some toxic substances within cells are supposed to be removed for the purpose of cell survival. It is reported that vesicle plays essential roles in the delivery of components, and because the organelle membranes are mainly made up of phospholipids as well, it is easy for vesicle to fuse with other membranous materials. There are three common intracellular vesicles in mammal, including secretory vesicle, lysosome and transport vesicle. Different vesicles have different specific functions. Secretory vesicle carries materials which will be secreted out of cells, and these materials could be the wastes produced by cells or chemicals secreted by the specialized cells. For example, the neurotransmitter is carried by synaptic vesicle. Lysosome contains many enzymes that mainly digest the destroyed organelles and degrade useless or harmful components. Transport vesicle, we are focus on in this thesis, is represented as those vesicles that move molecules, such as protein or lipid, from the donor membrane to the receptor membrane inside cells, and the term “vesicle” will be represented as transport vesicle unless otherwise specified.

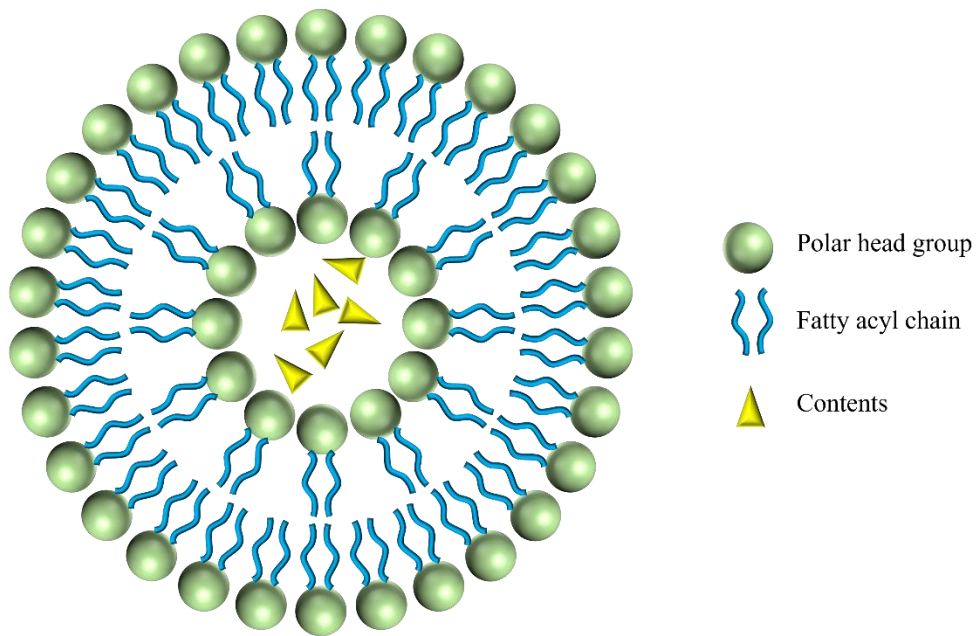


Figure 1-9. Diagram of vesicle structure

1.4.2 The functions of transport vesicle

Complex endomembrane within eukaryotic cells functions as secretory or endocytic pathways, playing an essential role in cellular physiology. The secretory process encompasses the biosynthesis of proteins or lipids in ER and the delivery of the cargo from ER to their next destinations, such as different organelles (Golgi apparatus, endosome, and lysosome) and extracellular environment. The communications between different organelles and between the plasma membrane and intracellular organelles rely on the vesicular transport in the secretory pathway. Transport vesicles ferry protein and lipid from the donate organelle to the target organelle, which maintain the

composition and function of the target organelles. The incorrect delivery of protein could induce the defects in immunity and the membrane polarity. Likewise, the inappropriate shipment of lipid will change the lipid composition of both donor and acceptor membranes and thus these membrane structure will be destroyed and interfere the proper functions of membrane. Coat protein complex II (COPII) vesicle mainly plays a role in the transport of the cargo from ER to Golgi, and this transport is called anterograde transport. Coat protein complex I (COPI) vesicle mediates the intra-Golgi transport and also the retrograde transport from Golgi or ER-Golgi intermediate compartment (ERGIC) to ER.

1.4.3 Vesicular transport of lipid

Similar to proteins, lipids are predominantly synthesized in ER, thereby providing the membrane lipids to the organelles along the secretory/endocytic pathways. Different organelles have different functions depending dominantly on different lipid composition of different organelle membranes. Lipid composition not only influences the category of sorting protein, but the membrane properties of organelle such as curvature and fluidity.^{64, 65} Therefore, lipid composition has a remarkable function in organelle function in eukaryotic cells. The primary lipid classes in the organelle membrane are

glycerophospholipids, sphingolipids, and sterols.⁵ Charge-paired phospholipid (PC, PE, and the anionic phospholipid, PI) are most abundant in the early secretory compartments, such as ER and cis-Golgi. However, low levels of other negatively charged phospholipids (PS and PA) accompanied with sphingolipids and sterol could be found in these organelle membranes. Therefore, slightly loose and thin membrane structure could be observed in ER, but plasma membrane (PM) is much tighter and thicker than the ER membrane due to the high level of sphingolipids and sterols in PM.⁶⁶ In addition, two ER membrane leaflets show symmetrical lipid distribution, whereas asymmetric distribution of lipid is displayed for plasma membrane, Golgi, and endosomal membrane with relatively high abundance of PS and PE in the cytosolic leaflet and SM and GSL on the non-cytosolic side of membranes.^{67, 68}

It has been well studied that membrane lipids can be transported from ER to Golgi, PM, endosome and lysosome through a vesicle-independent mechanism, requiring either lipid transfer proteins or the membrane contact site, which is typically located between the acceptor membrane and the ER membrane. It is reported that lipids can be transported to the endomembrane systems not only by means of non-vesicular pathways but by a vesicle-based

mechanism upon their synthesis within ER. As the major structural components of the vesicle membranes, a bulk of lipids could be transported bidirectionally along the secretory pathway accompanied with newly synthesized proteins via the vesicular pathway. Being far from just the structure backbone of the vesicle membrane or just the server as the protective cover, lipids play an essential role in vesicular trafficking. For example, lipids are able to propel cytosolic proteins to be recruited for the formation of transport machinery and correlate with the membrane curvature to facilitate vesicle fission, budding and fusion.

However, the vesicular transportation of lipid is unexplored and there is still a niche in this research field. The enrichment of SL and sterols can be achieved by prior inclusion of these two lipids in the secretory vesicles or preferentially exclusion of them out of the vesicle membrane involved in the retrograde pathway, which is driven by the vesicular-based lipid trafficking. Compliant with this, lower abundances of SLs and cholesterol are observed in the retrograde vesicles originated from the Golgi apparatus than that of the Golgi membrane in mammalian cells. The transport of different ER-synthesized lipids shows various dependence on COPII-based trafficking. Although it is reported that cholesterol and *de novo* synthesized ceramides are

mainly transported from ER through a vesicle-independent mechanism, and in mammals, the delivery of most ceramides with very long fatty acyl chain (e.g., C24) from ER to the cis-Golgi to serve as the precursor of the synthesis of sphingolipids relies on the COPII-dependent vesicles, whereas ceramides with shorter fatty acyl chain are most probably exported to the cis-Golgi via a non-vesicular pathway. Besides, COPII vesicles are involved in the synthesis of cholesterol as well. Furthermore, the membrane bending to capture the cargo is a challenge for COPII, the difficulty can be solved if the ER membrane can be mechanically deformed easily with the help of alteration of lipid composition, which can determine the membrane bending rigidity. The membrane flexibility can be improved when the specific lipids are accumulated at the ER exit site (ERES), thereby enabling the deformation of COPII-regulated membrane. These specific lipids are identified as lysophospholipids, which are conical lipids credited to contribute to the enhancement of membrane curvature and fluidity and then facilitate the formation of vesicles. In addition, it is illustrated that conical lipids were linked with the function of COPII machinery.

1.5 Outline of this thesis

The PhD projects targeted on the study of the differentiation of ischemic stroke (IS) based on LC-MS and DART-MS, and the lipid composition of the vesicle under hypoxic condition will be explored as well.

In chapter 1, the basic knowledge of lipid and the field of lipidomics will be elaborated first and then analytical techniques such as LC-MS and AIMS will be introduced. Finally, the lipid trafficking within the intracellular environment will be discussed.

A LC-MS-based differentiation model of IS and health control (HC) will be established in chapter 2 for rapid diagnosis of IS. First, a discovery dataset will be employed to discover the potential biomarkers, and an additional dataset will be used for the validation of the level of these potential markers. Second, the prediction ability of these validated potential markers for IS will be tested by a binary logistic regression model. Finally, the differentiation capability of the potential markers will be evaluated by the receiver operating characteristic (ROC) curve analysis. The development of this method could shorten the diagnosis time of IS, and then tPA could be treated within the golden window to minimize disability and mortality of IS.

DART-MS, a simple and rapid technique for direct analysis of plasma, will be utilized to develop a method for rapid differentiation of IS and HC in chapter 3. Experimental parameters such as the temperature and the grid voltage will be optimized. And an Orthogonal projection to Latent structures-discriminant analysis (OPLS-DA) model will be established based on the discovery dataset, the differentiation ability of the established OPLS-DA will be assessed using the validation dataset. The established approach could achieve more rapid diagnosis of IS and be employed with LC-MS method complementarily.

In chapter 4, lipid composition of semi-intact (SI) cell or vesicle under hypoxic condition will be studied and then compared with the corresponding lipid composition under normal culture condition. The quantitative categories will include glycerophospholipid, glycerolipid, and sphingolipid. The length of fatty acyl chain will be also calculated in this chapter to reflect the influence of hypoxia in lipid trafficking. Additionally, nanoparticle tracking analysis and transmission electron microscope will be utilized for vesicle characterization. The study of hypoxic effect on vesicle trafficking could provide a new prospective for cancer treatment.

Finally, the summary for the results of all the projects will be shown in chapter 5.

Chapter 2: Lipidomics analysis of plasma from ischemic stroke patients

2.1 Introduction

Stroke is one of the leading causes of death and a major cause of disability in the world.^{69, 70} In Hong Kong, stroke is the No. 4 cause of death, with a mortality rate of 20-25% in the first year after stroke and about 3000 deaths per year, according to the data from the Centre for Health Protection in Hong Kong in 2017.⁷¹ Stroke includes hemorrhagic stroke (HS) and IS. More than 87% of stroke cases are IS,⁶⁹ which is most prevalent among the elderly but has shown increasing incidence in working-aged individuals in recent years.^{72, 73} IS, also referred to as cerebral infarction, is a group of clinical syndromes caused by a sudden interruption of blood supply to brain tissues that results from various pathophysiologic processes, which finally leads to hypoxic-ischemic necrosis and causes a loss of normal neurological function in the focalized area.⁷⁴ The lack of blood supply to the brain is caused by a blood clot, or plaque buildup, or their combination. IS is considered to result from atherosclerosis and thrombosis, a condition characterized by the buildup of lipids and plaque on the blood vessel walls. Most of the time atherosclerosis and thrombus do not cause any symptoms, but for IS patients, some specific

characteristics are common across most of the IS, such as face drooping and arm weakness, which often occur on just one side of the body.⁷⁵ Since the blockage in the blood vessel decreases blood flow and oxygen supply to the brain, brain tissue damage and death occur. And if the blood circulation is not recovered promptly, the damage or death of brain cells cannot be reversed, and that is the reason why long-term disability is a very common outcome of IS. IS brings both physical and psychical harm to patients. In addition to its high prevalence, it is also a disease known to bear high morbidity, mortality, and reoccurrence rate. Stroke survivors are handicapped across many domains, especially in their physical independence and occupation, which in turn, become a tremendous burden to family and society. With the aging population, we are facing a problem of increasing severity.

Tissue plasminogen activator (tPA) is the only FDA-approved medicine for IS, which could be used to dissolve blood clots and restore the blood supply, but such treatment should typically be made within 3-4.5 hours of the onset of stroke.^{69, 73, 76} Rapid and reliable diagnosis of IS is thus extremely important, and is currently one of the major problems in neurology. In current clinical practice, neurovascular imaging, computed tomography (CT) and magnetic resonance image (MRI) are usually employed for IS diagnosis.⁷⁷ CT is

difficult to diagnose acute IS, but it could be detected by MRI, which has higher sensitivity and specificity, so these two techniques are commonly used together. However, because the diagnosis decision is mainly made by doctor by analysis of the CT or MRI result, these IS diagnostic techniques are time-consuming, subjective and sometimes give unreliable results,⁷⁸ which might cause the delay of treatments and aggravation of brain damage. Therefore, developing new tools for early prognosis and diagnosis of IS is crucial. Discovery of IS biomarkers and development of corresponding detection techniques can be the key to solve this problem. A lot of efforts have been made to identify potential blood biomarkers of IS.^{79, 80} However, most of these identified biomarkers are proteins or peptides, like S100b and neuron-specific enolase, which are hydrophilic and do not readily cross the blood-brain barrier (BBB) to cause a significant increase in blood within the 3~4.5-hour “golden window”, and thus cannot fulfill the requirement.⁸¹⁻⁸³ These problems can be overcome by using lipids as biomarkers, since lipids are hydrophobic and can readily cross the BBB.^{83, 84} Lipids account for about 10% wet weight of brain, and brain lipids alter immediately once ischemic injury happens.^{85, 86} Five major lipid categories are reported to be associated with IS risk, namely fatty acyls, glycerolipids, glycerophospholipids, sphingolipids, and sterol lipids. An investigation with a US cohort of Whites revealed a significant positive

association between plasma saturated and monounsaturated fatty acids and IS.⁸⁷ Another investigation also revealed that composition of phospholipid fatty acids in plasma was associated with stroke risk in a Korean population, with higher proportions of elaidic acid and arachidonic acid and lower proportion of lignoceric acid in cardioembolic stroke, a subtype of IS.⁸⁸ A recent study using the mouse model also demonstrated that plasma unesterified fatty acids were dramatically and acutely altered at early stages of IS.⁸³

The searching of lipid biomarkers of IS has been conducted using metabolomics and lipidomics approaches.⁸⁹ In two earlier studies, the fatty acid compositions in plasma phospholipids were found to be significantly different between the non-stroke controls and IS patients, and content of docosahexaenoic acid in plasma was suggested to be an indicator of the risk of IS.^{90, 91} A sphingolipid-targeted study in 2015 demonstrated sphingolipids as potential plasma biomarkers of acute brain injury.⁸⁴ Metabolic profiling of human serum in 2017 identified some lipids, including PC(5:0/5:0) and LPE(18:2), as potential markers of IS.^{92, 93} A recent lipidomics analysis of plasma samples from patients with lacunar infarction, one of the IS subtype, and health controls also determined 13 differential lipid species.⁹⁴ According

to the previous studies, the metabolism of glycerophospholipids has a strong association with atherosclerosis and thrombosis which are processes that involve lipids and plaque build up on the blood vessel walls and then result in IS. During the process of thrombus formation, negatively charged aminophospholipids were the major component of the procoagulant platelet surface, and served to anchor vitamin K–dependent coagulation factors and enhanced the thrombin generation.⁹⁵ The change in the distribution of PLs in the plasma membrane of activated platelets stimulated by the combined action of thrombin and collagen was considered as a result of transbilayer movement.⁹⁶ It was also found that the increase in PS in the membrane of platelets was accompanied by an increased ability to enhance the conversion of prothrombin to thrombin which was related to the occurrence of IS. As the most frequently studied LPL, LPA was thought to be the primary platelet-activating lipid of atherosclerotic plaques in LDL.^{97, 98} LDL could activate endothelial cell and platelets, so it is generally considered as a critical risk factor in the pathogenesis of atherosclerosis and its thrombotic complication of IS in clinical practice.⁹⁸ Siess and Tigyi revealed that LPA could trigger platelet aggregation and intra-arterial thrombus generation and that the amount of LPA within human carotid atherosclerotic lesion was the highest in the lipid-rich core.⁹⁹ A large number of studies have also demonstrated the

occurrence of membrane PLs degradation in human cerebral ischemia.¹⁰⁰⁻¹⁰²

A recent study showed that TMEM16F-mediated platelet membrane phospholipid scrambling contributed significantly to hemostasis and thrombosis.¹⁰³ Direct profiling of rat brain tissue by matrix-assisted laser desorption/ionization mass spectrometry demonstrated the significant changes of PLs and LPLs after ischemic injury.^{104, 105} Differential PLs and LPLs between health controls and IS patients have also been reported in previous metabolomics and lipidomics studies.^{84, 92-94}

As introduced above, lipids are ideal biomarkers for IS due to their ability to cross the BBB and their rapid changes upon ischemic injury. However, although some metabolomics and lipidomics studies have already been carried out for identifying potential lipid biomarkers, those studies mainly focused on comparative studies of patients of one of the IS subtypes and health controls, or only targeted individual classes of lipids, such as fatty acids and sphingolipids. So far, there have been no lipidomics studies of plasma to differentiate IS patients and health controls. Mass spectrometry (MS) is rapid, sensitive, and is the technique of choice for metabolomics and lipidomics studies.^{106, 107} With the assistance of tandem mass spectrometry (MS/MS) and high resolution MS, the specific lipid molecule species could be identified.¹⁰⁸⁻

¹¹⁰ Some state-of-the-art mass spectrometers, such as Orbitrap Fusion Lumos, have never been used in the reported metabolomics and lipidomics studies of IS.^{83, 84, 88-94} The higher resolution of these instruments may allow us to obtain more accurate results and new insights into the biomarkers and related mechanism.

The objectives of this project are to seek the marker that can be used for differentiation of IS patients and the health controls and to develop a new, accurate, stable, non-invasive, and cost-effective method for IS detection.

2.2 Experimental section

2.2.1 Materials and chemicals

Isopropanol, chloroform, acetonitrile, and methanol (HPLC grade) were purchased from Duksan (ANSAN-SI, South Korea). Distilled water was purified using a Milli-Q water purification system (Millipore, Bedford, MA, USA). Formic acid and ammonium acetate were purchased from Sigma-Aldrich and Acros Organics respectively. Deuterated internal standards consisting of 1-palmitoyl-d31-2-oleoyl-sn-glycero-3-phosphocholine and 1,3 (d5)-dihexadecanoyl-2-octadecanoyl-glycerol were purchased from Avanti Polar Lipids (Alabaster, AL, USA).

2.2.2 Sample collection

All participants of the IS group were recruited from the Accident & Emergency Department of the First Affiliated Hospital of Wenzhou Medical University (Zhejiang, China). Age- and sex-matched health controls were recruited from the medical examination center of the same hospital. Blood samples were taken immediately from patients who showed local neurologic deficits with acute onset. After confirming with CT scan and MRI scan, only patients diagnosed with IS were included into the IS group, and healthy volunteers without any IS symptoms were recruited to health control (HC) group. In total, 53 cases of IS and 28 cases of HC were recruited. The participants aged between 30 and 80 and had no history of acute coronary syndrome, tumors, severe liver or kidney dysfunction or autoimmune diseases. Baseline characteristics (age, sex, body mass index, triglyceride level, HDL-cholesterol level, LDL-cholesterol level, etc.) of the studied subjects were recorded. Venous blood samples (5 mL from each participant) were collected in heparin-treated tubes and centrifuged at 3600 rpm under 4 °C for 10 min. Supernatant plasma specimens were separated and stored at – 80 °C before further processing. These samples were separated into two datasets for discovery and validation analysis. The discovery lipidomics analysis was

conducted by an untargeted search for plasma lipids related to IS via a case-control study (discovery dataset, 36 IS cases and 19 controls). Furthermore, a validation dataset of 17 IS cases and 9 controls was analyzed for assessing the prognosis ability of potential lipid markers that were produced from the discovery dataset.

2.2.3 Lipid extraction

The lipid extraction procedure was optimized based on a modified Folch method. Specifically, 50 μ L clinical plasma samples spiked with internal standard (1-palmitoyl-d31-2-oleoyl-sn-glycero-3-phosphocholine and 1,3 (d5)-dihexadecanoyl-2-octadecanoyl-glycerol) were pipetted into a glass tube and extracted using 2.5 ml of a mixture of chloroform/methanol (2:1, v/v) and vortexed for 5 min. The mixture was then sonicated in ice water for 1 hour, 200 μ L of water was added to the mixture and then vortexed for 5 min. Finally, the lower layer was isolated following the centrifugation at 2000 rpm for 10 min. The upper layer was re-extracted with another 2.5 mL of chloroform and methanol mixture, and the procedure mentioned above was repeated for a more thorough extraction of lipids. Two organic phases were combined and dried under nitrogen prior to storage at -80 °C. Equal aliquots of 5 μ L of all the plasma samples were pooled and mixed to serve as the quality control (QC)

samples for the evaluation of the stability of the LC-MS system during data acquisition and the QC samples were conserved under -80°C until use. Seven continuous injections of QC samples were run at the very beginning of the LC-MS analysis and one QC sample was injected into the instrument after every seven samples were analyzed throughout the whole analysis.

2.2.4 Untargeted lipidomics analysis

UPLC-ESI-MS analysis was performed using an Agilent 1290 LC system coupled to a 6540 Q-TOF-MS with an electrospray ionization source (Agilent Technologies, Santa Clara, CA, USA). Lipid extracts were injected into a reversed-phase CSH column of 2.1×100 mm, 1.7 μm particle size (Waters, Milford, MA). The column was maintained at 55°C . The mobile phase comprised component A (10 mM ammonium acetate and 0.1% formic acid in water:acetonitrile (40:60, v/v)) and component B (isopropanol:acetonitrile (90:10, v/v) with 10 mM ammonium acetate and 0.1% formic acid). The mobile phase was pumped at a flow rate of 0.3 mL/min. The gradient elution program was as follows: 0-2 min, 30%-43% B; 2-2.1 min, 43%-50% B; 2.1-12 min, 50%-54% B; 12-12.1 min, 54%-70% B; 12.1-18 min, 70%-99% B; 18-18.1 min, 99%-30% B; 18.1-22 min, 30% B. Both positive and negative modes were performed with an acquisition time of 0.2 second per scan. The

scan range was from 100 to 1200 Da. The capillary voltage was set at 3 kV and 2 kV in positive ion mode and negative ion mode, respectively. The capillary temperature was set to 350 °C.

2.2.5 Data processing

Progenesis QI 2.3 (Nonlinear Dynamics, Newcastle upon Tyne, United Kingdom) was employed for peak picking and peak alignment. The data type was profile. Only those features detected in more than 80% individuals in at least one group were retained to minimize the effect of missing values. Missing values were replaced with one fifth of the minimum value found in the same study group. The normalization of detected ions in the dataset was conducted using total ion current in each sample. The normalized dataset was imported to SIMCA P software (version 17.0, Umetrics, Umea, Sweden) for further analysis.

Principal component analysis (PCA) and OPLS-DA were utilized for the differentiation of control group and disease group. Potential candidates were preliminarily selected according to the variable importance in projection (VIP) value of OPLS-DA. The markers were further identified with the accurate *m/z* and mass fragment and matched with the Lipid Maps

(<https://www.lipidmaps.org/>), METLIN (<https://metlin.scripps.edu/>) and Human Metabolome Databases (<https://hmdb.ca/>), and/or confirmed by the commercially available standards according to their mass, retention times and fragmentation patterns. The mass error windows for precursor ion and fragment were set as 5 ppm and 50 ppm separately.

2.2.6 Statistical analysis

After the abundance of exported ions were log₁₀-transformed, statistical differences were evaluated by Mann-Whitney test and false discovery rate (FDR) correction based on the Benjamini-Hochberg method was performed. Fold change (FC) of the intensity of ion was equal to the average abundance of the ion in the IS group divided by that in the control group, and the cut-off value was set as 1.2 or 1/1.2, i.e., $FC > 1.2$ indicates the ion upregulates in the IS group and $FC < 1/1.2$ means that the ion downregulates in the IS group. To identify lipid markers, odds ratios (ORs) of developing IS in lipid species were calculated by binary logistic regression models, after adjusting for age, BMI, smoking and drinking status, family history of IS, physical activity, and blood pressure (SBP). Model performance was presented as receiver operating characteristic (ROC) curves. All statistical analyses were performed using IBM SPSS statistics software (version 22.0, SPSS Inc., Chicago, IL, USA),

RStudio (version 1.1.463) and Metaboanalyst (version 5.0, <https://www.metaboanalyst.ca/>) and p value < 0.05 was considered as statistically significant.

Identified markers were imported to the Ingenuity Pathway Analysis software (Qiagen) and open database sources, including KEGG, MetaboAnalyst, and Human Metabolome Database, for pathway analysis, from which the biological pathways and molecular regulators involved were determined.

2.3 Results and discussion

2.3.1 Baseline characteristics of cohorts

To reduce the influence of confounders on the analysis, patients with acute coronary syndrome, tumors, severe liver or kidney dysfunction or autoimmune diseases were excluded and healthy subjects and IS subjects for both the discovery and validation sets were matched with respect to age, sex, smoking history, alcohol history, hypertension, diabetes, hyperlipidemia, and coronary disease.

Table 2-1 shows the baseline characteristics of both cohorts of 53 patients and 28 health controls. In the discovery dataset, a total of 55 subjects (36 patients

with IS: 21 males, aged 66.14 ± 9.81 ; 19 health control subjects: 12 males, age 65.15 ± 8.76) were recruited. There were no remarkable differences between the two groups in their age, sex, BMI, and other clinical characteristics. In the validation data set, a total of 26 individuals, 17 IS patients (7 males, age 67.41 ± 12.59) and 9 controls (1 male, age 61.25 ± 10.02), were recruited. There was no significant difference between HC and IS group for the validation set in all the clinical characteristics except hypertension. In total, there were 55 and 26 well matched subjects for discovery phase and validation phase, respectively.

Table 2-1. Information of the IS patients and health controls recruited in this project

Clinical parameters	Discovery dataset		Validation dataset	
	Patients (n=36)	Control (n=19)	Patients (n=17)	Control (n=9)
Age, years	66.14 ± 9.81	65.15 ± 8.76	67.41 ± 12.59	61.25 ± 10.02
Male	21	12	7	1
BMI	23.97 ± 4.17	23.02 ± 3.07	23.97 ± 4.17	23.02 ± 3.07
Smoking history	10	0	3	1
Alcoholism history	11	3	3	1
Hypertension	31	9	13 ^a	2 ^a
Diabetes	12	3	6	3
Hyperlipidemia	17	7	4	3
Coronary disease	1	0	1	0
Total cholesterol (mmol/L)	4.76 ± 1.00	4.62 ± 0.85	4.44 ± 0.60	4.64 ± 0.84
TG (mmol/L)	1.69 ± 0.92	1.72 ± 1.04	1.45 ± 0.65	2.15 ± 1.18
HDL-C (mmol/L)	1.08 ± 0.31	1.16 ± 0.42	1.06 ± 0.24	1.15 ± 0.45
LDL-C (mmol/L)	3.10 ± 0.10	2.77 ± 0.68	2.91 ± 0.63	2.56 ± 0.83

2.3.2 Stability and accuracy of the lipidomics platform

The analytical characteristics of lipid profiling were studied to assess the stability of LC-MS systems and to test the reproducibility of the sample preparation.

Because the plasma samples for discovery phase and validation phase were collected and analyzed in different time, the stability of LC-MS platform for both discovery phase and validation phase should be evaluated. QCd and QCv were two sets of mixed QC samples prepared by mixing aliquots of all plasma samples collected during the discovery phase and the validation phase, respectively. Then eight aliquots of 5ul of QCd and QCv were prepared, i.e., QCd1-QCd8 and QCv1-QCv8. First, QCd1 or QCv1 was analyzed after the analysis of every seven samples from the corresponding dataset, and the QCd1 and QCv1 measurements would be used for the testing of the stability of the LC-MS platform. Three representative total ion chromatograms of QCd1/QCv1 measurements that were acquired at the beginning, the middle and the end of the entire analysis were plotted together to test their consistency (**Figure 2-1**). The results suggested that these three chromatograms overlapped well in both positive and negative ion modes of the discovery and validation stages, which indicated that LC-MS platform was stable throughout

the analyses. Second, after the lipid extraction and LC-MS analyses of QCd2-QCd8 and QCv2-QCv8, the number of detected features and the signal of spiked lipid standards of these QC samples were obtained for the evaluation of the reproducibility of the lipid extraction for discovery and validation phases. The results which were related to the number of features were shown in **Table 2-2**. In the discovery phase, a total of 3690 (83.96%) out of 4395 features were found with relative standard deviation (RSD) ≤ 0.3 , 2995 features had RSD ≤ 0.2 and 1351 ions had RSD ≤ 0.1 after LC-MS analyses of QCd2-QCd8 in positive ion mode. After LC-MS analyses of the QCd2-QCd8 in negative ion mode, a total of 87.13% (2858 out of 3280) ions had RSD ≤ 0.3 , 2369 ions had RSD ≤ 0.2 , and 1183 ions had RSD ≤ 0.1 . In the validation phase, LC-MS analyses of QCv2-QCv8 in the positive ESI displayed 3810 (95.56%) out of 3987 variables with RSD ≤ 0.3 , 3428 variables with RSD ≤ 0.2 and 2174 variables with RSD ≤ 0.1 . LC-MS analyses of the QCv2-QCv8 in the negative ion mode showed 2783 (84.98%) out of 3275 ions with RSD ≤ 0.3 , and 2369 ions and 1336 ions had RSD ≤ 0.2 and 0.1, respectively. Moreover, the results of the reproducibility of signal of spiked internal standards, including PC (16:0-d31/18:1) and TG (16:0/18:0/16:0-d5) showed that the RSDs for all QC samples, except for d5-TG in the discovery stage with RSD of 0.3, were smaller than 0.15, indicating

that high reproducibility of the lipid extraction was obtained. The details were summarized in **Table 2-3**.

These results suggested the stability of the LC-MS platform throughout the entire analyses was good and the reproducibility of lipid extraction were basically good. It is namely that the analytical methods used in this lipidomics study were reliable.

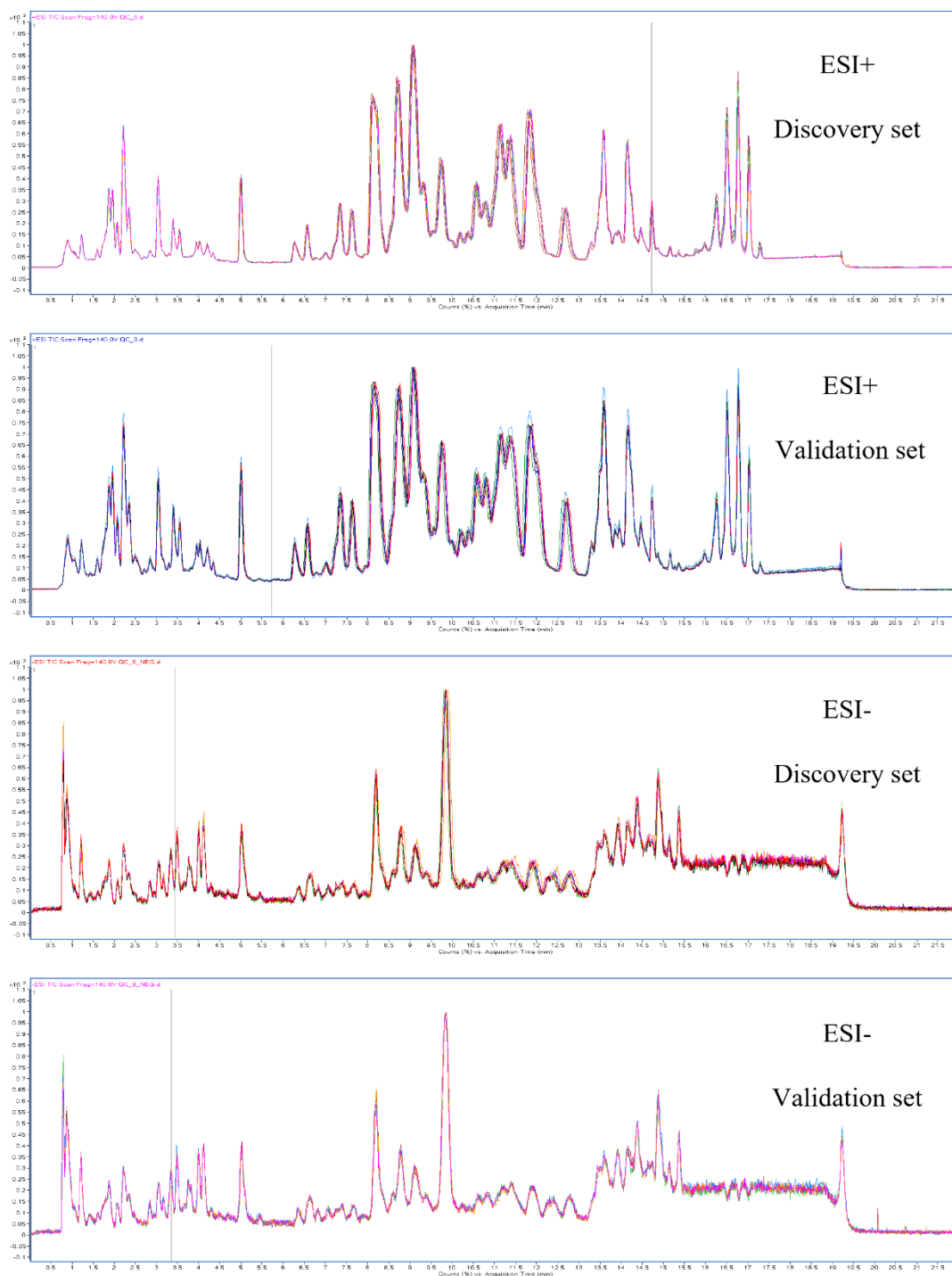


Figure 2-1. Superposed total ion chromatograms of pooled QC samples (QCd1 and QCv1) with three representative measurements for each ion mode

Table 2-2. The number of detected features of QC samples

Sample	Ion mode	Number of total ions	Number of ions with			
			RSD > 0.3	RSD ≤ 0.3	RSD ≤ 0.2	RSD ≤ 0.1
QCd2- QCd8 (Discovery Dataset)	+	4395	705	3690	2995	1351
	–	3280	422	2858	2369	1183
QCv2- QCV8 (Validation Dataset)	+	3987	177	3810	3428	2174
	–	3275	492	2783	2369	1336

Table 2-3. The stability of the signal of internal standards spiked in plasma samples

Internal standard	QC sample	Adduct	Retention time	<i>m/z</i>	RSD of abundance
16:0-d31-18:1 PC	QCd2-QCd8	[M+H] ⁺	10.78	790.7753	0.09
	QCv2-QCv8	[M+H] ⁺	10.80	790.7755	0.03
	QCd2-QCd8	[M+FA-H] ⁻	10.85	834.7661	0.09
	QCv2-QCv8	[M+FA-H] ⁻	10.87	834.7669	0.07
16:0-18:0-16:0-d5 TG	QCd2-QCd8	[M+NH ₄] ⁺	17.30	857.8348	0.30
	QCv2-QCv8	[M+NH ₄] ⁺	17.28	857.8337	0.14
QCd2-QCd8 were QC samples for discovery phase and QCv2-QCv8 represented QC samples for validation phase					

2.3.3 Selection and identification of potential biomarkers

The discovery dataset consisting of 36 IS plasma and 19 plasma samples from health controls were utilized for the screening of possible biomarkers. After the pretreatment procedure, a total of 2577 and 2892 ions were obtained from the plasma samples of the discovery dataset in positive and negative ion modes, respectively. Lipids were scrutinized before selected as biomarker candidates to reduce the risk of misreading and to ensure an accurate correlation with IS risk. Lipid classes including PC, PE, PS, PG, PI, PA, LPC, LPE, MG, DG, TG, Cholesterol, SM, Cer and FA were simultaneously detected in plasma samples of both health controls.

PCA was initially performed to study the distribution of plasma samples from health control group and IS group, and two clusters were separated distinctively, which indicated there was a significant difference of lipid levels between IS group and healthy group. The selection of potential markers in discovery stage was conducted as follows. First, OPLS-DA models were employed to identify important features between the disease groups and control groups, and the score plot of OPLS-DA model showed that there was also an obvious difference of lipid level between control subjects and IS patients (**Figure 2-2**). After OPLS-DA model was established, only those

variables with VIP values larger than one were retained, resulting in a total of 984 and 1025 features for the positive and negative ion modes, respectively. Then those ions that could not be searched and matched by databases were not considered for the subsequent analysis. Second, the Mann-Whitney U test was utilized to evaluate the statistical significance of each feature. After that, 182 and 215 features with p values < 0.05 were retained for positive and negative ion modes, respectively. Third, fold changes (FC) were calculated by dividing the normalized intensities of features in IS group by that in HC group, and 317 features with $FC > 1.2$ (48 features in positive mode and 56 in negative mode) or $FC < 1/1.2$ (95 features in positive mode and 118 features in negative ion mode) were included.

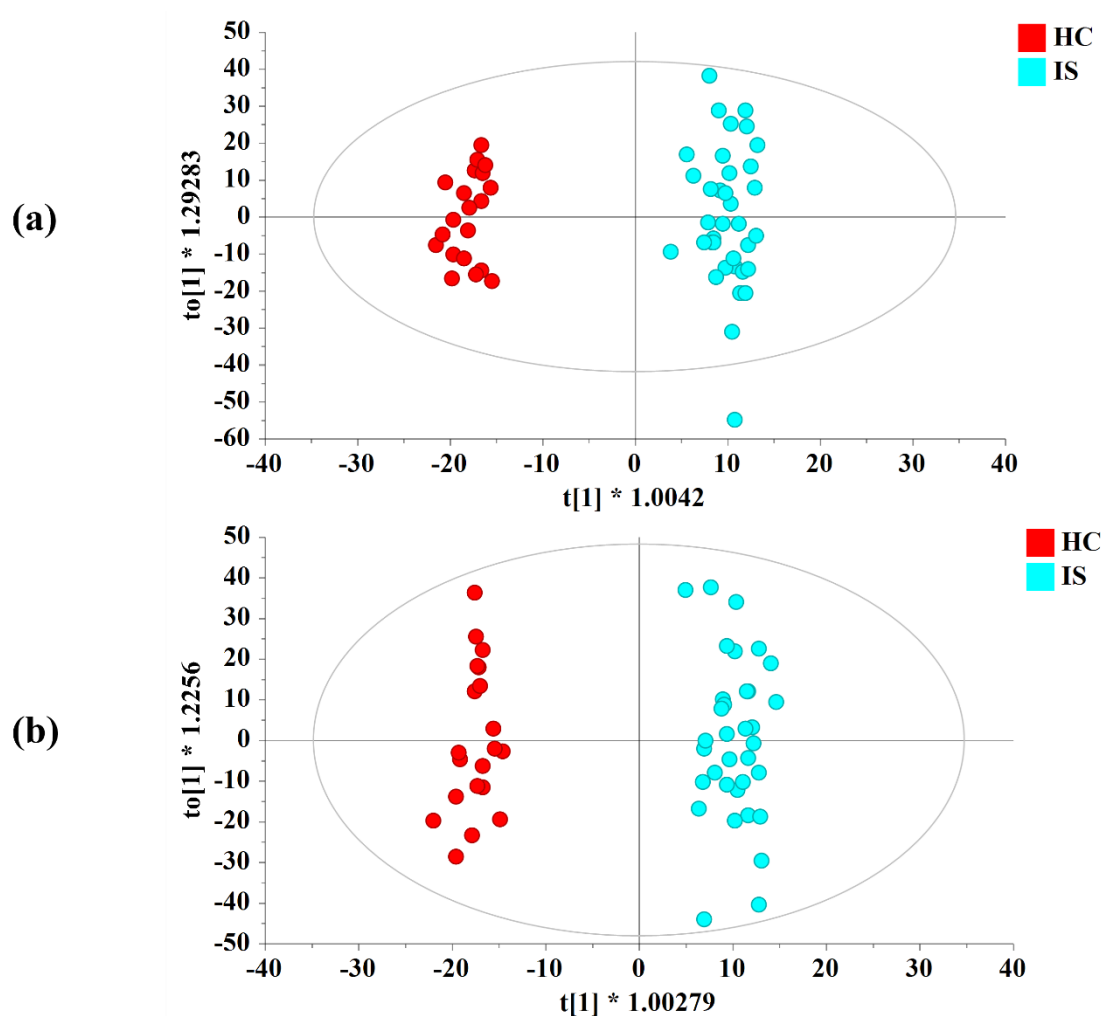


Figure 2-2. OPLS-DA score plots for HC and IS groups from the discovery phase in positive ion mode (a) and negative ion mode (b). Red dots represented the HC group and blue dots represented the IS patient group

2.3.4 Validation and evaluation of potential biomarkers

A validation dataset of 26 plasma samples from 17 patients and 9 health controls was collected and analyzed to validate the reliability of the 317 potential biomarkers of IS.

The processing of raw data was performed following the same procedure of discovery dataset, such as peak picking, missing value replacement, and normalization. After data processing, OPLS-DA models were employed as well to identify important features between the disease group and control group. And the score plot of OPLS-DA model showed that there was an obvious difference of lipid level between control subjects and IS patients (**Figure 2-3**). Data analysis for the validation dataset was performed similar to that of discovery dataset, but there were some adjustments of data analysis in validation dataset. For example, to decrease the false-positive risk, p values were calculated first using Mann-Whitney U test and then these p values were corrected via the false discovery correction using Benjamini-Hochberg method ($\text{FDR} < 0.05$) for the validation dataset. The list of p values was ordered and then ranked and the largest p value that is smaller than the corresponding adjusted p value was treated as the threshold and all the ions with adjusted p values above it (i.e., those with lower p values) were

considered significant and were retained. After that, the remaining features in the validation data set were matched with the 317 lipids found in the discovery dataset. The matching criteria were as follows: m/z (± 10 ppm), retention time (± 0.5 min) and same trend in FC. After data processing and data analysis, 297 lipids were removed from the candidate pool. Among the remaining 20 lipids that were successfully validated, 15 lipids were detected in positive ion mode and 5 lipids were analyzed by negative MS. Their abundance in each group, adjusted p values and FC were summarized in **Table 2-4**. FA(20:0), LPC(O-20:0), LPS(O-15:0), nordeoxycholic acid, 23:0 sterol, and 14:0 cholesterol ester were upregulated in the IS patients, while other lipids, such as n-butyl arachidonoyl amine, FA(20:3), TG(14:0/18:2/18:3), TG(15:1/19:1/20:5), PS(P-20:0/20:5), PI(17:0/22:1), PC(O-38:9), PI(O-20:0/15:1), and PE(P-16:0/20:3), were downregulated in the patient group. After establishing the above model for the differentiation of IS and health control, we went on to explore the level of those markers in the disease group. For those 20 lipids that had the same trend in FC for both the control group and IS group, the results showed that, in general, most of the differential ions were classified into the GP category, including PS, PC, PE, and PI, were downregulated in the IS group. Most of the phospholipids belonged to plasmalogen phospholipids, which were the ether phospholipids related to the onset of IS. The most

famous plasmalogen-PL is the platelet activating factor (PAF), a bulk of studies have illustrated that PAF contributes to neurological impairment by binding to the platelets in stroke patients, and PAF can be employed as an index to evaluate the severity and treatment outcome of stroke. Similar to phospholipids, the levels of TG and DG potential ions were decreased in the disease group as well. However, the lysophospholipid and the sterol, such as LPC(O-20:0), LPS(O-15:0), nordeoxycholic acid, 23:0 sterol, and 14:0 cholesterol ester, were obviously increased in the IS patient plasma, and this trend can also be found in some specific fatty acyls.

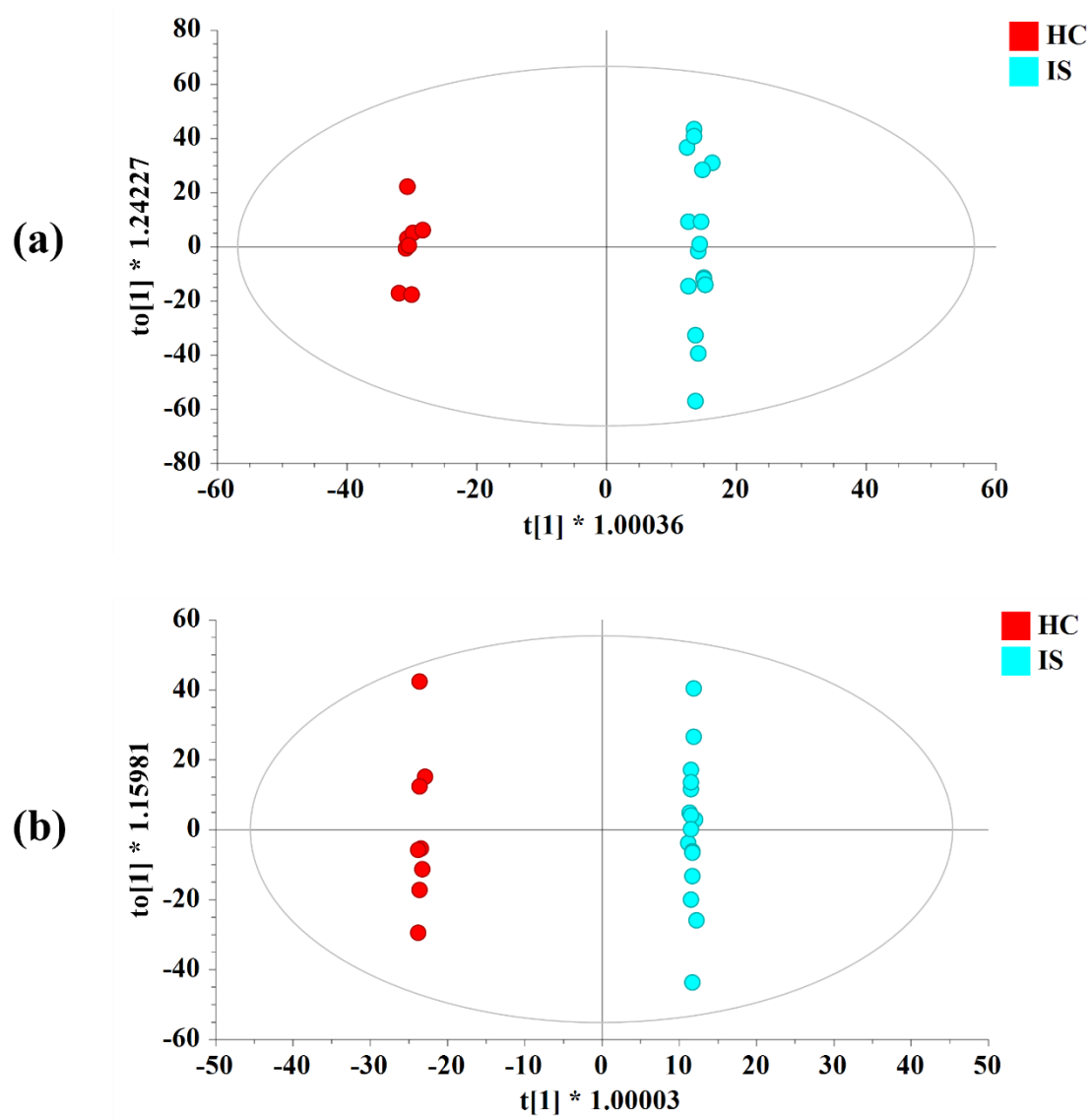


Figure 2-3. OPLS-DA score plots for HC and IS groups from the validation phase in positive ion mode (a) and negative ion mode (b). Red dots represented the HC group and blue dots represented the IS patient group

Table 2-4. 20 potential lipid candidates in validation dataset

Lipid species	Lipid class	m/z	Adduct	Normalized abundance ^a		Fold change ^b	Adjusted <i>p</i> values ^c
				Control (n=9)	IS (n=17)		
N-butyl arachidonoyl amine	FA	360.3244	[M+H] ⁺	0.0197±0.0038	0.0010±0.0021	0.4832±0.2886	5.2632E-04
FA(20:3)	FA	310.3113	[M+NH ₄] ⁺	0.0395±0.0110	0.0188±0.0047	0.4765±0.3721	1.5789E-03
FA(20:0)	FA	417.3436	[M+H-H ₂ O] ⁺	0.0417±0.0092	0.0720±0.0195	1.7273±0.3502	4.2105E-03
TG(13:0/18:4/19:1)	GL	842.7244	[M+NH ₄] ⁺	0.3918±0.1315	0.1838±0.0784	0.4690±0.5428	6.8421E-03
TG(15:1/19:1/20:5)	GL	894.7549	[M+NH ₄] ⁺	2.5304±1.1527	1.3057±0.5983	0.5160±0.6462	1.2105E-02
PS(P-20:0/20:5)	GP	804.5526	[M+H-H ₂ O] ⁺	0.7438±0.2175	0.3402±0.1557	0.4574±0.5430	2.6316E-03
PI(17:0/22:1)	GP	889.6191	[M+H-H ₂ O] ⁺	0.1105±0.0434	0.0450±0.0238	0.4068±0.6581	3.6842E-03
PC(O-38:9)	GP	789.5875	[M+NH ₄] ⁺	0.2686±0.0858	0.1330±0.0562	0.4951±0.5297	5.7895E-03
PI(O-20:0/15:1)	GP	837.5887	[M+H] ⁺	0.1050±0.0398	0.0501±0.0234	0.4767±0.6023	1.3158E-02
PE(O-18:0/18:4)	GP	726.5444	[M+H] ⁺	0.0780±0.0429	0.0352±0.0190	0.4517±0.7702	1.5789E-02
LPC(O-20:0)	GP	538.4243	[M+H] ⁺	0.0129±0.0082	0.0234±0.0110	1.8094±0.7920	2.0000E-02
PC(18:2/22:6)	GP	874.5623	[M+FA-H] ⁻	0.4219±0.1040	0.2384±0.0866	0.5652±0.4389	8.6957E-04
PS(O-18:0/19:1)	GP	834.5829	[M+FA-H] ⁻	0.0807±0.0155	0.0403±0.0236	0.4993±0.6173	1.3043E-03
PS(20:0/22:4)	GP	912.5965	[M+FA-H] ⁻	0.1679±0.0278	0.1039±0.0390	0.6185±0.4100	1.7391E-03
LPS(O-15:0)	GP	530.2713	[M+FA-H] ⁻	0.0097±0.0028	0.0157±0.0046	1.6242±0.4170	2.1739E-03
Glycinoprenol-9	PR	619.6149	[M+H-H ₂ O] ⁺	0.0043±0.0031	0.0006±0.0005	0.1278±1.1756	7.8947E-03
Cer(m18:1/24:1)	SP	649.6616	[M+NH ₄] ⁺	0.0067±0.0031	0.0022±0.0008	0.3307±0.5967	1.0526E-03
Nordeoxycholic acid	ST	396.3111	[M+NH ₄] ⁺	0.0081±0.0030	0.0153±0.0047	1.8915±0.4793	4.7368E-03
23:0 Sterol	ST	398.3271	[M+NH ₄] ⁺	0.0190±0.0067	0.0352±0.0109	1.8530±0.4707	6.3158E-03
14:0 Cholesterol ester	ST	595.5434	[M-H] ⁻	0.0300±0.0085	0.0617±0.0269	2.0536±0.5194	4.3478E-04

^a The normalized abundance indicates that the mean value of the HC and the IS samples were normalized to the total ion abundance.

^b Fold change was determined by dividing the average value of IS group by the average value of control group, with the cut-off value set as >1.2 (highlighted in yellow) or <1/1.2.

^c Adjusted *p* values were corrected using the Benjamini-Hochberg FDR method.

After the potential lipid biomarkers have been defined, ROC analysis was employed for evaluating the diagnosis potential of these 20 lipids. ROC curve was exploited based on the results of area under the curve (AUC) by using the ROC analysis in SPSS, and the results of ROC analysis for these 20 lipids in the discovery dataset were shown in **Table 2-5**. The result showed that AUC values of all the potential biomarkers were larger than 0.79. To further obtain the better prediction of the IS, the combination of these lipid biomarkers was evaluated by binary logistic regression which was conducted based on the forward stepwise (likelihood ratio) method, and the result for the binary logistic regression was presented in **Table 2-6**. The “Exp(B)” in this table was the exponential value of B. If Exp(B) is smaller than one, increasing the level of lipid corresponds to decreasing odds of the IS occurrence. When Exp(B) is greater than 1, increasing values of the variable correspond to increasing odds of the event's occurrence. As a result, a combination of FA(20:3), PS(O-18:0/19:1) and LPS (O-15:0) achieved an effective discrimination of IS and HC in the discovery dataset. As shown in **Figure 2-4a**, in comparison of one lipid biomarker, the combination of three lipids biomarkers has larger AUC values, specificity and sensitivity, the combination of FA(20:3), PS(O-18:0/19:1) and LPS (O-15:0) was thus defined as the “biomarker panel”. Specifically, the AUC value of biomarker panel in the discovery dataset was

1.000, and both of its sensitivity and specificity were 100% at the best cut-off value.

Besides, the diagnosis efficiency of this biomarker panel was validated in the validation dataset as well. The AUC value of the biomarker panel was 0.963, and the sensitivity and the specificity of the biomarker panel were 100% and 87.5%. The result shown in **Figure 2-4b** indicated that the IS patients could be diagnosed at the best cut-off in the validation dataset. Thus, a novel plasma biomarker model of 3 lipids was ultimately defined on the basis of a comprehensive screening and verification procedure.

Table 2-5. ROC analysis results for 20 potential markers in the discovery dataset

Lipid species	AUC	Standard error ^a	95% CI ^b		Sensitivity	Specificity
			Lower	Upper		
N-butyl arachidonoyl amine	0.9853	0.0192	0.9480	1.0000	0.8824	1.0000
Cer(m18:1/24:1)	0.9706	0.0327	0.9070	1.0000	1.0000	0.8750
FA(20:3)	0.9632	0.0392	0.8860	1.0000	1.0000	0.8750
14:0 Cholesterol ester	0.9485	0.0424	0.8650	1.0000	0.7647	1.0000
PS(P-20:0/20:5)	0.9412	0.0447	0.8540	1.0000	0.7647	1.0000
FA(20:0)	0.9338	0.0549	0.8260	1.0000	0.9412	0.8750
PI(17:0/22:1)	0.9338	0.0484	0.8390	1.0000	0.8235	1.0000
PC(18:2/22:6)	0.9265	0.0515	0.8250	1.0000	0.7059	1.0000
Nordeoxycholic acid	0.9265	0.0574	0.8140	1.0000	1.0000	0.7500
PC(O-38:9)	0.9191	0.0548	0.8120	1.0000	0.7059	1.0000
PS(O-18:0/19:1)	0.9191	0.0574	0.8070	1.0000	0.8235	1.0000
23:0 Sterol	0.9118	0.0566	0.8010	1.0000	0.7059	1.0000
TG(14:0/18:2/18:3)	0.9044	0.0630	0.7810	1.0000	0.7647	0.8750
PS(20:0/22:4)	0.9044	0.0592	0.7880	1.0000	0.7059	1.0000
Glycinoprenol-9	0.8971	0.0911	0.7190	1.0000	0.9412	0.8750
LPS(O-15:0)	0.8897	0.0768	0.7390	1.0000	0.9412	0.7500
TG(15:1/19:1/20:5)	0.8750	0.0686	0.7400	1.0000	0.7059	1.0000
PI(O-20:0/15:1)	0.8676	0.0792	0.7120	1.0000	1.0000	0.6250
PE(P-16:0/20:3)	0.8309	0.0881	0.6580	1.0000	0.9412	0.6250
LPC(O-20:0)	0.7941	0.1061	0.5860	1.0000	0.6471	0.8750

^a The standard error of the AUC values under the nonparametric assumption.

^b Asymptotic 95% confidence interval.

Table 2-6. Binary logistic regression results in the discovery dataset

Variables in the Equation						
	B	Standard error	Wald	df	p value	Exp(B)
FA(20:3)	-139.463	53.776	6.726	1	0.010	1.000
PS(O-18:0/19:1)	-85.963	36.820	5.451	1	0.020	1.000
LPS(O-15:0)	315.643	155.199	4.136	1	0.042	1.208 E+137
Constant	6.981	2.888	5.740	1	0.017	1010.463

B: coefficient for the constant in the model, it is also called the “intercept”

Wald: the Wald chi-square test to test the null hypothesis that the constant is 0

df: the degrees of freedom for the Wald chi-square test

Exp(B): the odds ratio

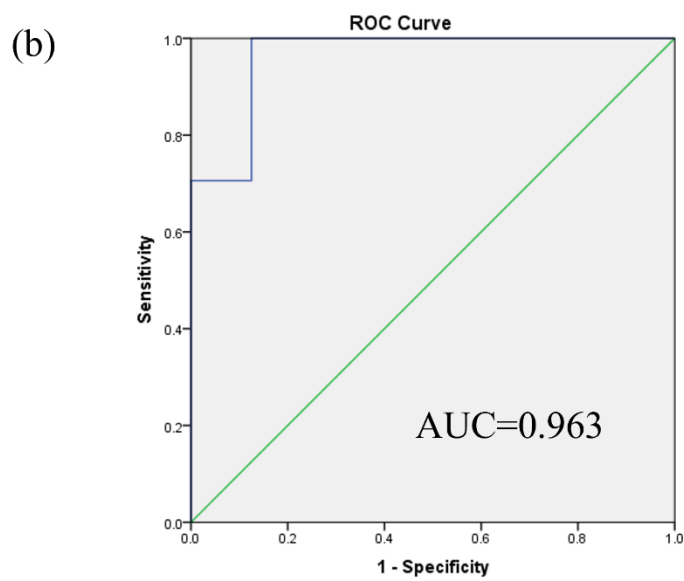
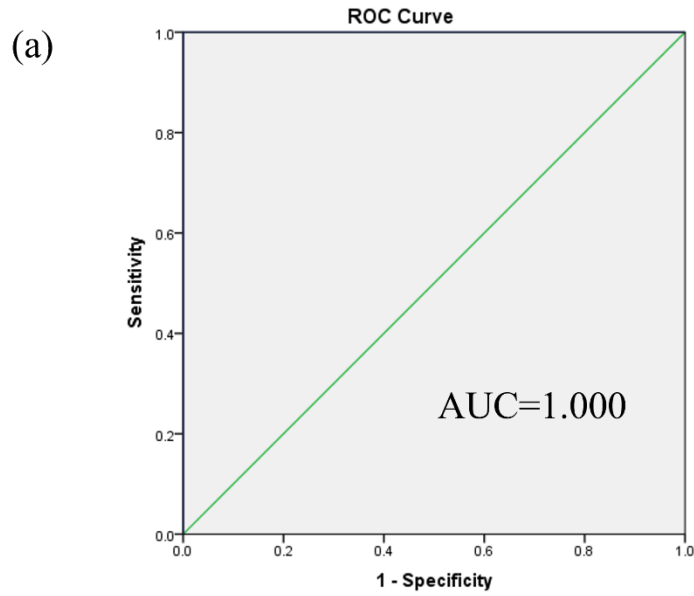


Figure 2-4. ROC curves of the discovery dataset (a) and validation dataset (b) based on the biomarker panel

2.3.5 Correlation and pathway analysis

To reveal the possible relationship between differentiated lipids, a correlation analysis was conducted for 20 differentiated lipids. The results shown in **Figure 2-5**~~Error! Reference source not found.~~ revealed that negative correlations between many lipids could be found in HC group, but these negative correlations were reversed to positive correlations in IS group. For example, PI(O-20:0/15:1) was negatively correlated with many other lipids in the HC group, such as LPC(O-20:0), PC(18:2/22:6), sterol(23:0) and FA(20:0), but these negative correlations could not be observed in the IS group. Besides, we could also find that many phospholipids in HC group were negatively correlated with each other except for the triangular network between PI(O-20:0/15:1), PI(17:0/22:1) and PC(18:2/22:6), however, most of these negative correlations turned to positive correlations in IS group. Furthermore, we also explore the related pathway of IS, The result presented in **Figure 2-6** showed that two major pathways related to IS were discovered, including glycerophospholipid metabolism and sphingolipid metabolism.

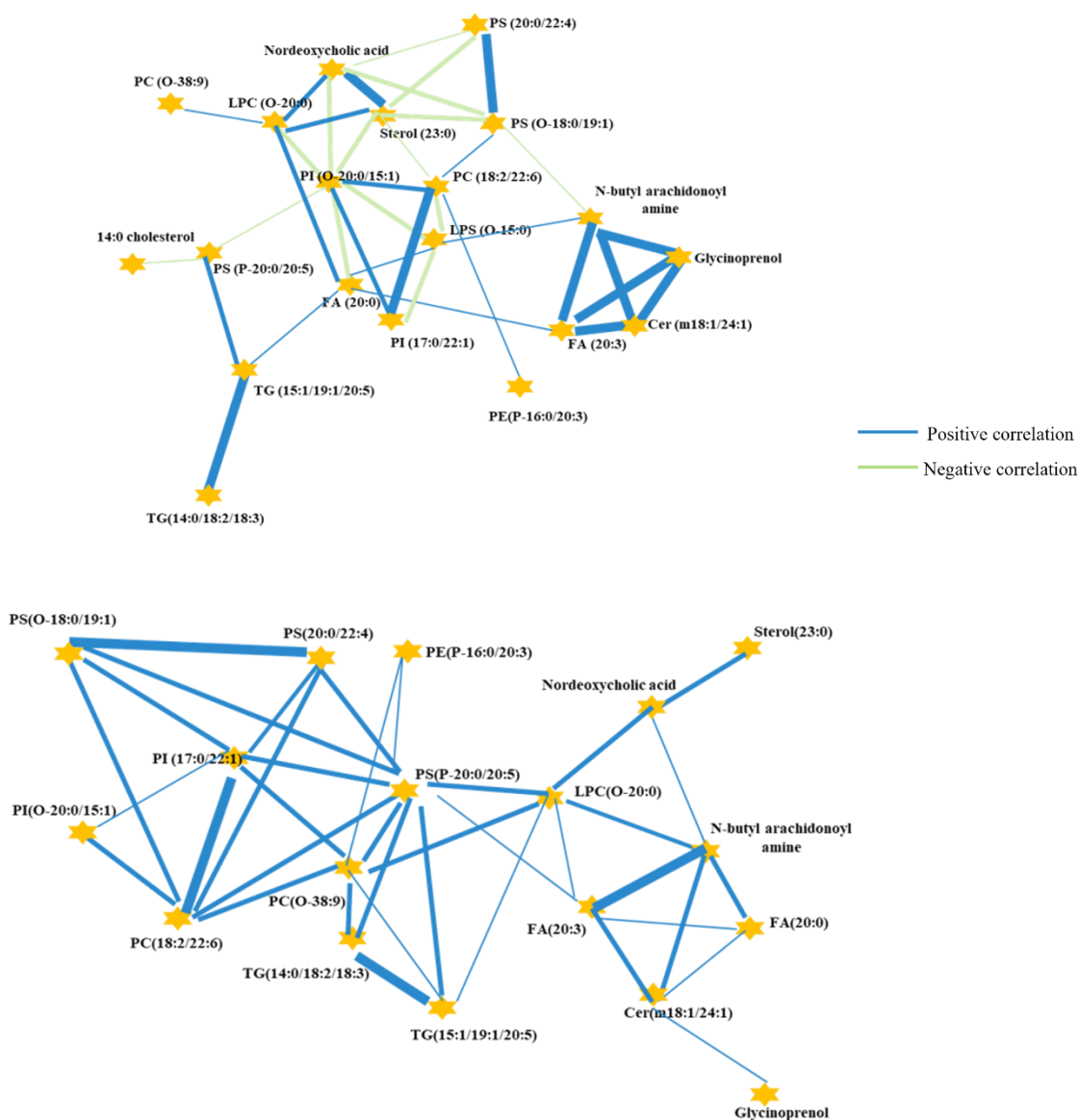


Figure 2-5. The correlation network of differentiated ions of IS, the upper figure indicated correlations between lipids in HC group and the lower figure indicated the lipid correlations in IS group. The positive correlation was highlighted in blue, and the green color was negative correlation.

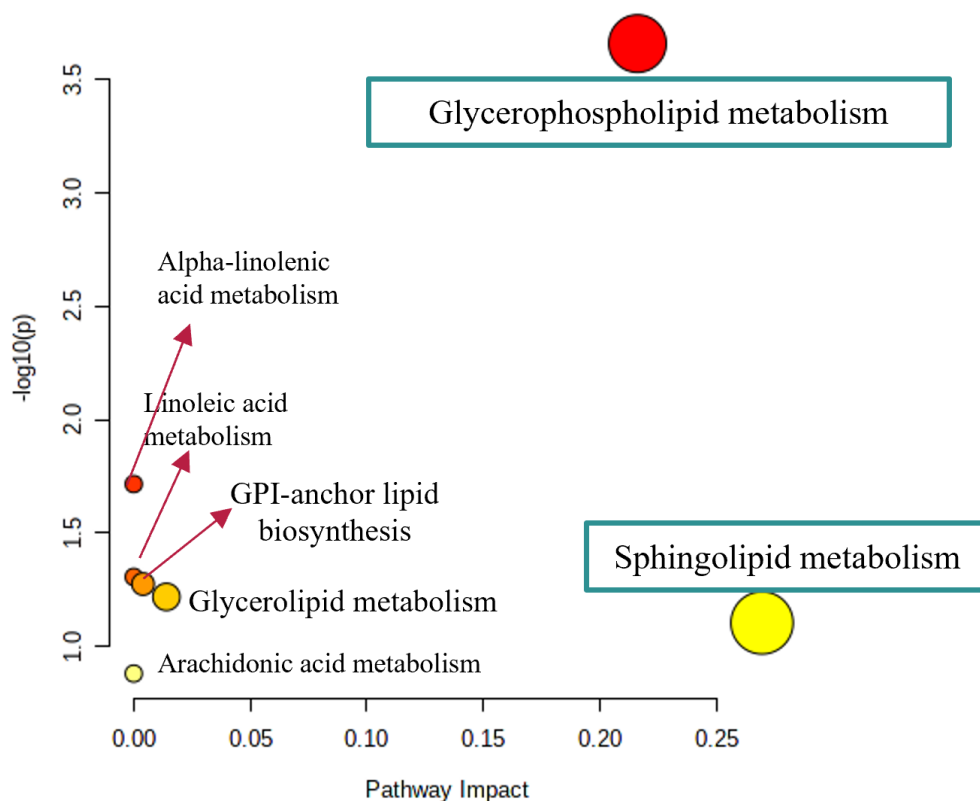


Figure 2-6. Pathway analysis of potential biomarkers

2.3.6 Biomarkers related to IS severity

In order to find the potential biomarkers for the prognosis of IS patients, lipidomics analysis of plasma samples from IS patients with different severity were conducted. The National Institute of Health Stroke Scale was commonly employed to evaluate the severity of IS patients, and IS severity was classified into three levels, including mild, moderate, and severe. Because no severe patient was recruited in this project, we only compared the lipid abundance in mild group with that of moderate group. Using the same criteria for biomarker

screening as the previous study, as shown in the volcano plot (**Figure 2-7**), we found that there were six lipids with significant difference between mild and moderate IS. In addition, the histogram shown in **Figure 2-8** indicated that compared with the mild group, the levels of GlcCer(t18:1/18:0), GlcCer(d16:2/22:0) and LPE(O-14:0) were significantly higher than those in the moderate group. But when we make a comparison of control group and the mild group, only GlcCer(t18:1/18:0) was significantly enhanced in the mild group, but there was no significant difference in the levels of GlcCer(d16:2/22:0) and LPE(O-14:0) between the control group and mild group. Instead, the levels of PG(O-20:0/22:0), PI(O-18:0/19:1) and PC(20:5/20:4) were significantly decreased in the moderate group when compared with those in the mild group, and there was an obvious decrease in PG(O-20:0/22:0) level in mild group compared with that in the control group, but PI(O-18:0/19:1) and PC(20:5/20:4) were not altered in the mild group when compared with the control group. Therefore, the relative abundance of these six biomarkers could help to identify the severity of IS patients to evaluate the prognosis of IS after treatment.

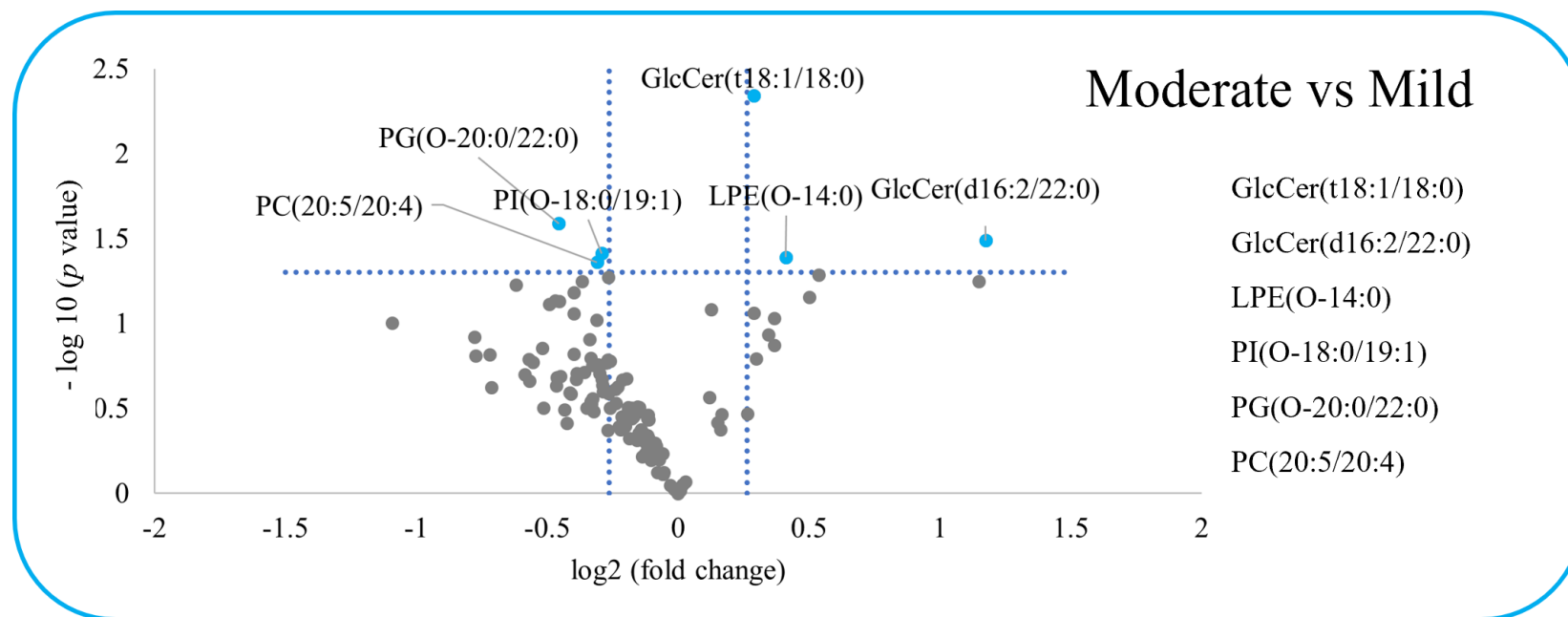


Figure 2-7. Volcano plot of potential biomarkers

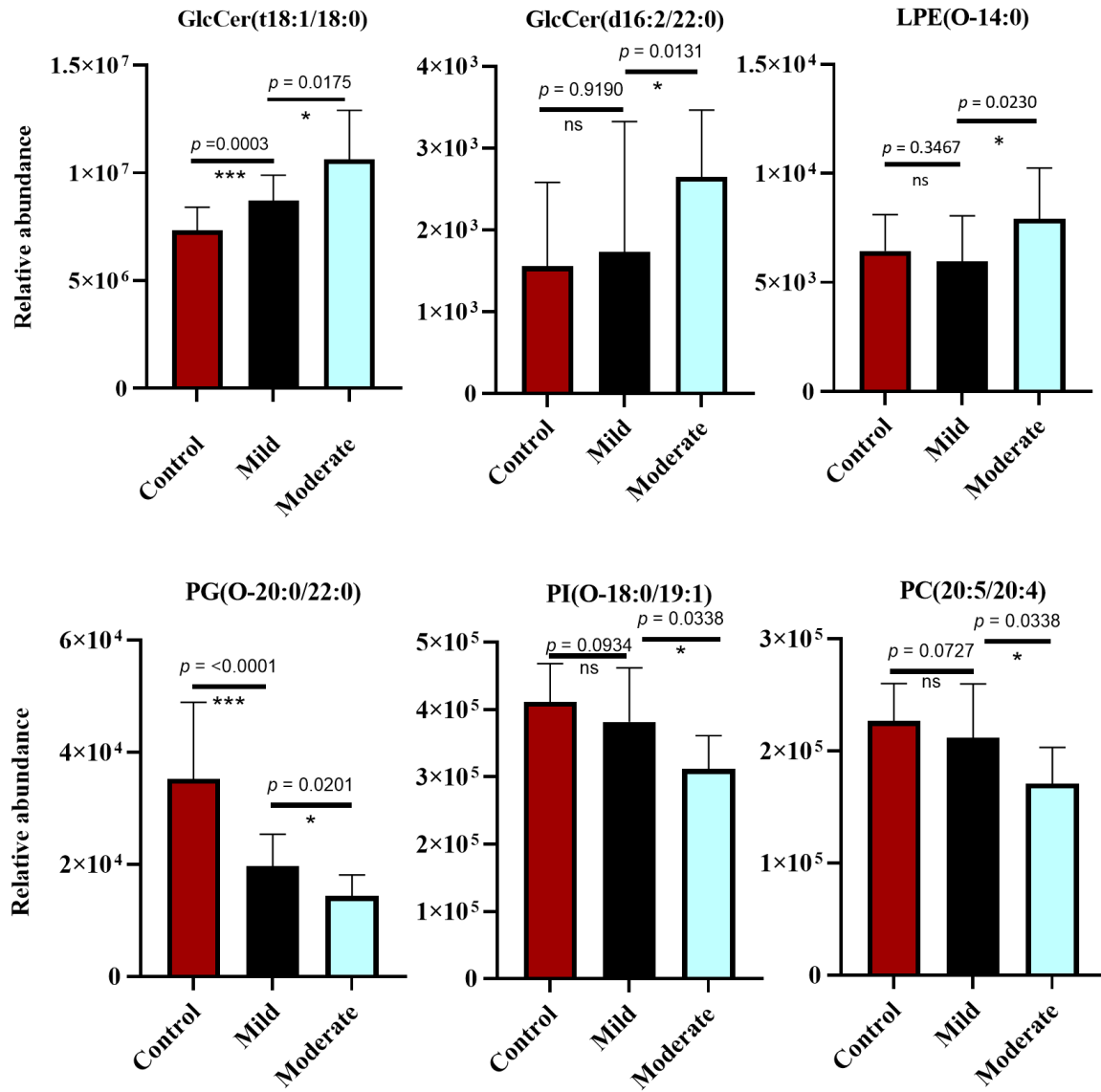


Figure 2-8. Comparison of lipid abundance between control group and IS patients from mild group and moderate group

2.4 Conclusions

IS is the disease with a high mortality and disability rate, and tpA should be administered within the “golden window” of IS onset (3-4.5 hours). CT and MRI are routine techniques for IS diagnosis. However, this is restricted because these two techniques are relative time-consuming and unreliable. Although some researchers have conducted lipidomics of IS, they only focused on lacunar infarction, one of the subtypes of IS, instead of including all the subtypes for the lipidomics analysis. However, the discovery of shared biomarkers for all subtypes is essential for rapid differentiation of IS and HC after IS onset. To date, there are no lipidomics studies of plasma to differentiate IS patients including all subtypes from health controls. In this study, a rapid, novel, accurate, stable and non-invasive method for the IS detection was developed. First of all, in order to discover the lipid markers with higher accuracy, the influence of confounders was reduced before the analysis and the plasma samples were divided into two datasets. The discovery dataset was used to screen potential markers and the validation dataset was employed to evaluate the reliability of the screened markers for IS diagnosis. The stability of the sample preparation procedure and LC-MS system was studied prior to the sample analysis to ensure the reliability of the acquired data. The analytical characteristics of lipid profiling were studied to

test the reproducibility of the sample preparation procedure and to assess the stability of LC-MS systems. In terms of the recovery of the detected features, more than 80% of the features could be found with $RSD \leq 0.3$ for both the discovery and validation datasets. Besides, the repeatability of the spiked internal standards was good as well with $RSD \leq 0.3$. Therefore, the analytical methods used in this lipidomics study were reliable.

PCA score plots demonstrated the good separation between the two groups, indicating that IS and HC could be differentiated by the lipid profile. A total of 20 lipids were screened as potential markers after the comprehensive analysis of the discovery dataset and the confirmation by the validation dataset. Among these 20 potential markers, the levels of 6 lipids, including FA(20:0), LPC(O-20:0), LPS(O-15:0), nordeoxycholic acid, 23:0 sterol, and 14:0 cholesterol ester, were found to be significantly increased in the disease group compared with their levels in the controls. Conversely, the remaining 14 lipids exhibited decreases in the IS patients, including n-butyl arachidonoyl amine, FA(20:3), TG(14:0/18:2/18:3), TG(15:1/19:1/20:5), PS(P-20:0/20:5), PI(17:0/22:1), PC(O-38:9), PI(O-20:0/15:1), PE(P-16:0/20:3), PC(18:2/22:6), PS(O-18:0/19:1), PS(20:0/22:4), glycinoprenol-9, and Cer(m18:1/24:1).

Furthermore, the differentiation ability of abovementioned 20 lipids between HC and IS was evaluated in the discovery dataset using binary logistic regression and a combination of only three lipids (i.e., FA(20:3), PS(O-18:0/19:1) and LPS (O-15:0)) achieved effective discrimination of IS and HC in the discovery dataset with the AUC value, sensitivity, and specificity of 1.000, 100% and 100%, respectively. Therefore, the combination of FA(20:3), PS(O-18:0/19:1) and LPS (O-15:0) was thus defined as the “biomarker panel”. Moreover, the differentiation ability of this biomarker panel was also good in the validation dataset, as the AUC value of the biomarker panel was 0.963, with sensitivity and specificity of 100% and 87.5%, respectively. The development of this new, rapid and accurate method could shorten the diagnosis time and improve the accuracy of IS diagnosis, so that the tpA treatment could be given in a shorter time, and the mortality and disability rate of IS patients could be reduced. In addition, we could find that most of the negative correlations between lipids in HC group were reversed to positive correlations upon IS onset, and the glycerophospholipid and sphingolipid metabolism were strongly related to IS development. All these lipid biomarkers related to the dysfunction of the metabolism pathways may be the targets of the prevention and treatment of IS. Furthermore, six potential lipid markers were significantly changed in the moderate group when compared

with mild group, including GlcCer(t18:1/18:0), GlcCer(d16:2/22:0), LPE(O-14:0), PG(O-20:0/22:0), PI(O-18:0/19:1) and PC(20:5/20:4). These lipids could be employed to assess the severity of IS and help to evaluate the prognosis of IS patients.

Chapter 3: Rapid differentiation of plasma from ischemic stroke patients using direct analysis in real time mass spectrometry

3.1 Introduction

The lipidomics studies have already been carried out for the discovery of potential lipid biomarkers in chapter two, and the result indicated that lipids are ideal biomarker for the differentiation of IS patients and health controls. Gas chromatography (GC) and liquid chromatography (LC) coupled mass spectrometry are two predominant techniques used for metabolomic workflow. Despite their prevalence in comprehensive omics analysis, GC-MS and LC-MS are time-consuming and the memory effects of the analytical columns are commonly found particularly when analyzing metabolites in complicated biological samples, such as plasma. Therefore, other MS-based approaches for rapid detection of IS should be established.

In recent year, many ambient ionization mass spectrometry (AIMS) techniques that allow rapid, real time, high-throughput and in situ analysis has emerged in many fields including biomedicine, drug and food analysis.¹¹¹ Direct analysis in real time (DART), one of ambient ionization methods for

MS, is a plasma-based ion generation technique that operates under ambient conditions. DART operates by first creating a glow discharge within a flowing stream of gas (N_2 or He) to form metastables that interact with water in the atmosphere to produce protonated water clusters. These clusters react by proton transfer with analytes desorbed by the heated gas stream. Since DART has many merits, it has been successfully applied to the analysis of pharmaceuticals, food, drug detection and several other interesting applications.¹¹² For example, samples are handled either manually or with the assistance of autosamplers to facilitate user operation of the instrument. Moreover, in most cases, immediate ionization can be achieved without sample preparation, so sample can be analyzed in a short time. Furthermore, DART is well suited for the detection of non-polar compounds.¹¹³

In this work, we reported a method for the rapid fingerprinting of human plasma from IS patients and health controls via DART-Q-TOF MS. Various experimental parameters were optimized, and an OPLS-DA model of the discovery data for differentiation of IS and control was established. After the model was developed, the top markers for distinguishing IS and HC were selected and identified. And these differentiated ions were further validated

using a validation dataset. To the best of our knowledge, this is the first report on the differentiation of IS and HC based on DART-MS.

The objectives of this project are to optimize the DART-MS parameters; to develop a model for rapid detection of IS; to identify primary IS markers.

3.2 Materials and methods

3.2.1 Reagents and materials

HPLC grade chloroform and methanol were from Duksan (ANSAN-SI, South Korea) and deuterated internal standards, including n-pentadecanoyl-D-erythrospingosine-d7 and 1,3 (d5)-dihexadecanoyl-2-octadecanoyl-glycerol, were purchased from Avanti Polar Lipids (Alabaster, AL, USA). Mesh sampling strips (74 × 74) were provided by Ion Sense, Inc. (Saugus, MA, USA). Ultra-high purity helium (99.999%) was purchased from Linde Industrial Gases Co., Ltd. (Hong Kong, China).

3.2.2 Sample collection

Clinical samples used in this chapter are the same as that in the chapter two. Briefly, the patients of the IS group were recruited from the Accident & Emergency Department of the First Affiliated Hospital of Wenzhou Medical

University, and health controls were recruited from the medical examination center of the same hospital. After confirming with CT scan and MRI scan, patients who were diagnosed as IS were recruited to IS group and volunteers who have no IS symptoms were recruited to HC group. In total, 53 IS patients and 28 health controls were recruited. 5 mL of blood samples from the participants were collected in heparin-treated tubes and were conducted a centrifugation at 3600 rpm under 4 °C for 10 min to obtain supernatant. The supernatants were stored at - 80 °C before further processing. All the participants were aged between 30 and 80 years old and they had no history of acute coronary syndrome, tumors, severe liver or kidney dysfunction or autoimmune diseases. Baseline characteristics, such as age, sex, body mass index, total cholesterol level and triglyceride level, of the participants were recorded. These plasma samples were separated into two datasets for the discovery and validation analysis. The discovery study was conducted to screen the plasma metabolites related to IS (training dataset, 36 IS cases and 19 controls) and to establish a classification model based on the screened metabolites for the differentiation between IS and HC. Furthermore, a validation dataset of 17 IS cases and 9 controls was subsequently analyzed for assessing the differentiation ability of the classification model mentioned above.

3.2.3 DART-MS analysis

The DART SVP ionization source (Ion Sense, Inc.), which was equipped with a transmission module (P/N JCL-2100-A), was utilized for loading and ionizing plasma samples. The transmission module functioned as the stage for holding the metal screen had ten successive holes onto which the plasma samples were pipetted. These holes were 7 mm in diameter and placed 14 mm apart in horizontal direction. The DART setup was shown in **Figure 3-1**. The autosampler module was set up on a programmable rail track, which could move between the ceramic ion transfer tube of the VAPUR[®] interface and DART source, the samples could be thus ionized in the area between the DART source and the transfer tube. The ion transfer tube was placed near the MS inlet, and it could transfer ions into the mass spectrometer. After samples were pipetted in the sampler, the solvent in them could evaporate in several seconds, then the sampler module started to run, and the samples could be ionized under the ionization areas. The speed of the autosampler could affect the peak shape of the chromatograms, so it should be adjusted from 0.2 mm s⁻¹ to 10 mm s⁻¹ to obtain a good peak shape. Helium was used as the ionization gas and nitrogen was employed as the carrier gas under standby state. The thermal helium gas spewed from the DART source and reacted with the

atmosphere and the samples, resulting in the ionization of the analytes in the samples.

The Agilent Q-TOF mass spectrometry (Agilent Technologies, Santa Clara, CA, USA) was employed for the analyses in the positive ion mode. The MS interface was replaced with a VAPUR[®] interface which was compatible with DART. The VAPUR was linked with a vacuum pump, a vacuum environment was thus provided inside the VAPUR interface to avoid the flowing of the discharged gas into the mass spectrometer atmospheric pressure inlet. The optimized MS parameters for the further experiment were selected as follows: drying gas temperature, 350 °C; capillary, 1 kV; skimmer, 75 V; RFVpp, 100 V. Mass spectra were recorded in the *m/z* scan range 50-1200 Da. The collision energy for MS/MS was from 20 V to 40 V for the mass range from 50 to 1200 Da. Before the change of the ESI source to DART, the MS was calculated using the reference solution from Agilent prior to the DART analysis. Data acquisition, peak picking, and background subtraction were achieved using MassHunter acquisition software (Agilent Technologies, Santa Clara, CA, USA).

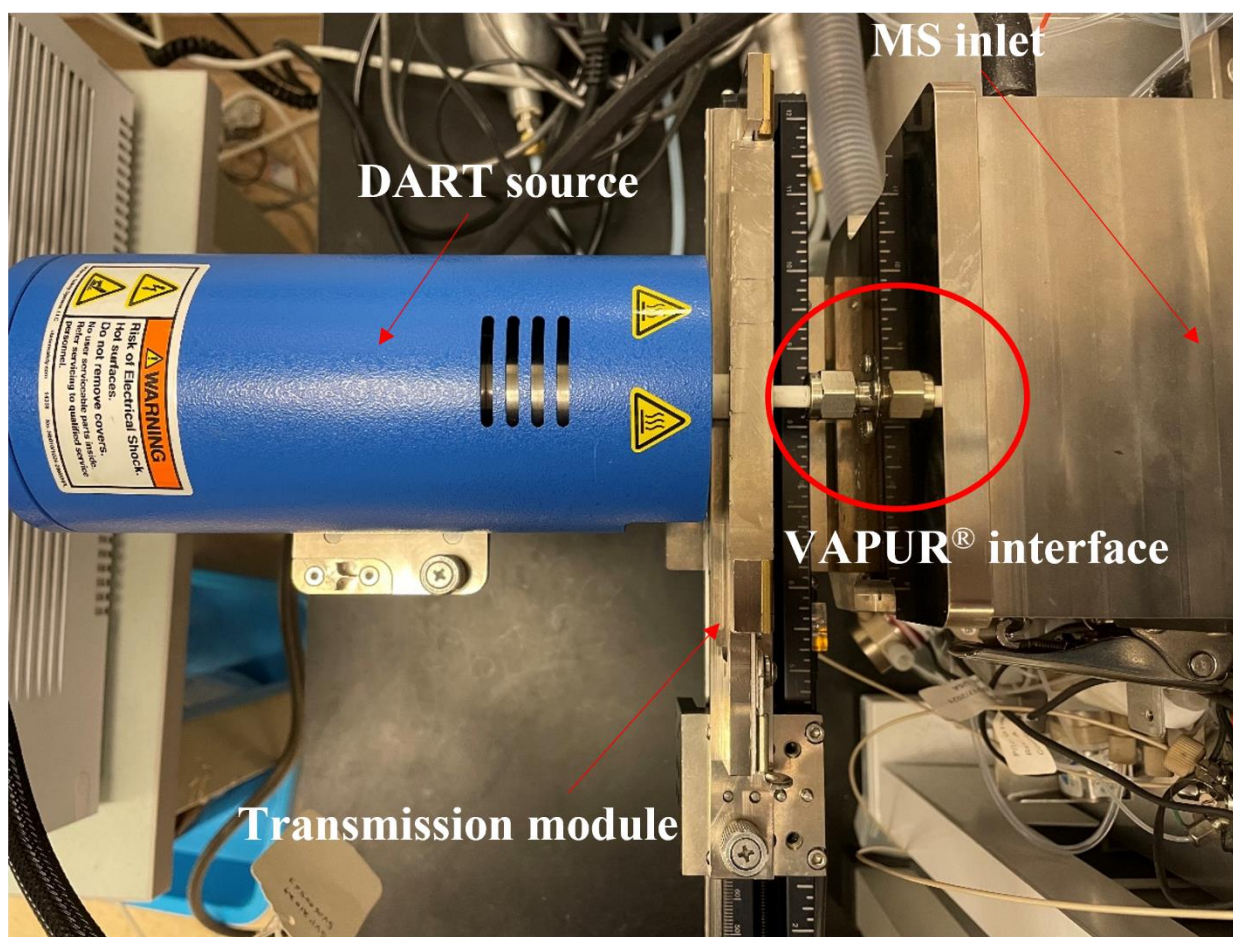
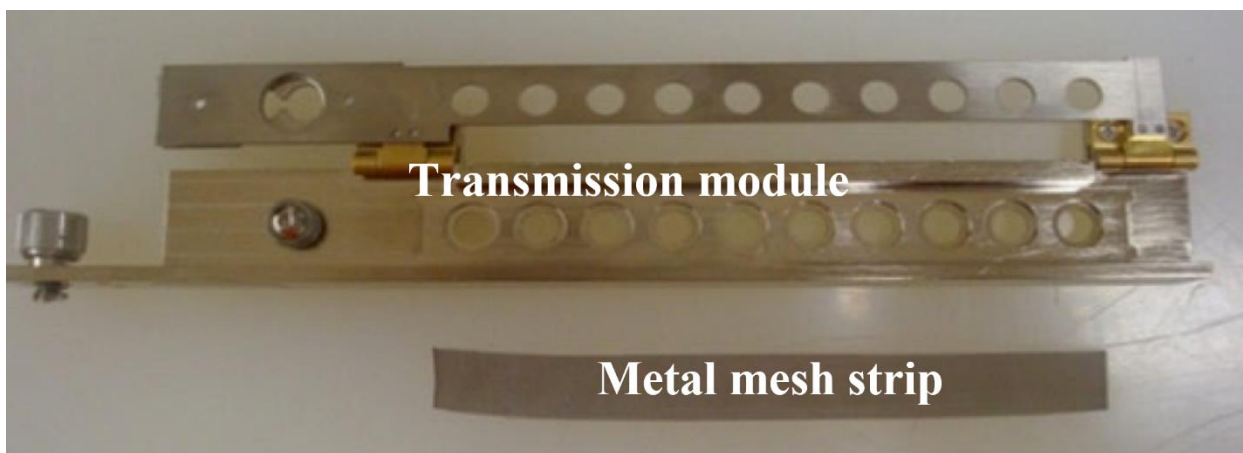


Figure 3-1. The DART-MS setup. The top panel shows the transmission module, and the metal mesh strips. The bottom panel displays the top view of the TM-DART-MS system.

3.2.4 Sample preparation and loading

In order to reduce carry-over from the wire screens which was used to load the plasma samples, those wire screens were first sonicated with distilled water and then with isopropanol for 1 hour before they were employed as the sampling support. 50 μL of each individual plasma sample was thawed on ice prior to analysis, followed by the centrifugation at $14\text{k} \times g$ for 1 minute. Then the supernatant was transferred to a new centrifuge tube for DART-MS analysis.

Transmission-mode DART, where the transmission module with ten holes was oriented vertically to both DART ion source and MS inlet, was utilized as the sampling strategy. To obtain more accurate semi-quantitative results, 2 μL of 2.5 $\mu\text{g/mL}$ TG and ceramide internal standards mixture, dissolved in acetonitrile and isopropanol mixture (1:1, v/v), were pipetted on the center of sampling area before plasma sample loading. The internal standard would be used for further normalization. After the internal standard mixture was dried at room temperature, 2 μL aliquots of each plasma sample were pipetted onto the same sampling area as that of internal standard, and five adjacent sample spots were pipetted onto the wire screen for replicate measurement and abundance average. Then the control program was activated to push the

individual samples through the ionization region between the DART source and the VAPUR inlet. The samples were analyzed continuously by pushing the module through the ionization region at a rate of 1.2 mm s⁻¹. The temperature of the DART metastable gas stream was set to 350 °C. Blank samples were prepared by pipetting 2 µL acetonitrile and isopropanol mixture directly onto the wire screen.

3.2.5 Data processing

All profile mass spectra were obtained by extracting the peaks in total ion chromatogram after each injection. Following DART-MS data acquisition, the background spectrum was subtracted, and profile spectra were exported in comma-separated format before the data analysis. A final mass list consisting of all the features from the samples was produced using Excel macro and then a data matrix with the final mass list and their corresponding abundance in different samples was produced using RStudio. Those ions with intensity lower than 10000 were removed, and then the abundance of the remaining compounds was subsequently normalized to the intensity of internal standard or scaled to total ion abundance. Subsequently, the intensity from 5 replicates was averaged and the compounds that could not be found in 80% of the samples of both IS and HC groups were excluded. Finally, the missing values

were replaced by 1/5 of the minimum of the ion abundance in the corresponding group.

Normalization is essential in MS-based omics analysis for calibrating data among different sample runs. Systematic variation during sample running, such as differences in sample loading, can then be corrected after normalization by a gain factor for each sample. The raw abundance of every individual feature in the sample is multiplied by the gain factor to produce its corresponding normalized abundance, and the calculation of normalized abundance was shown in **Equation 1**, where A_i is the measured ion abundance of ion i on sample k , α_k is the scalar factor for sample k , and A'_i is the normalized abundance of ion i on the sample k .

$$A'_i = \alpha_k A_i \quad (1)$$

This factor scalar can be estimated by several approaches. The commonly used method to determine α_k is normalization to the total abundance of all ions in the sample, i.e., the normalized signal is adjusted to the proportion of the total signal. However, this method could be inaccurate if the high abundant

ions that dominate the total abundance significantly fluctuate in samples due to instrumental variation or errors from sample preparation.

Internal standard (InterS)-based normalization is another common strategy applied for DART analysis. Internal standard should be spiked in the sample before instrumental analysis when this method is used for normalization. In this study, deuterated ceramide (d-ceramide), the mass of which is in the middle of the scanned mass range, was pipetted onto the mesh before sample loading and the abundance of the d-ceramide was used for the normalization of the detected ions. Five replicates of DART spectra for each sample were then averaged to increase the accuracy of ion abundance. Normalized data produced from the two normalization approaches mentioned above were compared and the optimal one was subjected to data analysis and modeling.

3.2.6 Data analysis

Relative log abundance (RLA) plots were obtained by removing the median of individual ion within IS or HC groups and then standardizing that ion. PCA and OPLS-DA were performed using SIMCA-P. After the abundance of exported ions was log₁₀-transformed, statistical differences were evaluated by Mann-Whitney test (p value < 0.05) and false discovery rate (FDR)

correction based on the Benjamini-Hochberg method was performed; Fold change (FC) of the intensity of ion was equal to the average abundance of the ion in the IS group divided by that in the control group, and the cut-off value was set as 1.2 or 1/1.2. After the univariate analysis mentioned above, the remaining lipids were treated as differentiated lipids. Accurate masses of candidate ions were searched in the Human Metabolome Database, the LIPID MAPS database and Metlin database, and the threshold of mass error was set as 0.01 Da. And those results with the highest matching score were taken as the tentative identifications which were further validated by MS/MS fragments. Identified markers were imported to the Ingenuity Pathway Analysis software (Qiagen) and open database sources, including KEGG, MetaboAnalyst, and Human Metabolome Database. All statistical analyses were performed using IBM SPSS statistics software (version 22.0, SPSS Inc., Chicago, IL, USA), RStudio (version 1.1.463), and Metaboanalyst (version 5.0, <https://www.metaboanalyst.ca/>).

3.3 Results and discussion

3.3.1 Optimization of DART-MS operating parameters

In order to obtain good signal of the compounds in plasma, some important parameters of DART, including gas temperature, the distance from DART ion

source to sample supporter (DS distance), and grid voltage, were optimized using a plasma mixture from IS patients and health control before the analysis,

The gas temperature ranges from 100 °C to 500 °C was investigated and it was found to be the prominent factor that influenced the metabolite signal. The mass spectra observed during the DART analysis of plasma at different gas temperatures were shown in **Figure 3-2**. The number of peaks with m/z lower than 400 Da increased as temperature increased from 100 °C to 350 °C and then dropped under rising temperature. Interestingly, larger m/z ions appeared obviously when the gas temperature was set to up to 350 °C, but the number of peaks with m/z over 600 Da decreased steadily when the temperature was continuously elevated, a result that might be attributed to the development of fragments at high temperatures. The total number of ions at various temperatures was also compared and the results were shown in **Figure 3-3a**. The results suggested that the number of ions increased when helium temperature increased from 100 °C to 350 °C and a total of 466 ions were detected when the plasma sample was analyzed under 350 °C. It was assumed that high temperature accelerated sample drying and facilitated desorption. Since most constituents in the plasma sample were non-volatile, a relative high gas temperature of 350 °C would be needed. However, decreased number

of ions was observed with increasing gas temperatures. Furthermore, two non-polar lipid standards, deuterated TG (m/z 857.8) and ceramide (m/z 531.5), were utilized for the optimization. The line chart (**Figure 3-3b**) showed that no signals were detected at 100 °C for both lipids and TG signals could not be observed although the gas temperature had reached 200 °C. The intensity of lipid standards gradually increased when the helium temperature was increased to 350 °C, but their amount started to decrease at temperatures over 350 °C. It was speculated that the compounds in the plasma samples were pyrolyzed under high temperature, as shown in **Figure 3-3c**. The sample in the wire mesh were found to be carbonized when the gas temperature was over 350 °C. Taking all the results illustrated above into consideration, 350 °C was employed as the optimal temperature.

The distance between DART source and sample supporter (DS distance) was optimized via comparing the quantity of detected peaks and the signal intensities of the two standards. The bar chart in **Figure 3-4a** illustrated the number of detected ions for the DS distance ranging from 2.5 cm to 4 cm. Compared with the signals that was obtained when the DS distance was 3 cm, other DS distances resulted in fewer ions. The effects of DS distance on the signal of the two lipid standards were studied and the results were

demonstrated in **Figure 3-4b**, which indicated that the intensities of both standards enhanced when increasing the DS distance from 2 cm to 3 cm. And the signals of both decreased when enlarging the DS distance from 3 cm to 4 cm. Thus, the distance of 3 cm would be utilized for further analysis of the plasma samples.

It has been reported that the grid voltage is a key factor to affect DART ionization, so grid voltage is another essential parameter for DART analysis in this study. We found that the ion number was significantly influenced by grid voltage and the maximum ion quantity was achieved when the voltage was set as 100 V (**Figure 3-5a**). When grid voltages of 50 V and 150 V were applied for the analysis of standards, comparable signals under these two voltages were observed. The abundance of two lipid standards under 100 V were obviously higher than those under other two voltages (**Figure 3-5b**), the optimal grid voltage 100 V was thus chosen. In summary, gas temperature of 350 °C, DART to sample supporter distance of 3 cm and grid voltage of 100 V were the optimal parameters for DART analysis, and they were employed for the further analysis.

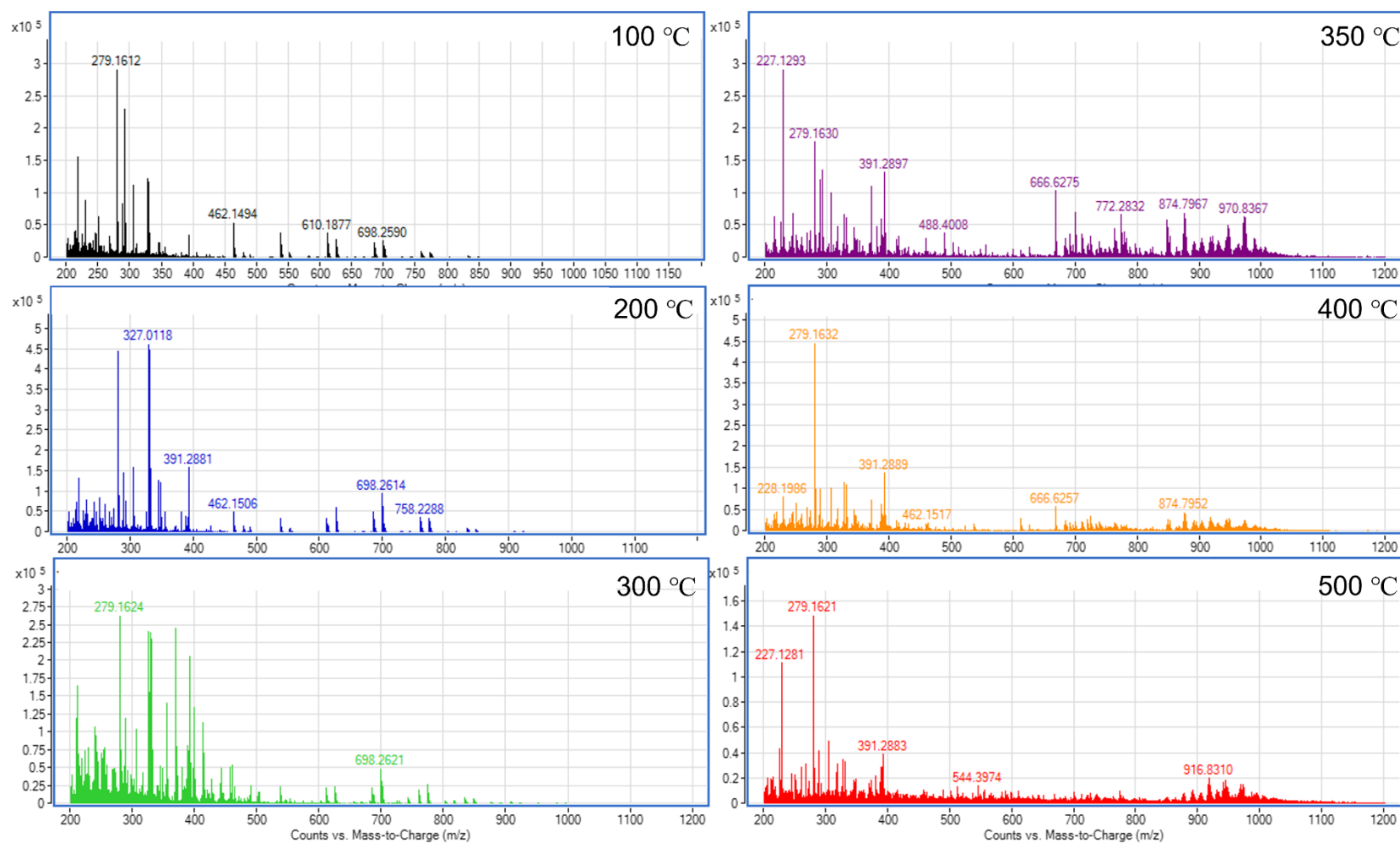


Figure 3-2. Mass spectra obtained at various helium temperatures

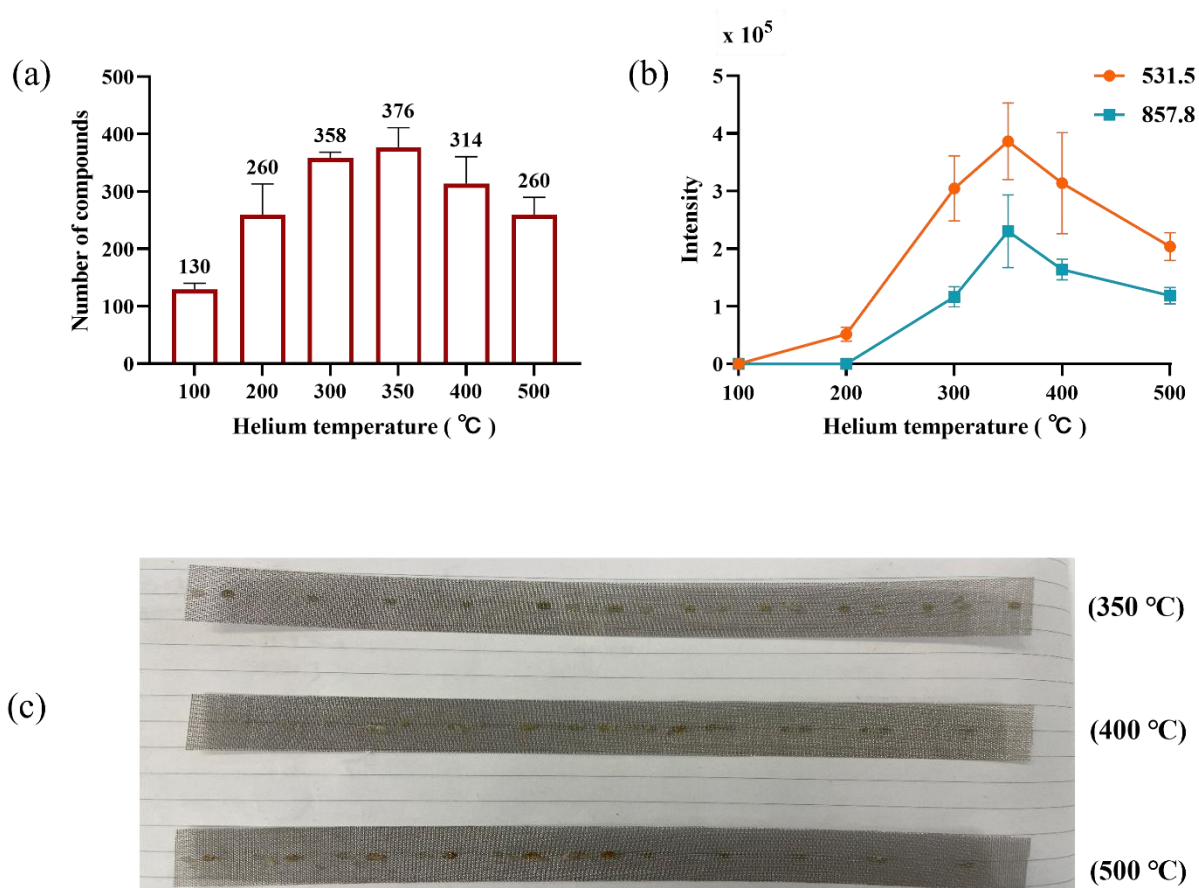


Figure 3-3. Helium temperature effect on DART-MS analysis of plasma and lipid standards (a) the number of ions detected at different helium temperatures, (b) intensity of internal standard at distinct helium temperatures, orange dots and light blue dots represented the signal for m/z 531.5 and 857.8, respectively, (c) plasma samples on the wire mesh after DART analysis under different temperatures

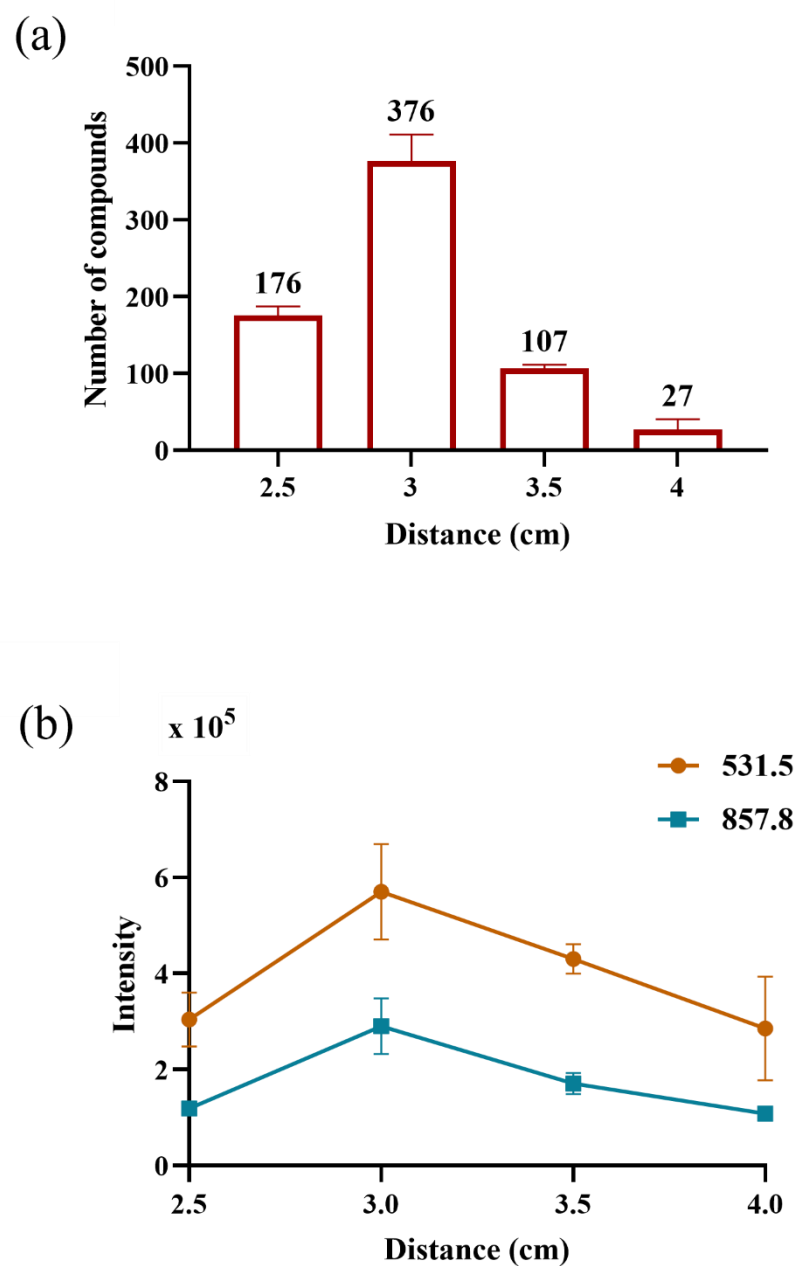


Figure 3-4. DS distance effect on DART-MS analysis of plasma and lipid standards. (a) the number of compounds detected at different DS distances, (b) intensity of internal standard at different DS distances, orange dots and light blue dots represented the signal for m/z 531.5 and 857.8, respectively

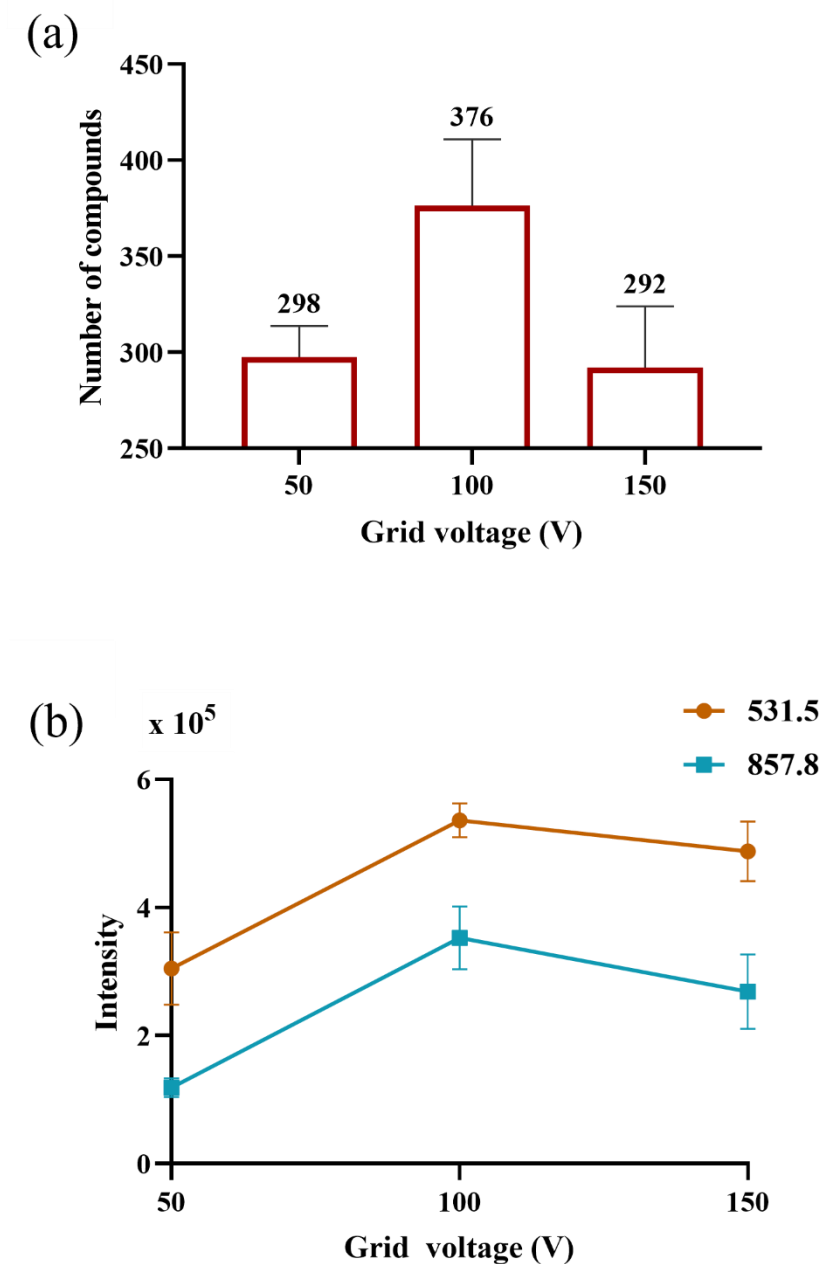


Figure 3-5. Grid voltage effect on DART-MS analysis of plasma and standards. (a) the number of ions detected at different grid voltages, (b) intensity of internal standard at different grid voltages, orange dots and light blue dots represented the signal for m/z 531.5 and 857.8, respectively

3.3.2 Selection of normalization method

The variation of deuterated-TG (d-TG), RLA and PCA were employed to evaluate the performance of different normalization methods, including total ion current (TIC)-based normalization and internal standard (InterS)-based normalization. 2 μ L Isotope-labeled TG was pipetted onto a wire screen before sample loading, and its relative abundance was subsequently calculated to assess different normalization methods. A plasma mixture from IS patients and health control was utilized here. The RSD of d-TG did not show significant improvement after TIC-based normalization for both HC and IS samples and the result was shown in **Table 3-1**. Compared with TIC-based normalization, the RSD of d-TG dropped from 0.8253 to 0.3860, and from 1.2727 to 0.5886 for HC samples and IS samples, respectively. Within-group RLA plots of TG signal could be used to assess the tightness of the replicates within groups, and they should have a median of 0 and low variation around the median. The RLA plots of both unnormalized raw data and different normalization methods for HC and IS patients were shown in **Figure 3-6**, which showed that the signal variation of TG in the HC samples was still visible in the TIC-based normalization and the distribution of the relative log abundance of internal standard was similar before and after TIC-based normalization. The normalization method relying on the internal standard

showed less variation than the TIC-based approach and its relative log abundance variation was within ± 0.5 for HC samples. For IS samples, box plots of the normalized values for internal standard were highlighted in red in **Figure 3-6**. From the plot, we can see that, in general, even with total ion normalization, the variation resembled that without normalization. However, InterS-based normalization decreased the variability within the IS samples. A QC sample was prepared by mixing ten HC samples with ten IS samples, which were randomly chosen. Ten consecutive injections of this QC sample were conducted on two different days to obtain two batches of data and the PCA score plots for these two datasets (**Figure 3-7**) were plotted to further assess the normalization approaches. Consistent with the results illustrated above, two datasets from two different days were significantly separated for the TIC-based method, which depicted TIC normalization could not reduce the instrumental variation. In contrast, the PCA plot for the InterS -based normalization method showed that half of the sample from the second batch were clustered with the first batch, indicating that batch variation was slightly removed via internal standard normalization. Therefore, d-ceramide was spiked into the samples as internal standard for data normalization.

Table 3-1. Comparison of the signal variation of d-TG (m/z 857.8) calculated by non-normalized method and normalized methods

Normalization method	RSD in HC samples	RSD in IS samples
None	0.8253	1.2727
TIC	0.8847	1.2125
InterS	0.3860	0.5886

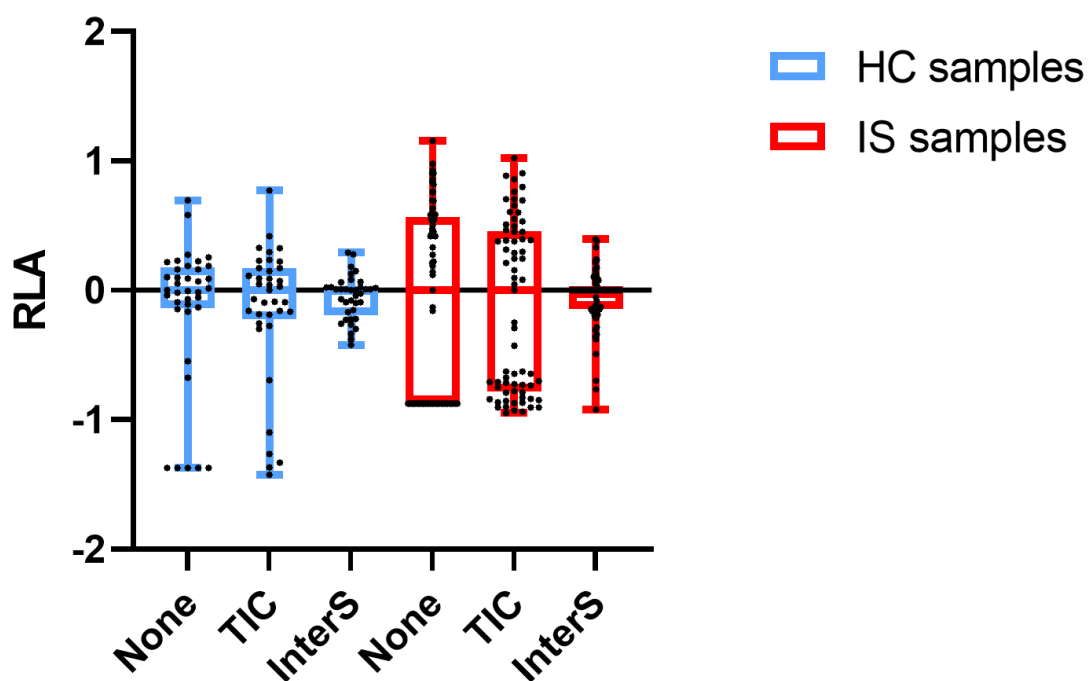


Figure 3-6. Within group RLA plots for HC sample (blue) and IS samples (red) before (None) and after different normalization methods

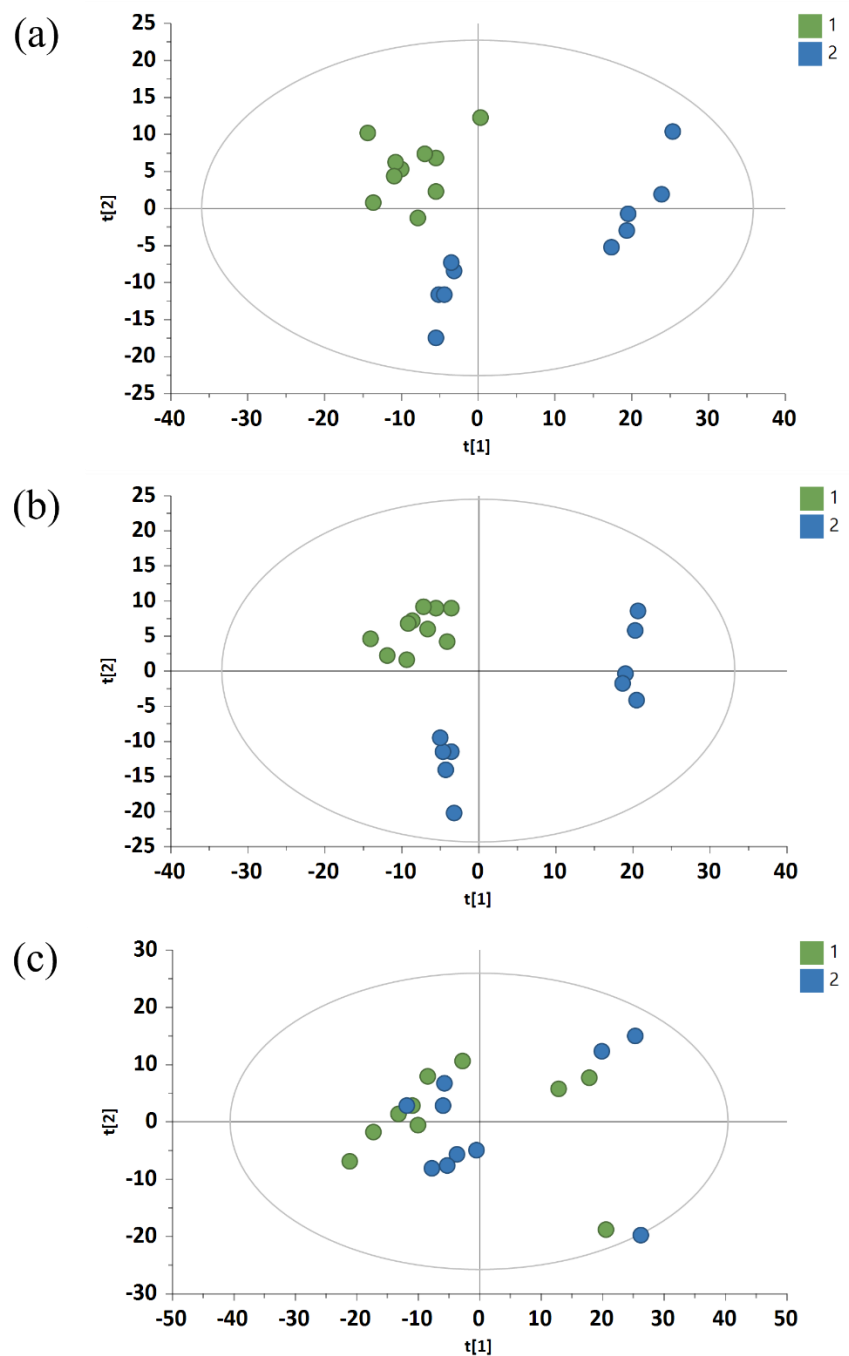


Figure 3-7. PCA plots before and after normalization for the QC samples detected on two different days. (a) Without normalization (None), (b) Normalization based on TIC, (c) Normalization based on one internal standard. Colors in the plots represent different days

3.3.3 Application of the DART-MS-based approach for IS diagnosis

A discovery dataset including 36 IS patients and 19 health controls was employed for the discovery of the potential markers. 11,664 ions in the plasma from discovery dataset were detected after the analysis by DART-MS, and only those ions detected in not less than 80% of at least one sample group were recruited to the dataset, and the missing values in the dataset were replaced by 1/5 of the minimum positive values of their corresponding variables, whereby 784 ions were retained. Afterwards, those ions with average raw abundance lower than 10,000 in both sample groups were subsequently ruled out, by which a total of 441 compounds were remained in the data matrix. This data matrix was then normalized by dividing the abundance of each variable by the abundance of internal standard spiked in the samples.

Then the result was subjected to the orthogonal projections to latent structures discriminant analysis (OPLS-DA) for distinguishing between IS patients and health controls. **Figure 3-8a** delineated the score plot of the OPLS-DA model, which showed good classification of plasma samples from HC and IS patients. The parameters R^2 and Q^2 , which were used for evaluation of the model performance, were 0.947 and 0.745, respectively. Besides, a response

permutation test was performed, and the obtained plot, shown in **Figure 3-8b**, indicated the OPLS-DA model established above was valid without overfitting, given that Q^2 was a negative intercept (-0.539) and R^2 in a response performance test was smaller than that of the original OPLS-DA model. Moreover, the prediction power of both the healthy group and IS group was 100% (**Table 3-2**).

Table 3-2. The classification result for discovery dataset

Group	Number	Predicted result		Correct classification
		HC	IS	
HC	19	19	0	100%
IS	36	0	36	100%
Total	55	19	36	100%

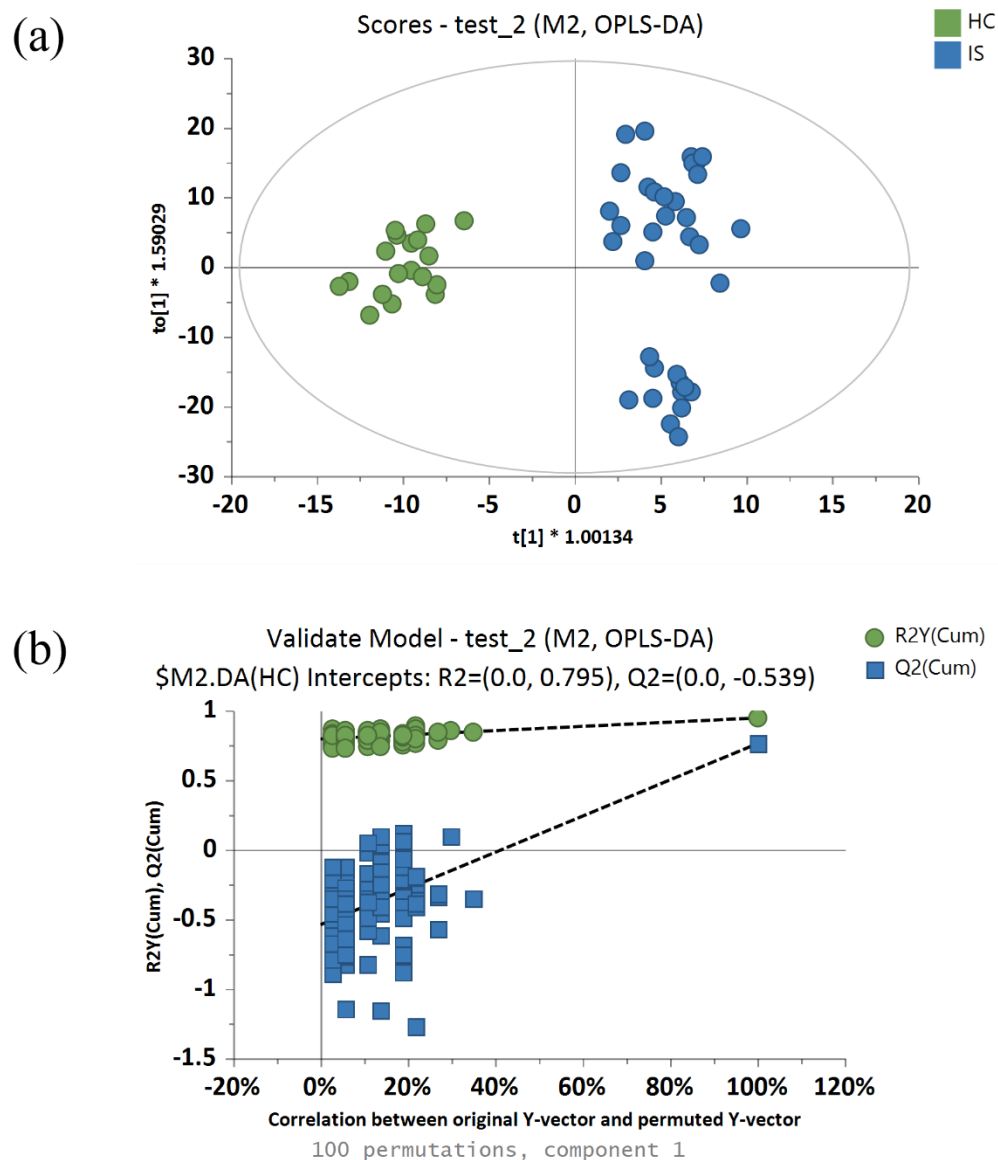


Figure 3-8. Score plot (a) and permutation plot (b) for the OPLS-DA model obtained upon DART-MS fingerprinting of plasma from IS patients ($n = 36$, blue) and health controls ($n = 19$, green)

3.3.4 Validation of OPLS-DA model

An additional dataset with data from 9 health control samples and 17 IS samples was used for the testing of the aforementioned OPLS-DA model. Using this external validation, the prediction abilities of 100% and 94.1% were achieved for the healthy group and diseased group, respectively. The classification results for the validation dataset were shown in **Table 3-3**, indicating that the rapid differentiation of IS was achieved using plasma analyzed by DART-MS together with the OPLS-DA.

Table 3-3. The classification result for validation dataset

Group	Number	Predicted result		Correct classification
		HC	IS	
HC	9	9	0	100%
IS	17	1	16	94.1%
Total	26	10	16	96.2%

3.3.5 Identification of differentiated ions

The representative DART-MS spectrum labeled with identified compounds in the mixed plasma from 7 IS samples and 7 HC samples was shown in **Figure 3-9**. Sphingonines, fatty acyls, steroids and other organic acids were detected within the lowest m/z range of the spectrum, followed by lysophospholipids and diacylglycerols with m/z range from 400 to 600 Da. And lipids with larger structure, including ceramides, phospholipids, sphingomyelins, cholesterol esters and triacylglycerols, occurred in the largest m/z range (m/z 600-1200 Da). Ions responsible for discrimination of IS and HC were selected based on the p value of student t test ($p < 0.05$), VIP value and Jackknifed confidence interval (not across zero), resulting in a total of 14 discriminant ions as listed in **Table 3-4**. Although other non-polar compounds could be detected by DART, the differentiated ions were basically identified as lipids, including fatty acyls, sterols and glycerolipids. Among these differentiated ions, those with higher VIPs suggested that they contributed more to the distinguishment of IS and HC by DART-MS. Different from the result of LC-MS, most sterols, instead of GP and GL, had large VIP in the differentiation model of DART-MS. In addition, we further compared the plasma cholesterol between IS patients and health control in our sample, showing that IS patients had significantly elevated plasma cholesterol (LDL-C) levels (**Figure 3-10**).

There is growing evidence that lowering cholesterol reduces the risk of IS.¹¹⁴

Therefore, DART-MS-based detection of sterol may be a potential approach for the IS diagnosis.

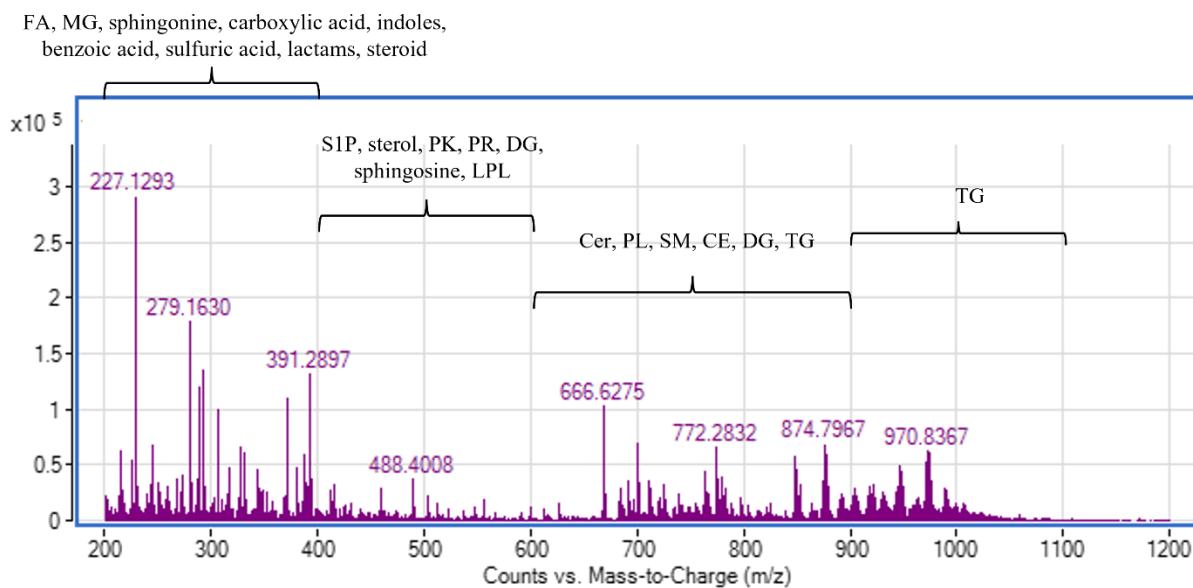


Figure 3-9. The DART-MS spectrum of a plasma mixture accompanying with the corresponding identifications.

Table 3-4. Fourteen differentiated ions for discrimination of IS and HC

Differentiated ions	<i>m/z</i>	VIP	Class
2-Methyl-dodecanedioic acid	227.1639	1.08	Fatty acyl
Eicosatrienoic acid	289.2521	1.01	Fatty acyl
Octadecenoic acid	301.2548	1.02	Fatty acyl
PGF2alpha-d4	341.2626	1.08	Fatty acyl
25-Hydroxy-cholesterol	423.4015	1.07	Sterol
Ergosterol	419.3265	1.02	Sterol
Obtusifoliol	425.3797	1.33	Sterol
5a-Androstan-3b-ol	259.2434	1.01	Sterol
24-Ethylidenelanost-9(11)-en-3-ol	455.4269	1.58	Sterol
Estramustine	462.1552	1.03	Sterol
CE(22:1)	707.6787	1.73	Sterol
TG(10:0/8:0/8:0)	499.4028	1.04	Glycerolipid
TG(18:0/20:0/20:0)	929.8918	1.30	Glycerolipid
MG(24:0)	465.3901	1.71	Glycerolipid

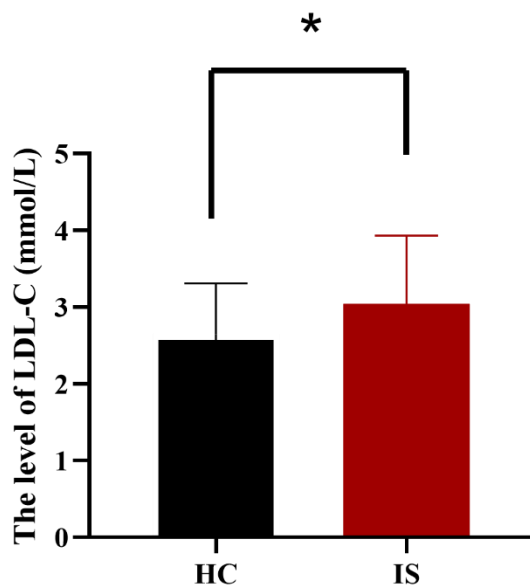


Figure 3-10. Comparison of LDL-C level between health control (n=28) and IS patients (n=53)

3.4 Conclusions

Treatment of IS should be carried out within 3-4.5 golden hours, otherwise it will damage the brain tissue and cells and then increase the disability and mortality. Therefore, rapid diagnosis of IS is very important. Although the LC-MS-based method in the previous chapter showed good differentiation of plasma from IS patients and health control, the total analysis time of LC-MS technique still required around 1.5 hours. This gap could be addressed by DART-MS, which is widely known for its fast analysis, simple sample pretreatment and less sample consumption. Therefore, in this project, a DART-MS-based method was established to rapidly differentiate IS patients

from health controls. Important parameters, namely gas temperature, DS distance and grid voltage, were optimized to obtain good signal of the metabolites prior to sample analysis. The temperature of 350 °C, the DS distance of 3 cm, and the grid voltage of 100 V were finally chosen as the optimal DART parameters for further experiment based on the total number of detected features and the signal of internal standard.

Normalization is required for the data process and has a great influence on the final analysis result, because appropriate normalization could provide us with more accurate results. Two normalization methods, InterS-based normalization and TIC-based normalization methods were compared. Generally, the InterS-based normalization produced more reliable result than the TIC-based normalization, given that the RSD of d-TG dropped significantly after the InterS-based normalization. Besides, the RLA analysis with smaller variation further validated the better performance of the InterS-based method than the TIC-based approach. Consistent with the results illustrated above, two batches of data acquired in two separate days formed two distinct clusters in the PCA plot by the TIC-based method, while the PCA plot for the InterS-based normalization method showed that half of the samples from the second batch were clustered with the first batch, indicating

that batch variation was slightly removed via internal standard normalization. Therefore, internal standard was spiked into the samples for data normalization.

The established DART-MS platform was then applied in the rapid differentiation of IS and HC. First, in the discovery phase, plasma samples from 36 IS patients and 19 health controls were analyzed by DART-MS to screen the differentiated ions for the distinguishment of IS and HC. The score plot of the OPLS-DA model for the discovery data showed good differentiation of plasma samples from HC and IS patients with R^2 and Q^2 of 0.947 and 0.745, respectively. Besides, a response permutation testing also indicated the OPLS-DA model established above was valid and overfitting was avoided. And the prediction power of both the healthy group and IS group in discovery dataset was 100%. Subsequently, an additional dataset of 9 health control samples and 17 IS patient samples was used for the validation of the established OPLS-DA model. The validation result showed that prediction abilities of 100% and 94.1% were achieved for the healthy group and diseased group, respectively. Therefore, the established OPLS-DA model could be used for the rapid differentiation of IS and health control after plasma sample was analyzed by means of DART-MS. Moreover, ions responsible for

discrimination were selected, and a total of 14 compounds were retained as potential markers, most of which were identified as fatty acyls, steroids and glycerolipids.

Compared with the previously established LC-MS method, DART-MS method required only simple pretreatment of plasma sample, which not only avoided sample degradation and contamination, but also saved time. In term of DART-MS method, the total time needed for the sample pretreatment and MS analysis was only around 5 minutes for one sample, but LC-MS method required around 1.5 hours for each sample. Therefore, the DART-MS-based approach was suitable for the rapid analysis of plasma from IS patients. Strikingly, DART-MS could even achieve a similar probability of IS prediction (96.2%) as the LC-MS-based approach in the previous chapter (96.3%). Furthermore, LC-MS-based method could be applied to IS diagnosis, mainly focusing on detecting relative levels of the potential biomarkers, but the DART-MS method was based on the full spectrum of the dataset, so data processing was simpler. But when compared with LC-MS method, DART-MS-based approach has lower sensitivity and stability, and mainly non-polar compounds could be detected by DART-MS. Therefore, LC-MS and DART-MS could be complementarily employed for the differentiation of IS and HC,

thus improving the speed and accuracy of IS diagnosis to achieve tPA treatment within the golden window to decrease the mortality and disability of IS patients.

Chapter 4: Investigation of the effects of hypoxia on vesicular lipid trafficking

4.1 Introduction

Lipid, a physiologically essential biological molecule, functions in various biological processes, such as energy storage and signal transduction. Additionally, it is noteworthy that lipid is also a fundamental structural component of the plasma membrane and other intracellular membrane from the endoplasmic reticulum (ER), the Golgi apparatus, and trafficking vesicles and so forth.¹¹⁵ Because different organelles have different functions, the composition of lipid differs among different organelles.¹¹⁶⁻¹¹⁸ Severe consequences could result from disordered alteration of lipid distribution or lipid metabolism. For example, diseases like diabetes and atherosclerosis are linked with irregular distribution of intracellular lipids. Membrane lipid composition could also lead to increased growth signaling, which is one of the major hallmarks of cancer.^{119, 120} Collectively, the lipid composition of membrane and intracellular trafficking of lipid are central to normal physiological processes within cells and organisms.

A vesicle is a structure with the surrounding lipid bilayers; it exists inside or outside of cells, which is called intracellular vesicle and extracellular vesicle, respectively. Vesicles can be basically classified as transport or secretory vesicles according to their functions. Vesicles are an indispensable part of the transport of biological components, for example, after proteins are produced in endoplasmic reticulum, proteins are sorted by vesicles and transferred to the Golgi apparatus for modification, and then the modified proteins are transported to next destination to exert their functions. The 2013 Nobel Prize in Physiology or Medicine was rewarded to James Rothman, Randy Schekman and Thomas Sudhof for their contribution to the discovery of the mechanism of vesicular transportation of cellular materials. The intracellular distribution of lipid depends on the intracellular trafficking of lipid. Similar to protein, it is reported that lipids and integral membrane proteins move between organelles in membrane-enclosed sacs that are called transport vesicles, and there is growing evidence that the transportation of lipids, similar to that of proteins, are based on transport vesicles.⁶⁶ However, we currently understand less about lipid trafficking than we do about protein trafficking, and it is of great significance to study lipid trafficking via vesicle transportation under different conditions.

The condition of oxygen deprivation was tested in this project given that the lack of oxygen, medically termed as hypoxia, is a commonly known feature of tumor microenvironment. Oxygen pressure is typically more than 20 mmHg in normal tissues, whereas oxygen tension is only approximately 10 mmHg in most of the tumor tissues as a result of the increased oxygen demand of rapid proliferation of cancer cell and less oxygen supply due to aberrant vascularization.¹²¹ Hypoxia has been reported to be linked with the progression, metastasis, immunosuppression and reduced therapeutic efficacy of cancer.¹²² Therefore, the study of hypoxic condition may facilitate the treatment of cancer. Hypoxia-inducible factor (HIF), a dimeric protein including an oxygen-sensitive subunit HIF- α and a HIF-1 β subunit, prominently regulate the essential hypoxia signaling pathway initiated by oxygen deprivation.^{123, 124} There is evidence that lipid accumulation could be induced by HIF1 upon hypoxia due to the dysfunction of lipid synthesis.¹²⁵⁻¹²⁸ In addition to lipid synthesis, some lipid mediators, the lipid metabolism and unsaturation (the number of C=C in fatty acyl chain) of lipid in membrane were also involved in the hypoxia stimulation. Bioactive lipid mediators, sphingosine 1-phosphate (S1P) and lysophosphatidic acid, in conjunction with hypoxic microenvironment inside cancer cells facilitate the development of pathological angiogenesis, which provides cancer cells with the oxygen and

nutrients they need for progression, proliferation, and metastasis.¹²⁹ It is reported that cancer cell facilitates fatty acid (FA) synthesis by de novo lipogenesis, thereby enhancing the levels of FA, triglyceride (TG) or phospholipids (PLs) to provide energy and maintain the function of organelle and plasma membrane.¹³⁰ Therefore, lipid metabolism is changed in cancer cells, which is named metabolic reprogramming and is a hallmark of tumor.¹³¹ The result from Victor Revin et al., demonstrated that insufficient oxygen contributed to a distinct change in the lipids composition of erythrocyte membrane.¹³² Ackerman et al., studied the viability of clear cell renal carcinoma (ccRCC) and their results showed that oxygen scarcity induced the saturation of FA and triglycerides principally activated the stearoyl-CoA desaturase (SCD) to synthesize unsaturated FA oleate for decreasing the FA saturation.¹³³

By understanding how cancer cells transport and distribute their lipids under hypoxic condition to fulfill metabolic requirements of exponentially growth, we may have some potential and innovative strategies for anti-tumor therapies. However, the influence of hypoxic condition on vesicular lipid trafficking is still unknown. Therefore, the objective of this chapter is to establish a platform

for lipidomic analysis of vesicles and to explore the effect of hypoxia on lipid trafficking and lipid composition within Hela cell.

4.2 Experimental section

4.2.1 Cell culture

Hela cell lines were provided by Prof. Zhao Qian's lab (The Hong Kong Polytechnic University, HK). Hela cells were cultured at an incubator at 37 °C with 5% CO₂ (v/v) and 24% O₂, and were maintained in Gibco Dulbecco's Modified Eagle Medium (DMEM) (Thermo Fisher, Waltham, MA) supplemented with 10% fetal bovine serum (FBS) (Thermo Fisher) and 1% penicillin streptomycin (Thermo Fisher). Cell counts were performed using a haemocytometer, which was employed for the viability determination accompanied with 0.1% (w/w) Trypan blue (Sigma-Aldrich). Hypoxic experiments were performed in a Hypoxic chamber with 1% oxygen at 37 °C in a 5% CO₂ humidified environment with the balance provided by nitrogen. Cells grown at around 90% confluence were passed into new sterilized 15-cm culture dishes, including 12 dishes for hypoxic group and 12 dishes for control group (2 dishes for each replicate). The control group and hypoxic group were simultaneously put into the normal and hypoxic environment for 24 hours, respectively.

4.2.2 In vitro vesicle formation assay

In vitro vesicle formation assay was modified based on a previously published method.^{134, 135} Hela cells were firstly suspended in ice-cold KAc buffer containing 110 mM potassium acetate, 2.5 mM magnesium acetate, and 20 mM Hepes (pH 7.2). Then 40 µg/mL digitonin was added into the cell suspension and incubated on ice for 5 minutes for permeabilization. The permeabilized cells, also called semi-intact cells, were obtained after centrifugation for 3 minutes at $300k \times g$ under 4 °C. Subsequently, the cell pellets resuspended in high salt KAc buffer (1 M potassium acetate, 2.5 mM magnesium acetate, 20 mM Hepes, pH 7.2) were incubated on ice for 5 min to remove cytosolic proteins. After the incubation with high salt KAc buffer, semi-intact cells were rinsed two times with KAc buffer and resuspended in 450 µL of KAc buffer, followed by the budding assay, i.e., the incubation of semi-intact cells at 32 °C with ATP regeneration system (4 mM creatine phosphate from Roche, 0.02 mg/ml creatine phosphokinase from Roche, and 100 µM ATP purchased from Sigma), 200 µM GTP (Wako, number SAH3766) and rat liver cytosol (RLC). After the incubation, cell debris and some large membranes, such as ER and Golgi membranes, were sedimented and eliminated by means of centrifugation for 20 minutes at $14k \times g$ under

4 °C, and the in-vitro formed vesicles were suspended in the supernatant. In respect of density gradient flotation assay, the vesicle solution was resuspended in 35% OptiPrep, followed by overlaying with 30% OptiPrep and 100 μ L KAc buffer. The OptiPrep gradients were then centrifuged at $300k \times g$ for 3 hours at 4 °C using a Hitachi Koki CP80WX ultracentrifuge with a P55ST2 rotor whereby the vesicle fraction was acquired by pipetting the supernatant slowly. Finally, the vesicle solution was analyzed by mass spectrometry after lipid extraction, nanosight nanoparticle tracking analysis, negative stain electron microscope, and western blot. The workflow for the vesicle budding assay and the procedure of SI cell was shown in **Figure 4-1**.

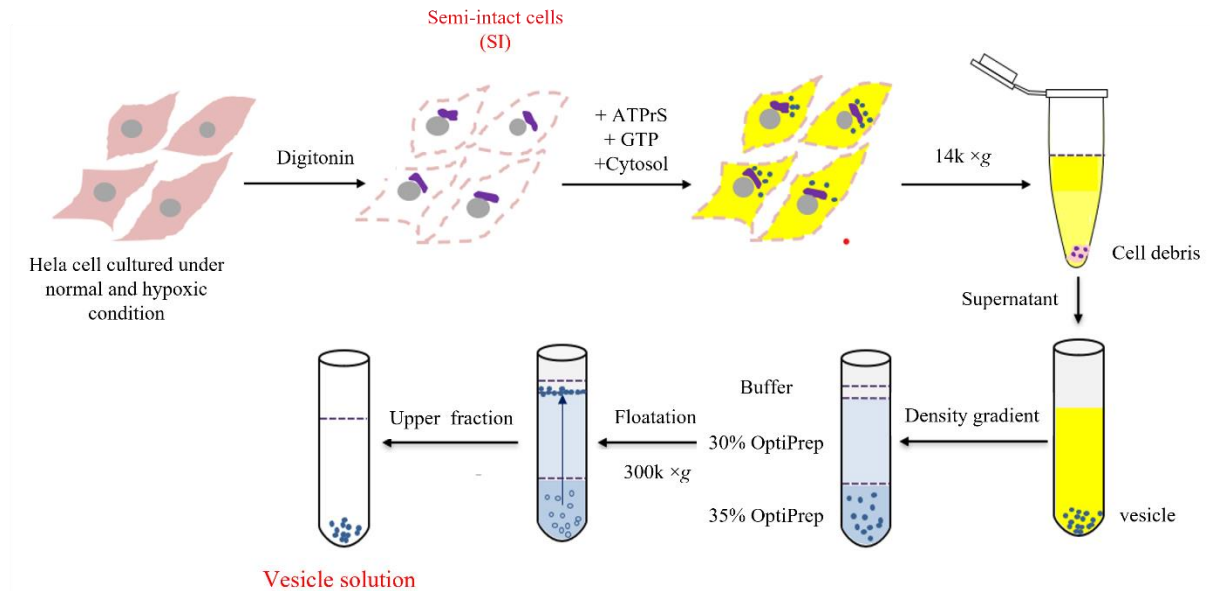


Figure 4-1. The diagram illustrating the experimental procedures for preparation of semi-intact cell and the vesicle budding assay

4.2.3 Nanosight nanoparticle tracking analysis

Nanoparticle (405 nm) tracking analyzer (Nanosight NS300HSBF) was utilized for the analysis of the released vesicles. Nanoparticles were illuminated by laser and their movements under Brownian motion were captured for 60 seconds. Video captures were subjected to nanoparticle tracking analysis (NTA) using the Nanosight particle tracking software to provide nanoparticle concentrations and size distribution profiles. At least three videos were captured for each individual sample to provide a representative concentration measurement, and all analysis settings were kept constant within each experiment. The size distribution profiles obtained from NTA were averaged within each sample across samples to provide representative size distribution profiles. These distribution profiles were then normalized to total nanoparticle concentrations or final cell counts.

4.2.4 Lipid extraction

The lipid extraction procedure has been optimized according to a previous method (**Figure 4-2**). Specifically, freeze-thawed 350 μ L vesicle solution or semi-intact (SI) cells suspension first and then added sample into a glass tube and spiked with internal standards, including deuterated internal standards consisting of 1-palmitoyl-d31-2-oleoyl-sn-glycero-3-phosphocholine, 1-

palmitoyl-d31-2-oleoyl-sn-glycero-3-[phosphor-L-serine], 1-palmitoyl-
 d31-2-oleoyl-sn-glycero-3-phosphoethanolamine, 1-palmitoyl-d31-2-
 oleoyl-sn-glycero-3-phosphoinositol, 1-pentadecanoyl-2-oleoyl(d7)-sn-
 glycero-3-phosphate, 1-hexadecanoyl-2-(9Z-octadecenoyl)-sn-glycero-3-
 phospho-(1'-rac-glycerol-1',1',2',3',3'-d5), N-pentadecanoyl-D-
 erythrosphingosine(d7), and N-pentadecanoyl-D-
 erythrosphingosylphosphorycholine-d9, 1,3 (d5)-dihexadecanoyl-2-
 octadecanoyl-glycerol purchased from Avanti Polar Lipids (Alabaster, AL,
 USA). Subsequently, the samples were extracted using 2.5 mL of a mixture
 of chloroform:methanol (2:1, v/v) and vortexed for 5 minutes. The mixture
 was then sonicated in ice water for 1 hour, then 200 μ L of water was added
 and vortexed for 5 minutes. Finally, the lower layer was isolated following
 the centrifugation at 2000 rpm for 10 minutes. The remaining was re-extracted
 with another 2.5 mL of chloroform and methanol mixture and the procedure
 mentioned above was repeated for a more thorough extraction of lipids. Two
 organic phases were combined and dried under nitrogen prior to storage at -
 80 °C. Prior to LC-MS analysis, the samples were reconstituted with
 isopropanol:acetonitrile:H₂O (8:1:1, v/v/v) and vortexed for 5 minutes and
 then centrifuged at 12k rpm for 5 minutes.

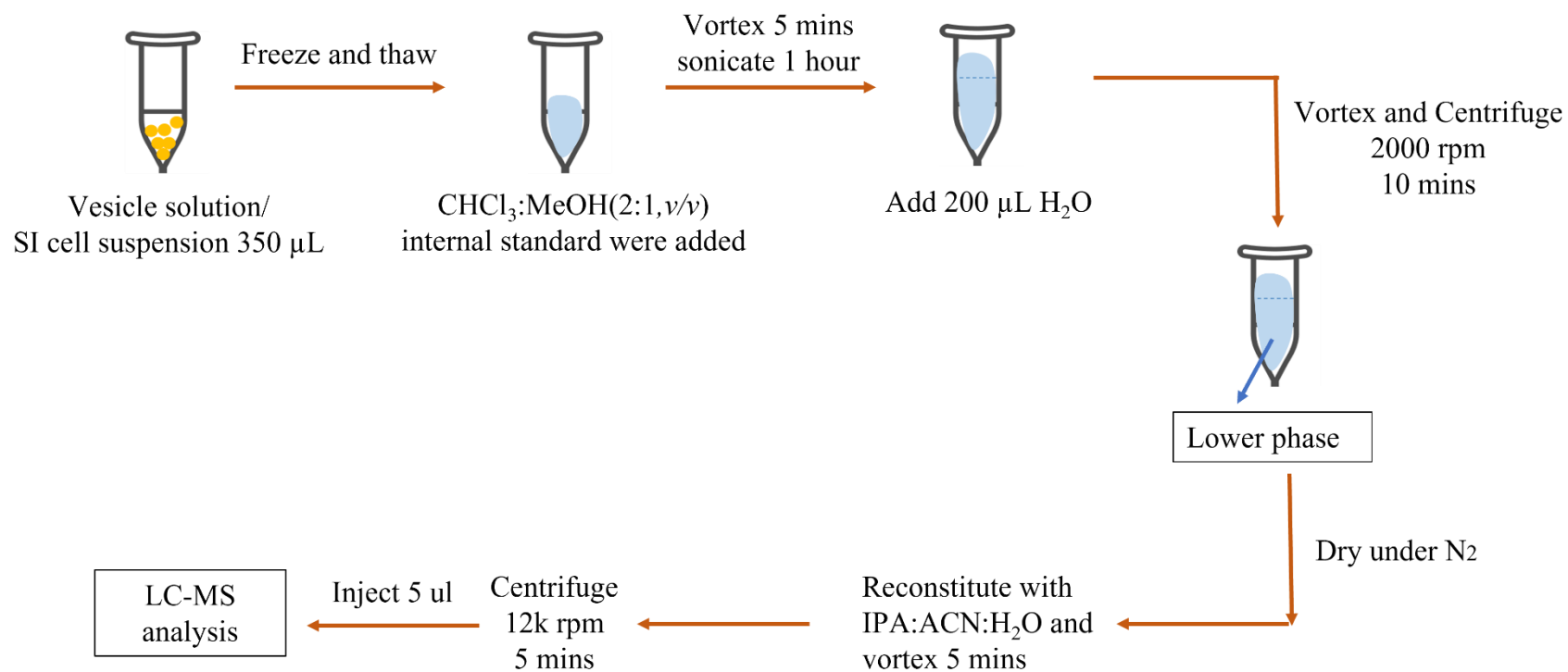


Figure 4-2. The diagram describing the experimental procedures of the lipid extraction for semi-intact cell and vesicle samples

4.2.5 LC-MS analysis

All the acetonitrile, methanol, isopropanol and chloroform were HPLC grade from Duksan (ANSAN-SI, South Korea). Milli-Q water was obtained from a Milli-Q water purification system (Millipore, Bedford, MA, USA). Formic acid and ammonium acetate were purchased from Sigma-Aldrich and Acros Organics respectively.

Mass spectra were acquired on a Thermo Scientific Orbitrap Fusion Lumos Tribrid mass spectrometer (Thermo Fisher Scientific, USA) equipped with a heated electrospray ionization probe in positive and negative ion modes. Mass spectrometric parameters for lipid extracts were as follows. In terms of positive ion mode, the spray voltage was set as 3.5 kV. The sheath gas, auxiliary nitrogen pressure and sweep gas were set as 35, 8 and 1 arbitrary units, respectively. Ion transfer tube temperature and vaporizer temperature were 325 °C and 300 °C, respectively. With respect to the negative ion mode, the spray voltage was set as 2.5 kV. The sheath gas, auxiliary nitrogen pressure and sweep gas were set as 40, 10 and 2 arbitrary units, respectively. Ion transfer tube temperature and vaporizer temperature were 325 °C and 275 °C, respectively. RF lens level was 30. Full scan, data-dependent MS/MS (top 10-ddMS2), and data-independent acquisition mode data were collected

at m/z 100-1200 Da for both negative and positive ion modes. Both MS and fragments were combined for the identification. DG and TG ions were monitored as ammonium adducts. Phospholipids, apart from PC for which acetate adduct was the highest, were detected as the deprotonated anions. Besides, ceramide was detected as acetate adducts, and cholesterol was monitored as the protonated ions.

An UPLC system (ACQUITY UPLC System, Waters, USA) was coupled to the Orbitrap Fusion Lumos Tribrid for the chromatographic separation of analytes. The autosampler temperature was maintained at 4 °C for all experiments. After lipid extraction, the extracts were injected into a reversed-phase CSH column, 2.1 × 100 mm, 1.7 µm particle size (Waters, Milford, MA). The column was maintained at 55 °C. The mobile phase comprised component A (10 mM ammonium acetate and 0.1% formic acid in acetonitrile:water (60:40, v/v)) and component B (isopropanol:acetonitrile (90:10, v/v) with 10 mM ammonium acetate and 0.1% formic acid). The mobile phase was pumped with a flow rate of 0.3 mL/min. The gradient elution program was as follows: 0-2 min, 30%-43% B; 2-2.1 min, 43%-50% B; 2.1-12 min, 50%-54% B; 12-12.1 min, 54%-70% B; 12.1-18 min, 70%-99% B; 18-18.1 min, 99%-30% B; 18.1-22 min, 30% B.

4.2.6 Data processing and analysis

The diagram in **Figure 4-3** was the systematic workflow of data processing. Particularly, Progenesis QI 2.3 (Nonlinear Dynamics, Newcastle upon Tyne, United Kingdom) was employed for peak picking and alignment. Low abundance peaks with abundance < 10000 were excluded. Then only those features detected in more than 80% individuals in at least one group were retained to minimize the effect of missing values (“80% Rule”). Quality screening was performed by removing those ions with RSD > 30% in all QC samples (“30% Rule”). Missing values were replaced with one fifth of the minimum value found in the same study group. The quantitative data were expressed in absolute units and the calculation of abundance was shown in **Equation 2**, where n_S is the mole of the analyte, n_{IS} is the mole of the internal standard, M_S is the molar mass of the compound, M_{IS} is the molar mass of internal standard, A_S is the abundance of the compound and A_{IS} is the abundance of the internal standard; the normalized ion abundance (mole) of all samples was imported to SIMCA P software (version 17.0, Umetrics, Umea, Sweden) for further analysis.

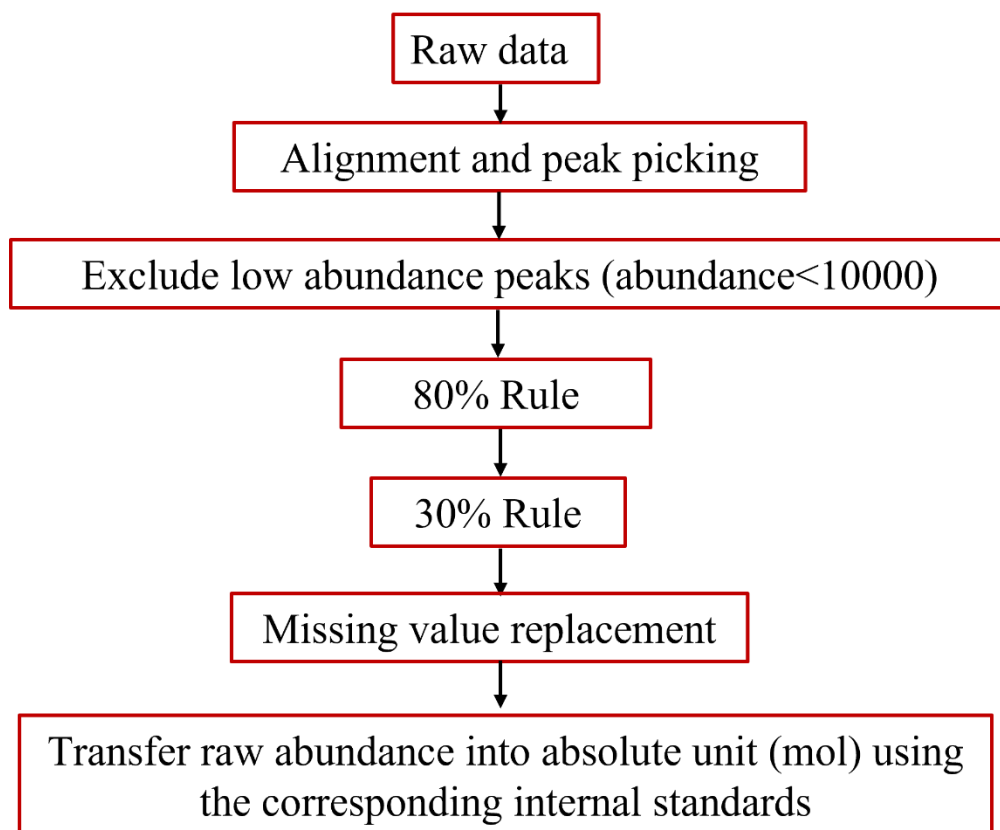


Figure 4-3. The diagram illustrating the procedures for data processing

$$n_S = n_{IS} \frac{M_{IS}A_S}{M_SA_{IS}} \quad (2)$$

The identifications of ions were conducted by matching their accurate mass with the Lipid Maps (<https://www.lipidmaps.org/>), METLIN (<https://metlin.scripps.edu/>), Human Metabolome Databases (<https://hmdb.ca/>), KEGG (<https://www.genome.jp/kegg/>), and/or confirmed by the commercially available standards based on their accurate mass,

retention times and fragmentation pattern. The mass error windows for precursor ion and fragment were set as 5ppm and 50 ppm separately.

4.3 Results and discussion

4.3.1 The chromatographic result of lipidomic analysis based on the established LC-MS platform

However, the lipidomic analysis pipeline of vesicle was underexplored, and the vesicle pellet produced after ultracentrifugation was difficult to dissolve in the extraction solution, even though vortex and sonication were employed to facilitate the dissolution. LC-MS analysis of extract from the vesicle pellet was performed to test if any lipids were extracted, but the chromatographic result turned out that no lipid was extracted and detected. Therefore, the vesicle solution, instead of vesicle pellet, was utilized for lipid extraction, which showed satisfactory results for further analysis (**Figure 4-4a-b**). Moreover, a similar extraction method was applied in the semi-intact cell samples, and the chromatograms of the lipid extracts of SI cell were delineated in **Figure 4-4c-d**. The LC-MS platform was established after the optimization of vesicle preparation steps, the lipid extraction method, and LC-MS parameters.

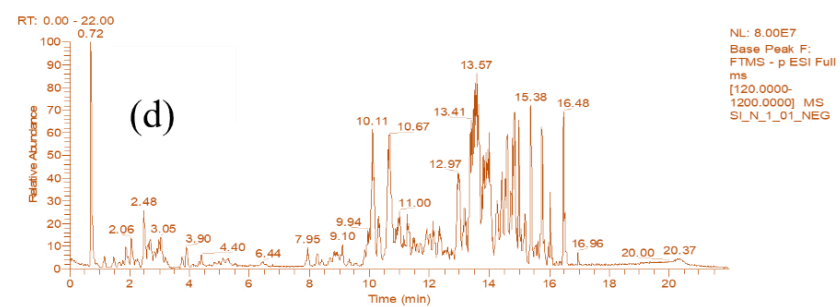
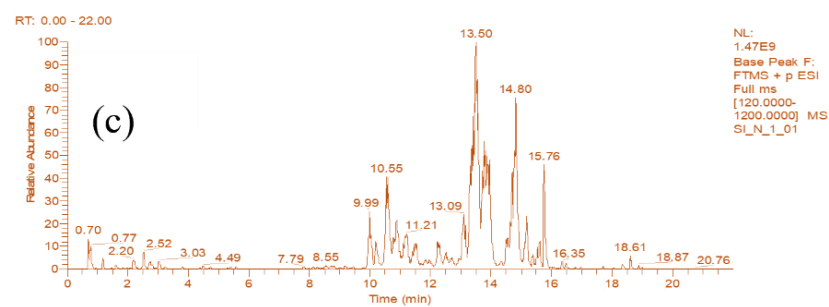
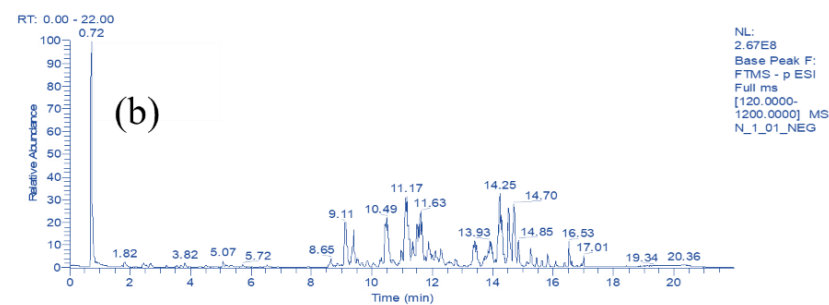
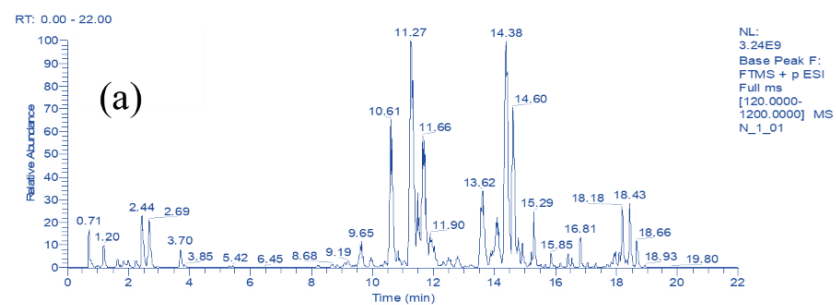


Figure 4-4. The chromatograms of vesicle solution (blue) and semi-intact cell (orange) in both positive (left panel) and negative (right panel) ion modes

4.3.2 The effect of oxygen deprivation on membrane lipids in semi-intact (SI) cells

Because the SI cells were treated with digitonin for permeabilization of plasma membrane, the cytosolic components were released from cells while keeping the organelles intact, therefore, SI cell were used for the study of the total membrane system based on lipidomic analysis in this study. As shown in **Figure 4-5a**, glycerophospholipids (GPs) are the major components of intracellular membrane system, followed by sphingolipid (SL). GPs were composed of two subclasses, i.e., phospholipid (PL) and their corresponding lyso-form (lysophospholipid, LPL), and PL accounted for the majority of GP. PC and PE were the most highly enriched lipids in the membrane system and were well known due to their roles in the membrane-mediated cell signaling and membrane fusion events. In order to investigate whether hypoxia influences the lipid composition of the membrane system, we calculated the lipid abundance in permeabilized semi-intact cells cultured under normal condition or hypoxic condition. Interestingly, as shown in **Figure 4-5b**, PC and PE were the two most abundant lipids of SI cell, but only the level of PC increased significantly due to oxygen insufficiency, and the level of PE did not change before and after hypoxic treatment, which suggested that metabolic disorder of PC was induced during the hypoxic process. In order to

investigate how the PC-related metabolism pathway changed after hypoxia, changes in the levels of PS, PG, DG and LPC, which participate in the metabolism of PC, were calculated. The result showed that the levels of PS, PG and LPC were decreased, and these three lipids were basically derived from PC by PTDSS1, PLD1-4/6 and PLA2, respectively. This indicated that the catabolism of PC was blocked after hypoxic stimulation. The metabolism pathway of PC with significantly altered lipids were shown in **Figure 4-6**. Interestingly, the levels of DG and PE, which are the main material to produce PC, were not altered by oxygen deprivation, which suggested that the anabolism of PC may not influenced by hypoxia.

In addition, although there were no significant difference of total PL and GL level between the control group and hypoxic group (**Figure 4-5a**), we found that the ratio of LPL was significantly decreased after hypoxic culture. Strikingly, the abundance of SL was dramatically improved under hypoxia due to the significant increase in sphingomyelin (SM), although the level of SM was much lower than that of GP. Of particular interest, lipidomes of vesicle membrane were previously shown that the levels of LPL and SL were essential for vesicle/COPII formation and transportation.^{136, 137} Thus, the lipidomic profile of SI cell that were distinctly associated with hypoxia

revealed that there may be a close relationship between oxygen deprivation and vesicle lipid compositions. We next wondered whether we could observe a change in lipid composition within vesicles.

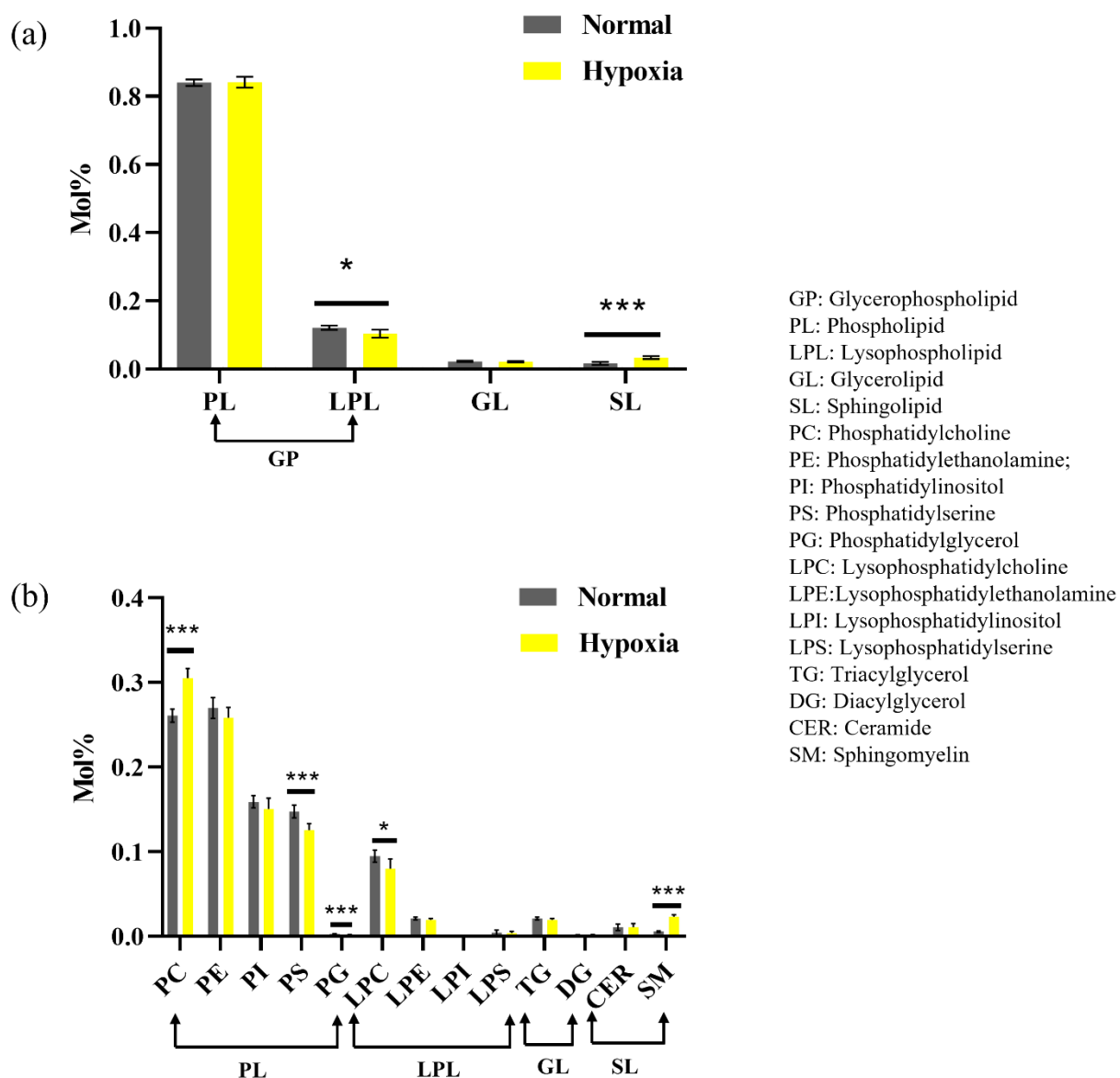


Figure 4-5. The lipid composition of semi-intact HeLa cells cultivated in normal (grey) and hypoxic (yellow) condition. (a) The composition of lipid class in mol% of total lipid contents, the lipids were organized according to their classes (b) Profile of lipid subclasses in mol% of all lipid categories. Each bar represents the average $\pm$ SD from $n=6$ independent trials. Statistical significance was tested by unpaired two-tailed t-test using Graphpad Prism, $*p < 0.05$, $***p < 0.001$.

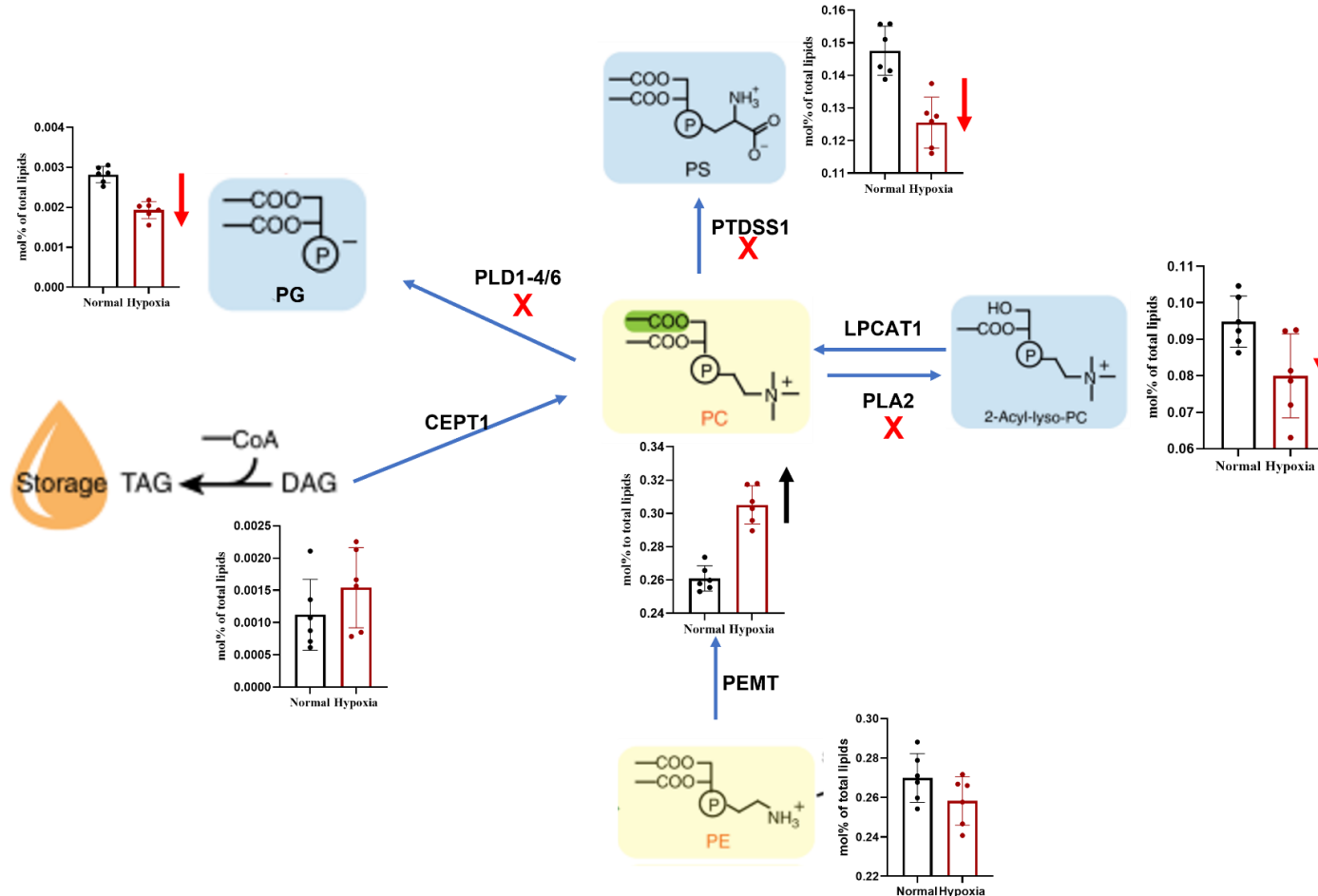


Figure 4-6. The metabolism pathway for phosphocholine. CEPT1: Choline/Ethanolamine Phosphotransferase 1; PLD1-4/6: phospholipase D1-4,6; PTDSS1: Phosphatidylserine Synthase; LPCAT1: Lysophosphatidylcholine Acyltransferase 1; PLA2: Phospholipase A2; PEMT: phosphatidylethanolamine N-methyltransferase

4.3.3 The effect of oxygen deprivation on membrane lipids in vesicle

4.3.3.1 The lipid composition in vesicle solution

The vesicle samples were analyzed by immunoblotting with antibodies against standard cargo proteins in COPII vesicles, Sec22B (a tSNARE) and ERGIC53, the result indicted the presence of COPII vesicles (**Figure 4-7**). Before the study of lipid levels in the vesicle under hypoxia, the lipid profile of vesicle was investigated. We found that our vesicle fraction was enriched in PLs, e.g., PC, PI, and PE, which occupied around 96 mol% of the total lipid contents (**Figure 4-8**). The level of PL was much larger than that of LPL of which abundance represented only around 2 mol%. The total level for PL and LPL was around 98% of the total lipid abundance, while there were much fewer GL and SL in our vesicle fraction. Interestingly, after the comparison of lipid composition in SI cell and vesicle sample, we found that the isolated vesicle sample were selectively enriched in PE, PI and PG (**Figure 4-9**). It was reported that PI was related to the signal transduction and membrane transport. PE is the most well-studied lipid, and it is known to produce intrinsic negative curvature of the membrane and facilitate to membrane fusion.

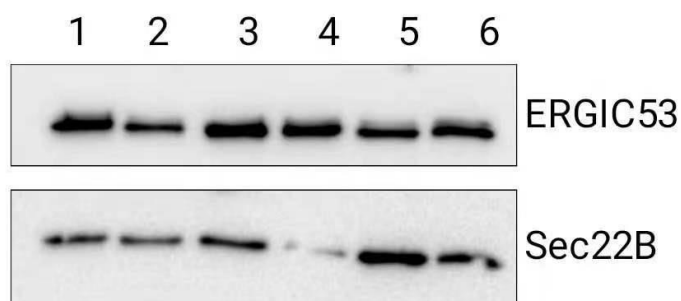


Figure 4-7. Vesicle fractions were analyzed by immunoblot using the indicated antibodies. 1, 2 and 3 represented three biological repeats for control group, and 4, 5 and 6 were three representative samples for hypoxic group

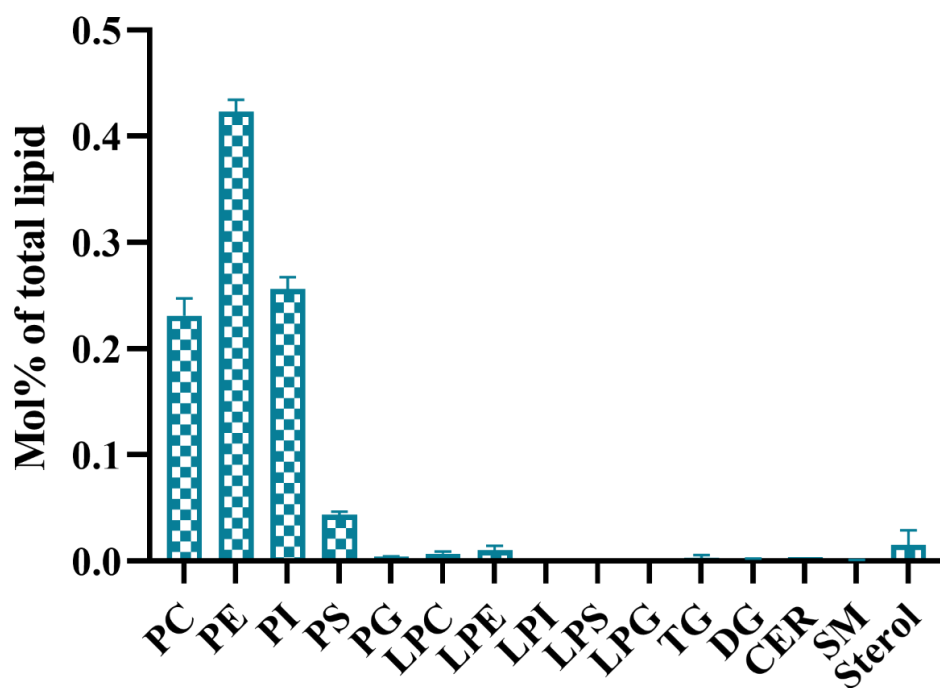


Figure 4-8. The composition of the quantitative lipid classes in mol% of all lipids in the vesicle samples

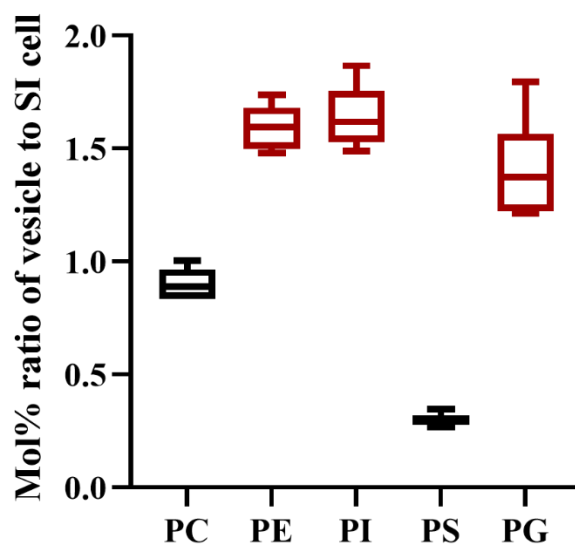


Figure 4-9. The ratio of lipid compositions (mol percentage) of vesicle to that of SI cell

4.3.3.2 Changes in lipid profile in vesicles under hypoxia

Because the levels of GP and SL were significantly altered in SI cells after oxygen deprivation, we mainly focused on these lipid classes to investigate the impact of hypoxia on the lipid composition of vesicle.

GP mainly comprises PL and LPL, the former of which includes PC, PE, PI, PS and PG, and the latter consists of LPC, LPE, LPI, LPS and LPG. We found that only PE which could induce the negative membrane curvature was decreased in vesicles from the hypoxic cells (**Figure 4-10**). Therefore, we speculated that the membrane rigidity of vesicle might increase under hypoxia.

To further validate our speculation, we studied the fatty acyl chains in vesicle fraction from both normal and hypoxic cells and calculated the abundance of the different fatty acyl chains. Because the fatty acyl chains were basically from GL and GP, the total carbon number of the fatty acyl chains from GL and GP were calculated to reflect the length of fatty acyl chain, and the combination of GL and GP was named GLP in this study. We found that the length distribution of the fatty acyl chain was equal between the vesicles and semi-intact cells under normal culture condition, with the length between 30 and 40 carbons being the most abundant, followed by fatty acyl chains of > 40 carbons and those of < 30 carbons (**Figure 4-11**). Strikingly, our data revealed that GLPs with very long fatty acyl chains ($C > 40$) were much more prominently found in the hypoxic vesicles than the normal vesicles, while the level of the corresponding fatty acyl chains in SI cells decreased after hypoxia. Moreover, there was significant decrease of level for medium fatty acyl chains ($30 \leq C \leq 40$) in the vesicles with the stimulation by low oxygen. This result suggested more very long fatty acyl chains were recruited in the vesicle after hypoxia, which further validated that hypoxia increase the rigidity of the vesicle membrane. Because it has been reported that the larger length of fatty acyl chain induced the higher rigidity of membrane and then reduced the membrane curvature.¹³⁸

To further validate this hypothesis, the Nanosight and transmission electron microscope (TEM) were used to characterize vesicles, and when we performed the vesicle budding reaction for the HeLa cells cultivated with low level of oxygen, the average diameter of the vesicle produced in the absence of oxygen was significantly increased, so we speculated that the membrane rigidity of vesicle was increased by hypoxia. Moreover, TEM indicated that membrane of hypoxic vesicle was thicker than that of normal vesicle (**Figure 4-12**), which was consistent with the result that longer FA chain was observed in hypoxic vesicle. Overall, Membrane rigidity could be increased by longer fatty acyl chains and membrane fusion could be limited by the decreased level of PE, which thereby hindered the fission, budding and fusion of the membrane. Our result showed that not only the decreased level of PE and smaller size of vesicle, but also the longer length of fatty acyl chains were seen in the hypoxia-stimulated vesicles. Therefore, all these results indicated that the vesicle membrane might turned out to be more rigid under hypoxia.

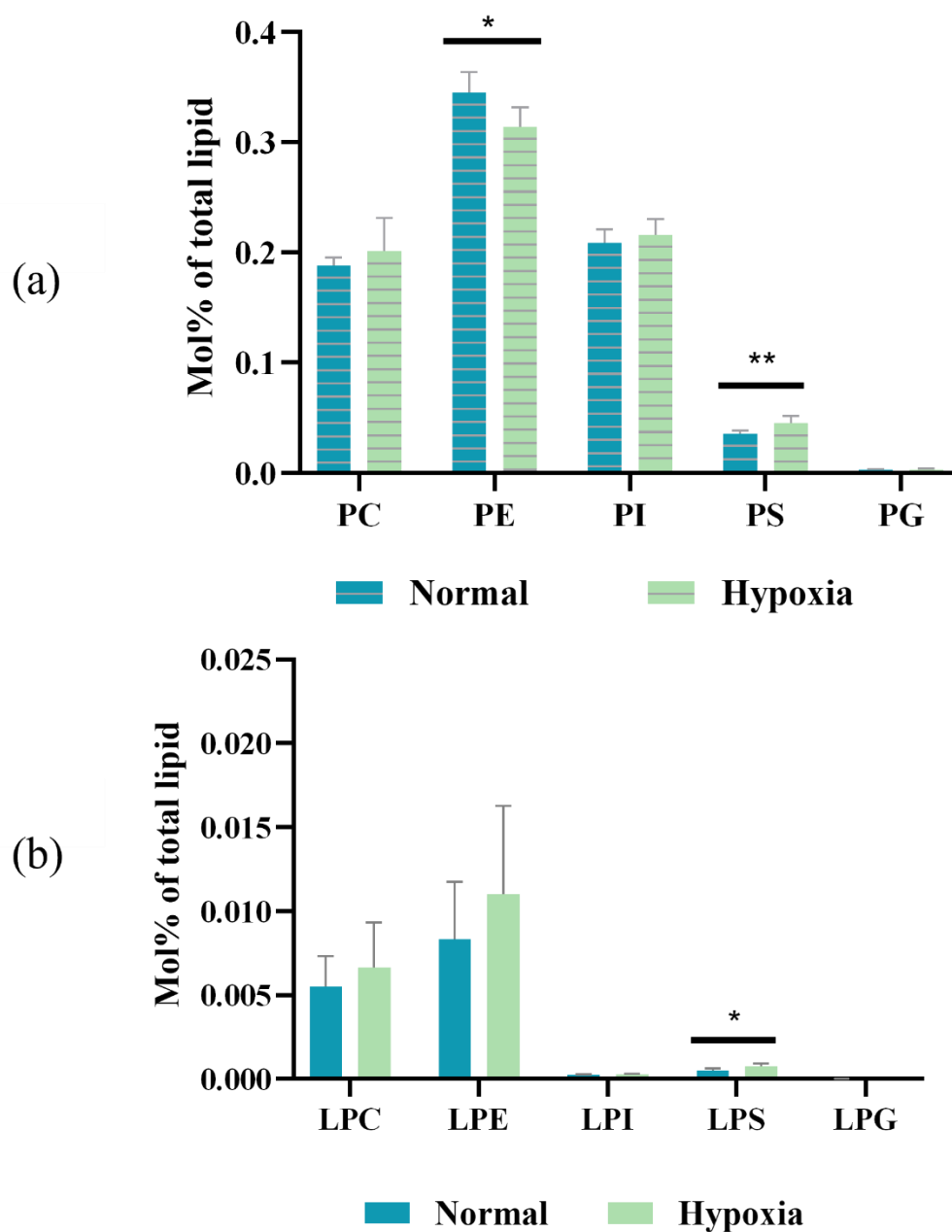


Figure 4-10. The profile of the main phospholipids together with the lyso-phospholipids detected by MS (mol%). Error bars represented SD. (a) Phospholipid and (b) lysophospholipid profiles of vesicles before and after oxygen deprivation (n=6). * $p < 0.05$, ** $p < 0.01$.

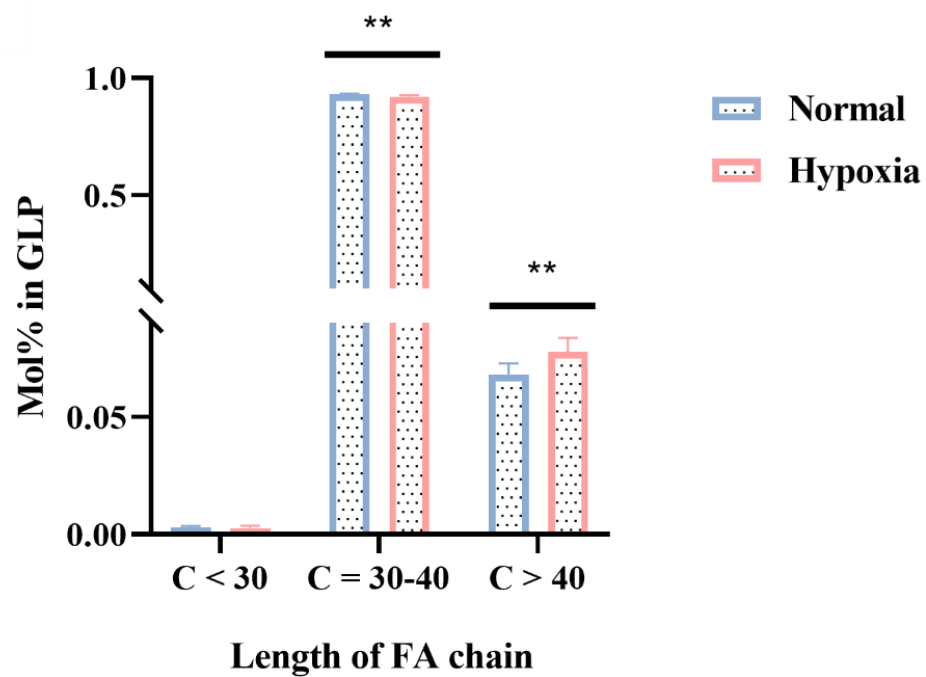


Figure 4-11. The profile of length of fatty acyl chains from vesicle fraction cultured in either normal (blue) or hypoxic (pink) conditions. The total length is calculated in mol% of the GLP

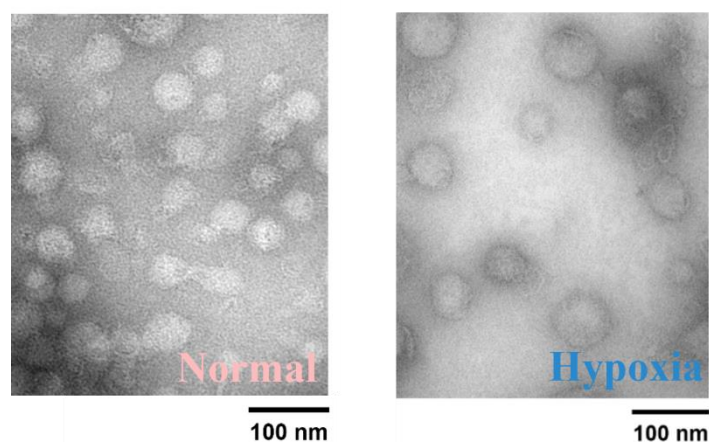


Figure 4-12. Structure of vesicle under normal (pink) and hypoxic condition (blue)

In order to gain more insight into the behavior of unsaturation of fatty acyl chain upon oxygen deprivation, we studied the number of double bonds in GLPs from vesicle samples. We found the almost exclusive polyunsaturated fatty acyl chains with ≥ 3 double bonds, and these polyunsaturated fatty acyls were selectively enriched in the vesicles when compared to SI cells under normal culture condition (**Figure 4-13**). This may be because polyunsaturated FA could facilitate the membrane fusion, which is required for the vesicle trafficking. However, the UI(unsaturation index) of fatty acyl chains were remarkably altered in vesicle when less oxygen was provided. The polyunsaturated fatty acyl chains with more than three C=C were incredibly increased, but the trend was reversed for those with fewer double bonds. Because the desaturation of *de novo* synthesized lipids by stearoyl-Coenzyme A (CoA) desaturases (SCDs), such as SCD1, requires oxygen. O₂ deprivation inhibits this enzymatic reaction, reducing the UI of lipid within cells, that might be the reason of the increased production of FA with larger UI in vesicle to maintain the intracellular balance of saturation of FA chains after hypoxia.

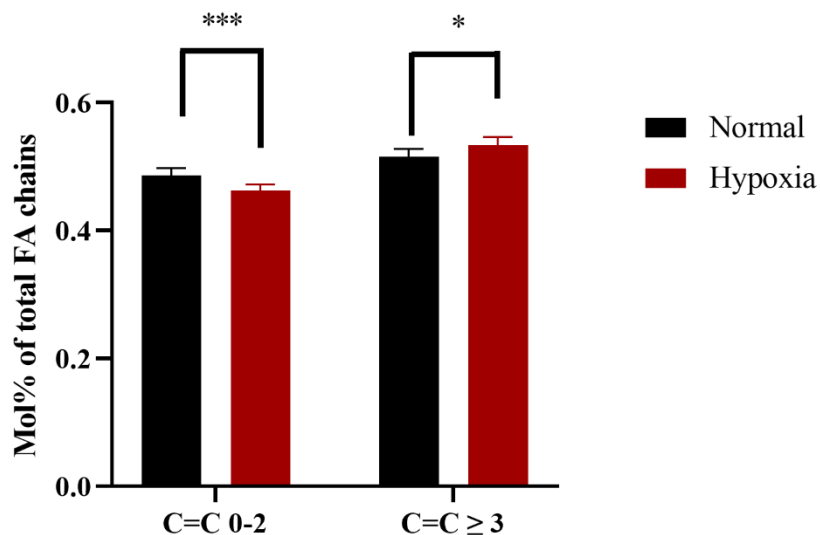


Figure 4-13. The profile of the saturation of fatty acyl chains from vesicle cultured in both normal and hypoxic condition. Total number of double bonds in GLPs was given as the total number of double bonds in fatty acyl chains in mol % of GLPs.

SLs mainly comprised SM and less abundant ceramide (**Figure 4-14**), and they are regarded as bioactive lipids in the biological organism. No marked difference of SM level could be observed between vesicles of normal cells and vesicles of hypoxic cells. Different from SM, the level of ceramide was significantly higher in vesicles from cells that cultivated under low oxygen when compared with those vesicles from Hela cells with sufficient O₂. And the result was shown in **Figure 4-14**. The level of (3'-sulfo)Galbeta-Cer(d18:1/16:0(2OH)), GlcCer(d18:1/26:0) and PI-Cer(d18:0/16:0(2OH)) were increased greatly in vesicle after hypoxia.

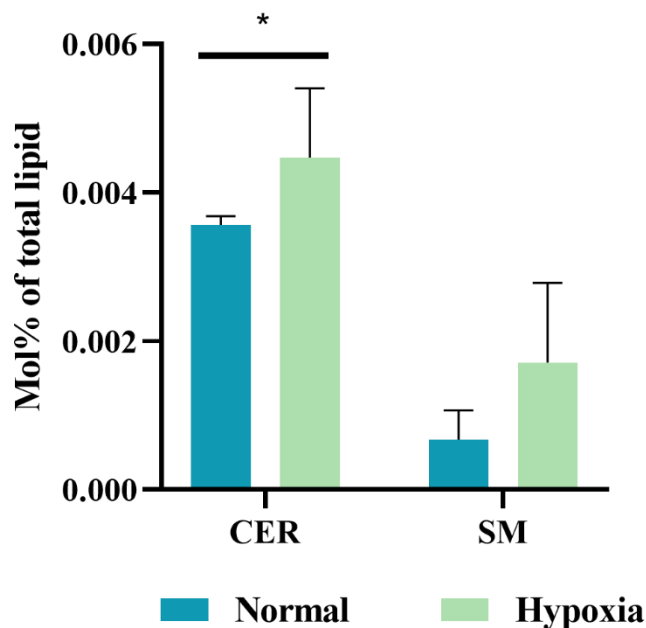


Figure 4-14. A box plot demonstrating the relative level of sphingolipids to the total lipid abundance for control (blue) and hypoxia-stressed (green) vesicle fraction. The superscript (*) shows significant differences of hypoxic group with respect to control group at $p < 0.05$ (n=6)

4.4 Conclusions

Hypoxia is the main characteristic of the microenvironment in tumor cell, and many studies were targeted on hypoxia to search for effective treatment of cancer. Vesicle trafficking was increasingly found to be closely related to the progress of diseases, but most of those studies focused on the transport of protein with few researchers aimed at the lipid trafficking within the membrane system. Therefore, in this project, SI cells from the permeabilized cells were used for the study of total membrane system. Vesicle samples were

budded in vitro for extraction, and LC-MS based-lipidomics was successfully established for analysis of the SI cell samples and vesicle samples after the optimization of preparation step, the lipid extraction method, and LC-MS parameters.

Membrane lipids, including GP and SL, are the major components of the plasma membrane and membrane system within cells. Interestingly, the level of PC, the most abundant membrane PL, increased significantly in SI cell due to insufficient oxygen. Although PC was the major synthesis material of PG, PS and LPC, when the level of PC increased after hypoxia, the levels of PG, PS and LPC were markedly decreased. In addition, PC was mainly produced by the conversion of DG and PE via CEPT1 and PEMT, respectively, and the levels of DG and PE were not changed in SI after hypoxic cultivation. These results suggested that hypoxia inhibited the catabolism pathway of PC, instead of the pathway of PC anabolism. In addition, we found that the ratio of LPL was significantly decreased, and the level of SL in SI cell was significantly enhanced under hypoxia, suggesting that there may be a close relationship between hypoxia and vesicular lipid compositions.

Our vesicle fraction was enriched in PLs, e.g., PC, PI, and PE, which occupied around 95 mol% of the total lipid contents. The level of phospholipid was much larger than that of LPL and sterol, while there was much less GL and SL in our vesicle fraction. Interestingly, after the comparison of lipid composition in SI cell and vesicle sample, we found that the isolated vesicle sample were selectively enriched in PE, PI and PG. The enrichment of these lipids might be closely related to the specific function of vesicle. Different from other GP, only the level of PE which could induce the negative membrane curvature were decreased in vesicles from the hypoxic cells, and the vesicle size was significantly decreased under hypoxia, so we speculated that the curvature and membrane fusion of vesicle might be decreased by hypoxia. Strikingly, our data showed that more very long fatty acyl chains ($C > 40$) were found significantly in vesicles cultivated under hypoxia compared to normal condition. However, under hypoxia stimulation, the level of relative short fatty acyl chain with carbon number between 30 and 40 in vesicles were markedly reduced. This result suggested that vesicle contained more very long fatty acyl chain after hypoxia, which further confirmed that hypoxia might increase the stiffness of vesicle membrane. Besides, the Nanosight and TEM result showed that the vesicle size was increased, and the vesicle membrane became thicker in hypoxic vesicles, which provided additional information to

prove that hypoxia induced higher rigidity of vesicle membrane. Moreover, the level of polyunsaturated fatty acyl (PUFA) chains with more than three C=C increased incredibly, but the trend was reversed for saturated FA and those FA with one or two double bonds. This suggested that vesicle might produce more polyunsaturated FAs as a response of hypoxia and to maintain the balance of saturation within the hypoxic cells. Furthermore, the levels of ceramide was significantly elevated in vesicles under low oxygen when compared with those vesicles with sufficient O₂. The level of (3'-sulfo)Galbeta-Cer(d18:1/16:0(2OH)), GlcCer(d18:1/26:0) and PI-Cer(d18:0/16:0(2OH)) in vesicle greatly increased after hypoxia.

Chapter 5: Overall conclusions and prospects

IS patients should be treated with tPA within the golden hours (3-4.5 hours) to avoid the increasing risk of disability and mortality. However, this is restricted by the development of rapid and reliable techniques for IS diagnosis. Therefore, two methods to differentiate plasma samples of IS patients from health controls were developed for the IS diagnosis by using LC-MS and DART-MS, respectively. In addition, lipidomics analysis of vesicles was also conducted as well to study the lipid compositions of vesicles under hypoxia stimulation in order to understand the general process of lipid trafficking.

The LC-MS platform used in the lipidomic study of clinical plasma samples was tested to be highly reproducible prior to sample analysis. PCA score plot of the normalized data demonstrated good distinction between IS group and HC group, suggesting that IS could be distinguished by lipid profiles. After comprehensive analysis of the discovery dataset and evaluation of the validation dataset, a total of 20 lipids were selected as the potential biomarkers. Among these 20 potential markers, the levels of FA(20:0), LPC(O-20:0), LPS(O-15:0), nordeoxycholic acid, 23:0 sterol, and 14:0 cholesterol ester were significantly increased in the plasma of IS patients compared with their levels in the plasma of the healthy controls. In contrast, the remaining 14 lipids

showed reductions in IS patients, including n-butyl arachidonoyl amine, FA(20:3), TG(14:0/18:2/18:3), TG(15:1/19:1/20:5), PS(P-20:0/20:5), PI(17:0/22:1), PC(O-38:9), PI(O-20:0/15:1), PE(O-16:0/20:3), PC(18:2/22:6), PS(O-18:0/19:1), PS(20:0/22:4), glycinoprenol-9, and Cer(m18:1/24:1). Furthermore, the prediction ability of the above 20 lipids for IS was assessed in the discovery dataset using binary logistic regression. The results showed that a combination of three lipids (FA(20:3), PS(O-18:0/19:1) and LPS (O-15:0)) effectively discriminated IS and HC in the discovery dataset with AUC value, sensitivity, and the specificity of 1.000, 100% and 100%, respectively. Therefore, this combination of FA(20:3), PS(O-18:0/19:1) and LPS (O-15:0) was thus defined as a “biomarker panel”. Moreover, the differentiation ability of this biomarker panel was also good in the validation dataset, and the AUC value of the biomarker panel was 0.963, with sensitivity and specificity of 100% and 87.5%, respectively. The development of this new, rapid and accurate method could shorten the detection time (around 1.5 hours for each sample) with high accuracy of IS diagnosis, thereby shortening the time of tpA treatment and reducing the mortality and morbidity of IS patients. In addition, potential IS-related lipid markers may be the target for early detection and timely treatment of IS. However, the sample size recruited for this LC-MS-based method was limited due to the outbreak of covid-19, so more samples

should be recruited to increase confidence of the diagnosis result of IS in the future.

In order to further shorten the analysis time for IS diagnosis, DART-MS-based approach was developed for the direct analysis of plasma from IS patients and health controls. After systematic optimization, the temperature of 350 °C, the DS distance of 3 cm, and the grid voltage of 100 V were chosen as the optimal DART parameters for the experiment. Compared with the TIC-based normalization, the InterS normalization produced more reliable results and was thus selected as the normalization method to reduce the signal variation of DART-MS analysis. The established DART-MS platform was then applied in the differentiation of IS patients (n=53) and HC (n=28). In the discovery phase, an OPLS-DA model with good differentiation of HC and IS patients was established (R^2 and Q^2 of 0.947 and 0.745). And the prediction power of the OPLS-DA model for both the HC group and IS group in discovery dataset was 100%. Subsequently, in the validation phase, an additional dataset of 9 health control samples and 17 IS patient samples was used for the validation of the established OPLS-DA model. The validation result showed that prediction abilities of 100% and 94.1% were achieved for the HC group and IS group, respectively. Therefore, the established OPLS-

DA model could be used for the rapid differentiation of IS and health control after plasma sample was directly analyzed by means of DART-MS. Compared with the previously established LC-MS method, DART-MS method required less time for sample preparation and sample analysis (~5 minutes for each sample). With less detection time, DART-MS could still achieve a comparable overall probability of IS prediction (96.2%) to the LC-MS-based approach (96.3%). Therefore, the DART-MS-based approach was suitable for the rapid analysis of plasma from IS patients. In terms of detected compounds, LC-MS could detect lipids from non-polar to polar lipids, but DART-MS primarily detected the non-polar lipids. Moreover, the sensitivity and reproducibility of DART were lower than those of LC-MS. Therefore, these two approaches could be employed complementarily for rapid and accurate differentiation of IS and HC, which could accelerate tPA treatment to reduce mortality and disability rate of IS patients. However, restricted sample size was also a limitation in this study, so plasma samples from more subjects should be recruited in the further study to make the result more convincing.

Hypoxia is the main characteristic of the microenvironment in tumor cell, and many studies were targeted on hypoxia to search for effective treatment of cancer. Vesicle trafficking was increasingly found to be closely related to the

progress of diseases, but most of those studies focused on the transport of protein with few researchers aimed at the lipid trafficking within the membrane system. Therefore, in this project, SI cells from the permeabilized cells and vesicle solution budded in vitro were used for the study of the total membrane system and vesicle membrane, respectively. And LC-MS based-lipidomics platform was successfully established for analysis of the SI cell samples and vesicle samples after the optimization of sample preparation steps, the lipid extraction method, and LC-MS parameters.

Glycerophospholipid is the major component of the membrane system of cells. Interestingly, the level of one of the most abundant membrane PL, i.e., PC, was increased significantly in SI cells treated with insufficient oxygen. However, the levels of PG, PS and LPC decreased markedly after hypoxia, indicating that the hypoxia inhibited the catabolism pathway of PC. By contrast, the levels of DG and PE were not changed in SI after hypoxic cultivation, which suggested that pathway of PC anabolism was not influenced by oxygen deprivation. Since we found that the levels of SL and LPL, which affect vesicular transportation, were significantly altered in SI cells under hypoxia, we further explored the relationship between hypoxia and vesicular trafficking. Interestingly, comparing the lipid composition in SI cells

and vesicle samples, we found that the isolated vesicle samples were selectively enriched in PE, PI and PG, which might be closely related to the specific function of vesicle. Different from other GPs, the level of PE, which could increase the negative membrane curvature, was decreased in vesicles from the hypoxic cells, so we presumed that the rigidity of vesicle membrane might be increased by hypoxia. Strikingly, our data showed that significantly more very long fatty acyl chains with C=C of more than 40 were observed in vesicles cultivated under hypoxia compared to normal condition, which further confirmed our speculation that hypoxia increase the membrane stiffness to decrease the vesicle membrane curvature and membrane fusion. Moreover, we could also observe that the vesicle size was significantly increased, and vesicle membrane was thicker under hypoxia, which further provided the evidence that hypoxia induced the increased membrane rigidity of vesicle. This study disclosed the impact of hypoxia on lipid composition of intracellular membrane system and vesicle, and might provide a new perspective on the mechanism study of cancer. In addition, although the vesicular transport of proteins has been reported in recent years, the mechanism of vesicular lipid trafficking remains elusive. Therefore, a general investigation of vesicle transportation of lipids is of great importance. However, there are some limitations in this study. First, different ratios of PE

and lipids with PUFA chains should be added when vesicle budding assay is performed to validate the effect of PE and PUFA on rigidity of vesicle membrane. Second, since COPII vesicle is mainly responsible for the transport substance from ER to Golgi apparatus, further lipidomic studies of both ER membrane and Golgi membrane are required.

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